

BOOK 6

DEFORMATION

6 Deformation of solids

6.1 Stress and strain

Candidates should be able to:

- 1 understand that deformation is caused by tensile or compressive forces (forces and deformations will be assumed to be in one dimension only)
- 2 understand and use the terms load, extension, compression and limit of proportionality
- 3 recall and use Hooke's law
- 4 recall and use the formula for the spring constant $k = F/x$
- 5 define and use the terms stress, strain and the Young modulus
- 6 describe an experiment to determine the Young modulus of a metal in the form of a wire

6.2 Elastic and plastic behaviour

Candidates should be able to:

- 1 understand and use the terms elastic deformation, plastic deformation and elastic limit
- 2 understand that the area under the force–extension graph represents the work done
- 3 determine the elastic potential energy of a material deformed within its limit of proportionality from the area under the force–extension graph
- 4 recall and use $E_p = \frac{1}{2}Fx = \frac{1}{2}kx^2$ for a material deformed within its limit of proportionality

1 9702/1/M/J/02/Q22

Which of the following correctly defines the terms *stress*, *strain* and *Young modulus*?

	stress	strain	Young modulus
A	(force) x (area)	(extension) x (original length)	(stress) / (strain)
B	(force) x (area)	(extension) / (original length)	(stress) x (strain)
C	(force) / (area)	(extension) / (original length)	(stress) / (strain)
D	(force) / (area)	(extension) x (original length)	(stress) x (strain)

2 9702/1/M/J/02/Q23

A wire is stretched by 8 mm when a load of 60 N is applied.

What will be the extension of a wire of the same material having four times the cross-sectional area and twice the original length, when the same load is applied?

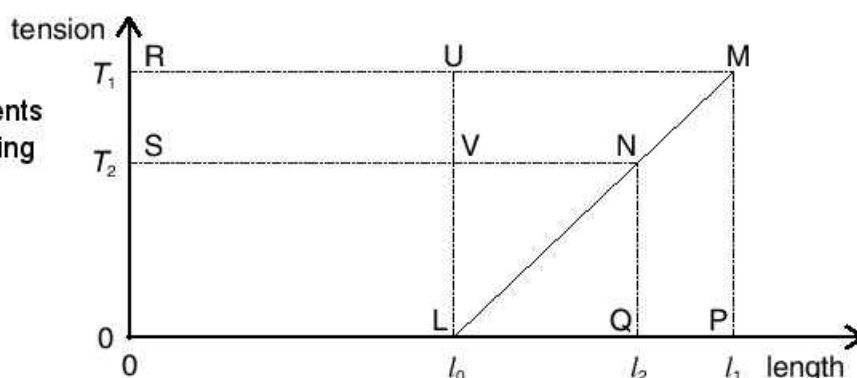
- A 2 mm B 4 mm C 8 mm D 16 mm

3 9702/1/M/J/02/Q24

The tension in a spring of natural length l_0 is first increased from zero to T_1 , causing the length to increase to l_1 . The tension is then reduced to T_2 , causing the length to decrease to l_2 (as shown).

Which area of the graph represents the work done by the spring during this reduction in length?

- A MLP B MNQP
C MNSR D MPLU



4 9702/01/M/J/04/Q19

Comparing the properties of solids, liquids and gases, which option is correct?

	property	solids	liquids	gases
A	ordering of molecules	high	not so high	random
B	spacing of molecules	close	far	far
C	translation of molecules	no	no	yes
D	vibration of molecules	no	yes	yes

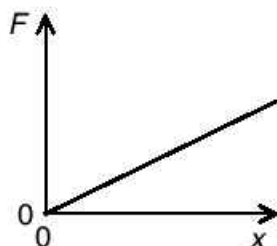
5 9702/01/O/N/03/Q21

What is the Young modulus of a metal?

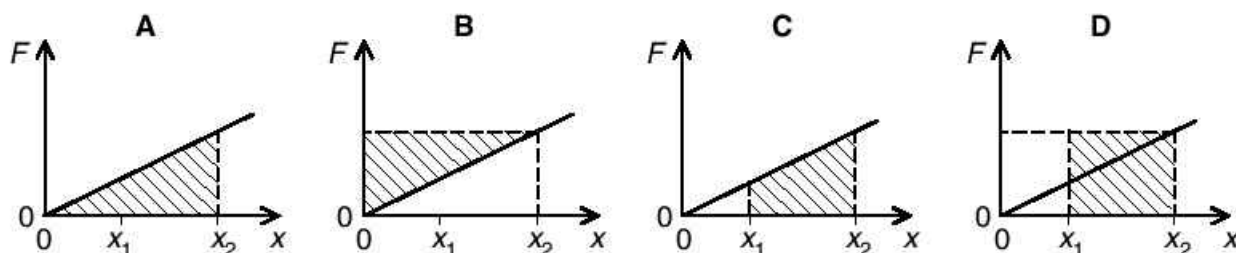
- A extension / force C strain / stress
B force / extension D stress / strain

6 9702/1/O/N/02/Q23

The variation of the extension x of a spring with applied force F is shown.



Which shaded area represents the work done when the extension is increased from x_1 to x_2 ?



7 9702/1/O/N/02/Q24

Two springs P and Q both obey Hooke's law. They have spring constants $2k$ and k respectively.

The springs are stretched, separately, by a force that is gradually increased from zero up to a certain maximum value, the same for each spring. The work done in stretching spring P is W_P , and the work done in stretching spring Q is W_Q .

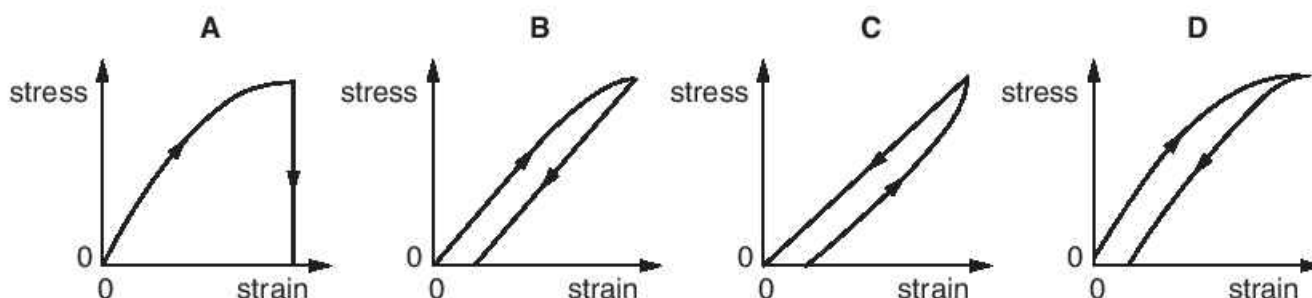
How is W_P related to W_Q ?

- A $W_P = \frac{1}{4}W_Q$ B $W_P = \frac{1}{2}W_Q$ C $W_P = 2W_Q$ D $W_P = 4W_Q$

8 9702/01/M/J/03/Q19

A suspended copper wire is gradually loaded until it is stretched just beyond the elastic limit, and it is then gradually unloaded.

Which graph (with arrows indicating the sequence) best illustrates the variation of the tensile stress with longitudinal strain?



9 9702/01/M/J/03/Q21

What is the ultimate tensile stress of a material?

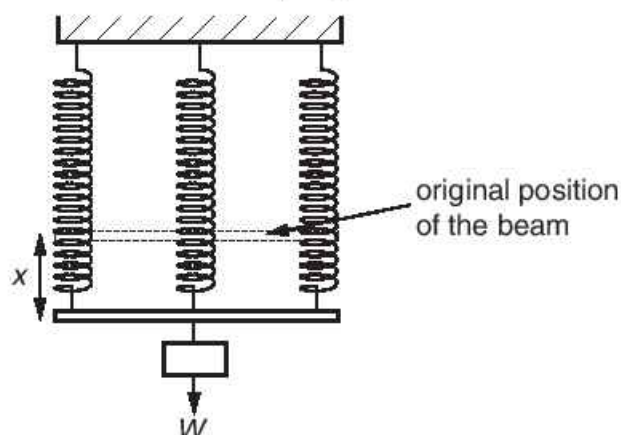
- A the stress at which the material becomes ductile
B the stress at which the material breaks
C the stress at which the material deforms plastically
D the stress at which the material reaches its elastic limit

10 9702/01/M/J/03/Q22

A beam, the weight of which may be neglected, is supported by three identical springs. When a weight W is hung from the middle of the beam, the extension of each spring is x .

The middle spring and the weight are removed.
What is the extension when a weight of $2W$ is hung from the middle of the beam?

- A $\frac{3x}{2}$ B $\frac{4x}{3}$
C $2x$ D $3x$

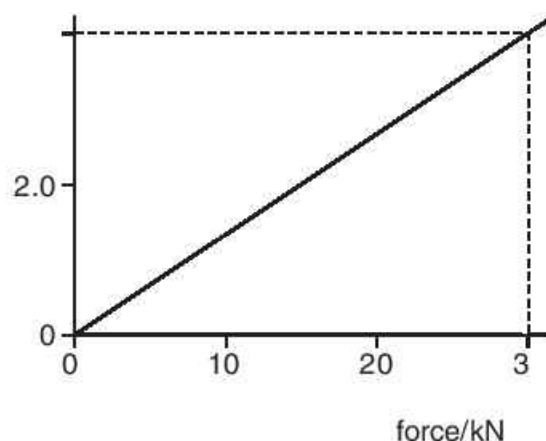


11 9702/01/O/N/03/Q22

The graph shows how the extension of a spring varies with the force used to stretch it.

What is the strain energy stored in the spring when the extension is 4.0 cm?

- A 60 J B 120 J
C 600 J D 1200 J



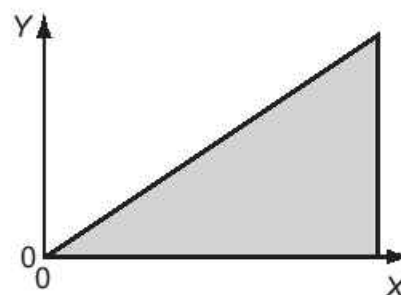
12 9702/01/M/J/04/Q22

The graph shown was plotted in an experiment on a metal wire.

The shaded area represents the total strain energy stored in stretching the wire.

How should the axes be labelled?

	Y	X
A	force	extension
B	mass	extension
C	strain	energy
D	stress	strain



13 9702/01/M/J/04/Q23

Nylon breaks when the stress within it reaches 1×10^9 Pa.

Which range includes the heaviest load that could be lifted by a nylon thread of diameter 1 mm?

- A 2 N to 20 N C 200 N to 2000 N
B 20 N to 200 N D 2000 N to 20 000 N

14 9702/01/O/N/04/Q22

The table shows a load applied to four wires and the cross-sectional area of each. Which of the wires is subjected to the greatest stress?

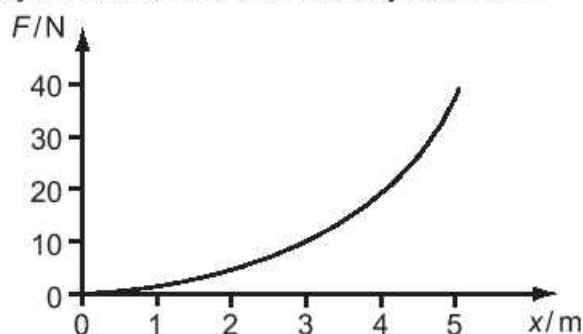
	load / N	cross-sectional area / mm ²
A	1500	0.25
B	2000	1.0
C	3000	0.56
D	5000	2.3

15 9702/01/O/N/04/Q23

The force F required to extend a sample of rubber by a distance x is found to vary as shown.

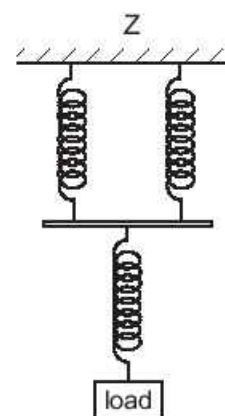
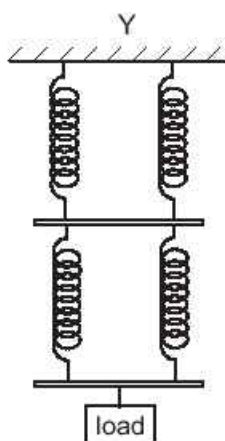
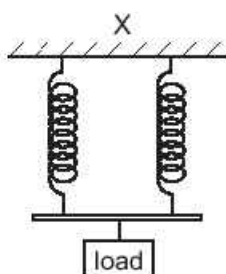
The energy stored in the rubber for an extension of 5 m is

- A less than 100 J.
- B 100 J.
- C between 100 J and 200 J.
- D more than 200 J.



16 9702/01/M/J/05/Q20

A number of similar springs, each having the same spring constant, are joined in three arrangements X, Y and Z. The same load is applied to each.



What is the order of increasing extension for these arrangements?

	smallest	→	largest
A	X	Y	Z
B	Z	X	Y
C	Z	Y	X
D	Y	X	Z

17 9702/01/O/N/06/Q23

What is the unit of the Young modulus?

- A Nm⁻¹
- B Nm
- C Nm⁻²
- D Nm²

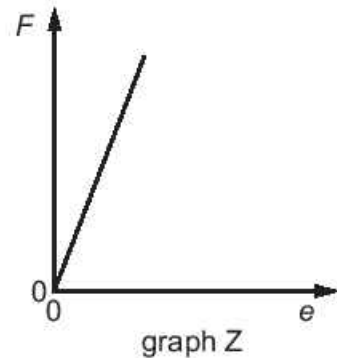
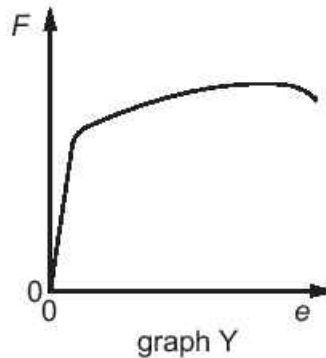
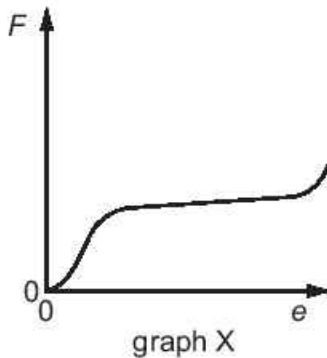
18 9702/12/O/N/13/Q21*

Which two substances are normally both crystalline?

- A copper and diamond C diamond and glass
B copper and glass D diamond and rubber

19 9702/13/M/J/14/Q24

Cylindrical samples of steel, glass and rubber are each subjected to a gradually increasing tensile force F . The extensions e are measured and graphs are plotted as shown below.



Which row correctly relates the graphs to the materials?

	steel	glass	rubber
A	X	Y	Z
B	X	Z	Y
C	Y	X	Z
D	Y	Z	X

20 9702/01/M/J/05/Q22*

Two steel wires P and Q have lengths l and $2l$ respectively, and cross-sectional areas A and $\frac{A}{2}$ respectively. Both wires obey Hooke's law.

What is the ratio $\frac{\text{tension in P}}{\text{tension in Q}}$ when both wires are stretched to the same extension?

- A $\frac{1}{4}$ B $\frac{1}{2}$ C $\frac{2}{1}$ D $\frac{4}{1}$

21 9702/01/O/N/05/Q19*

When white sugar granules are heated, they melt. When the melt is cooled quickly, a brittle solid form of toffee is produced.

How does the structure of the sugar change?

- A amorphous to polymeric C crystalline to polymeric
B crystalline to amorphous D polymeric to amorphous

22 9702/01/M/J/06/Q21

In describing the behaviour of a spring, the spring constant is used.

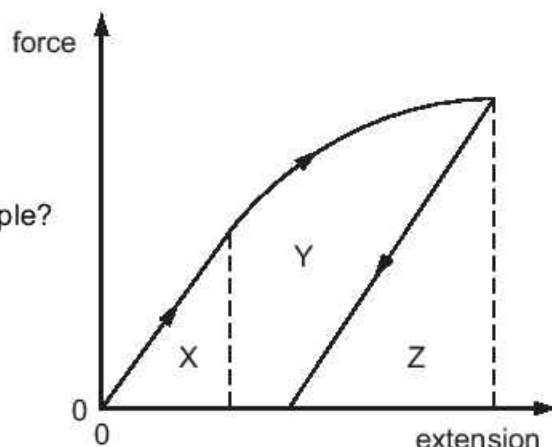
Different loads are used to extend the spring by different amounts.

To find the spring constant, which quantities are required?

- A the elastic limit and the loads
B the elastic limit, extensions and the length of the spring
C the loads and the extensions of the spring
D the loads and the length of the spring

23 9702/01/O/N/05/Q20

A ductile material is stretched by a tensile force to a point beyond its elastic limit. The tensile force is then reduced to zero. The graph of force against extension is shown below.



Which area represents the net work done on the sample?

- A X B X + Y
C Y + Z D Z

24 9702/01/O/N/05/Q21

A wire stretches 8 mm under a load of 60 N.

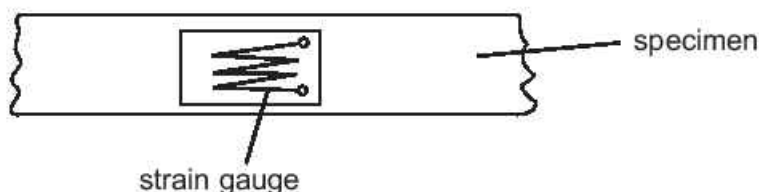
A second wire of the same material, with half the diameter and a quarter of the original length of the first wire, is stretched by the same load.

Assuming that Hooke's law is obeyed, what is the extension of this wire?

- A 1 mm B 4 mm C 8 mm D 16 mm

25 9702/01/O/N/05/Q33

Tensile strain may be measured by the change in electrical resistance of a strain gauge. A strain gauge consists of folded fine metal wire mounted on a flexible insulating backing sheet. The strain gauge is firmly attached to the specimen, so that the strain in the metal wire is always identical to that in the specimen.



When the strain in the specimen is increased, what happens to the resistance of the wire?

- A It decreases, because the length decreases and the cross-sectional area increases.
B It decreases, because the length increases and the cross-sectional area decreases.
C It increases, because the length decreases and the cross-sectional area increases.
D It increases, because the length increases and the cross-sectional area decreases.

26 9702/01/M/J/06/Q20

In an experiment to demonstrate Brownian motion, smoke particles in a container are illuminated by a strong light source and observed through a microscope.

The particles are seen as small specks of light that are in motion.

What causes the Brownian motion?

- A collisions between the smoke particles and air molecules
B collisions between the smoke particles and the walls of the container
C convection currents within the air as it is warmed by the light source
D kinetic energy gained by the smoke particles on absorption of light

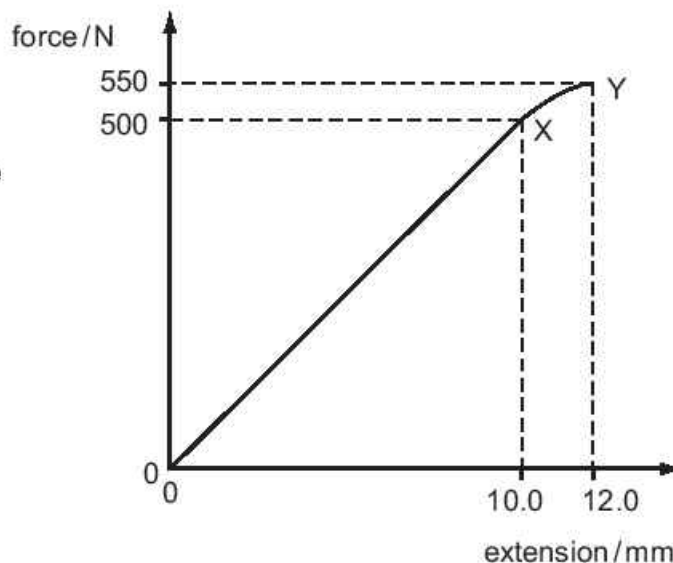
27 9702/01/M/J/06/Q22

The graph shows the behaviour of a sample of a metal when it is stretched until it starts to undergo plastic deformation.

What is the total work done in stretching the sample from zero extension to 12.0 mm?

Simplify the calculation by treating the region XY as a straight line.

- A 3.30 J B 3.55 J
C 3.60 J D 6.60 J



28 9702/01/O/N/06/Q22

What is represented by the gradient of a graph of force (vertical axis) against extension (horizontal axis)?

- A elastic limit B spring constant C stress D the Young modulus

29 9702/01/M/J/07/Q17

A piece of copper is drawn into a continuous wire.

What behaviour is the copper exhibiting?

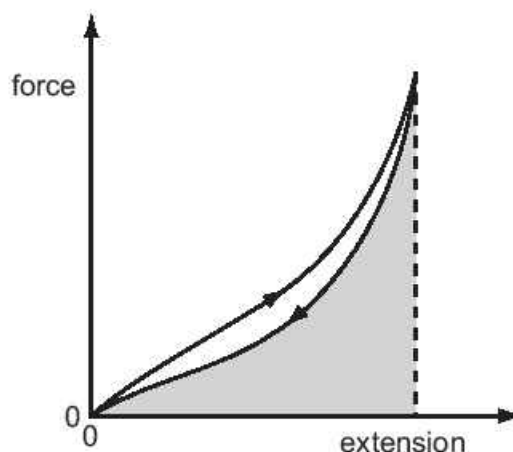
- A brittle only B elastic only C plastic only D both brittle and elastic

30 9702/01/M/J/07/Q18

The force-extension graph of a particular sample of rubber as a load is applied and then removed is shown.

What does the shaded area represent?

- A the energy transformed into heat during the complete cycle
B the recoverable elastic potential energy stored at maximum extension
C the work done on the sample while loading
D the work done on the sample while unloading



31 9702/12/O/N/13/Q23*

Which properties best describe modelling clay?

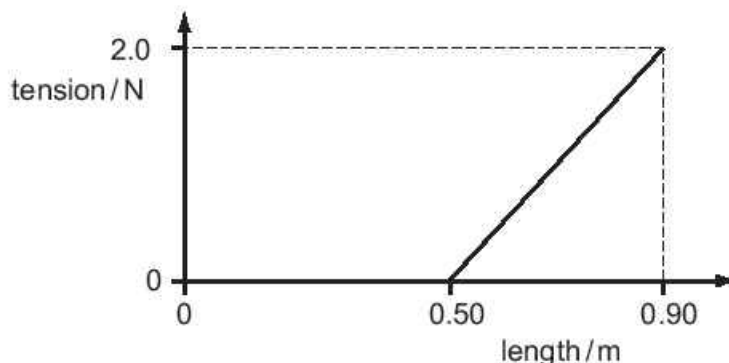
- A brittle and ductile C elastic and plastic
B ductile and elastic D plastic and ductile

32 9702/01/M/J/07/Q19

A spring of unextended length 0.50 m is stretched by a force of 2.0 N to a new length of 0.90 m. The variation of its length with tension is as shown.

How much strain energy is stored in the spring?

- A 0.40 J B 0.80 J
C 0.90 J D 1.8 J



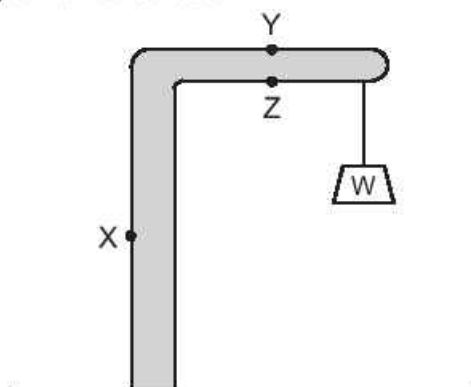
33 9702/01/M/J/07/Q20

A simple crane consists of a rigid vertical pillar supporting a horizontal beam.

A weight W is lifted by a rope at the end of the beam.

What are the forces at points X, Y and Z due to the weight W ?

	force at X	force at Y	force at Z
A	tension	compression	tension
B	tension	tension	compression
C	compression	tension	compression
D	compression	compression	compression



34 9702/01/O/N/07/Q19

What is plastic deformation?

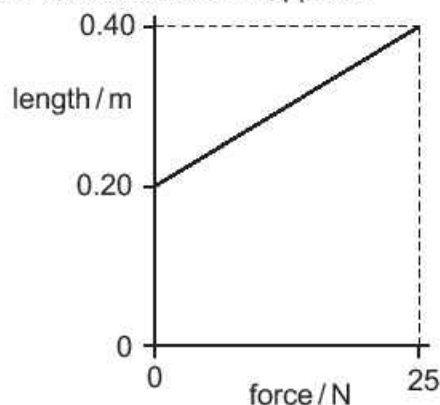
- A Plastic deformation occurs when strain is not proportional to stress but when the load is removed the material returns to its original length.
B Plastic deformation occurs if, when the load is removed, the material contracts but a permanent stretching has occurred.
C Plastic deformation occurs until the extension is no longer proportional to the load.
D Plastic deformation occurs when the material extends so that strain is directly proportional to stress.

35 9702/01/O/N/07/Q20

The graph shows how the length of a particular rubber cord varies as force is applied.

What is the maximum strain energy in this deformed rubber cord?

- A 2.5 J
B 5.0 J
C 7.5 J
D 10 J



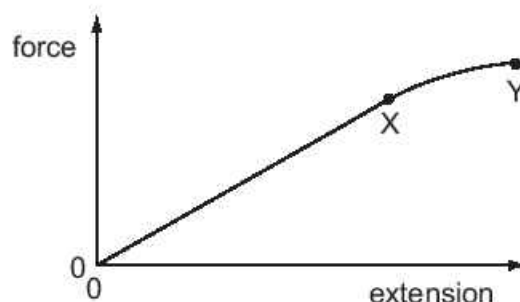
36 9702/01/M/J/08/Q22

A sample of metal is subjected to a force which increases to a maximum value and then decreases back to zero. A force-extension graph for the sample is shown.

When the sample contracts it follows the same force-extension curve as when it was being stretched.

What is the behaviour of the metal between X and Y?

- A both elastic and plastic
- B elastic but not plastic
- C plastic but not elastic
- D not elastic and not plastic

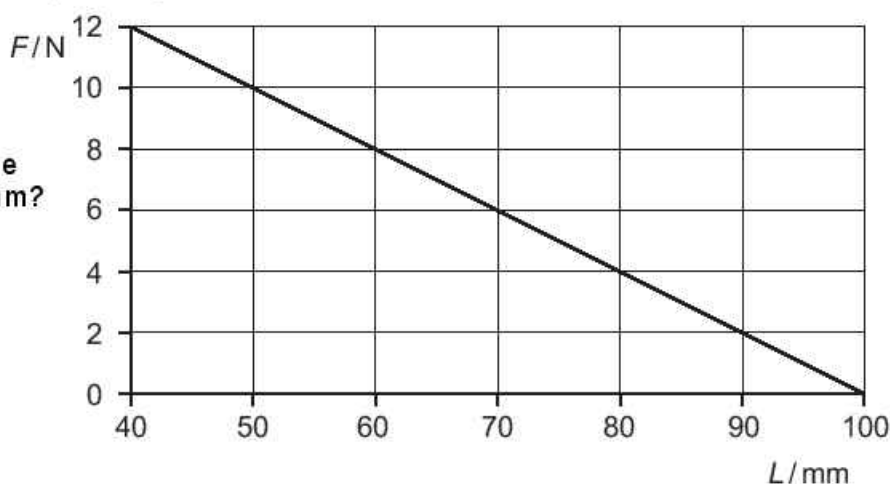


37 9702/01/M/J/08/Q23

A spring of original length 100 mm is compressed by a force. The graph shows the variation of the length L of the spring with the compressing force F .

What is the energy stored in the spring when the length is 70 mm?

- A 0.090 J B 0.21 J
- C 0.27 J D 0.63 J



38 9702/01/M/J/08/Q24

The Young modulus of steel is determined using a length of steel wire and is found to have the value E .

Another experiment is carried out using a wire of the same steel, but of twice the length and half the diameter.

What value is obtained for the Young modulus in the second experiment?

- A $\frac{1}{4}E$ B $\frac{1}{2}E$ C E D $2E$

39 9702/01/O/N/08/Q23

For a wire, Hooke's law is obeyed for a tension F and extension x . The Young modulus for the material of the wire is E .

Which expression represents the elastic strain energy stored in the wire?

- A $\frac{1}{2}Ex$ B Ex C $\frac{1}{2}Fx$ D Fx

40 9702/01/M/J/09/Q19

Four materials are formed into rods of the same dimensions.

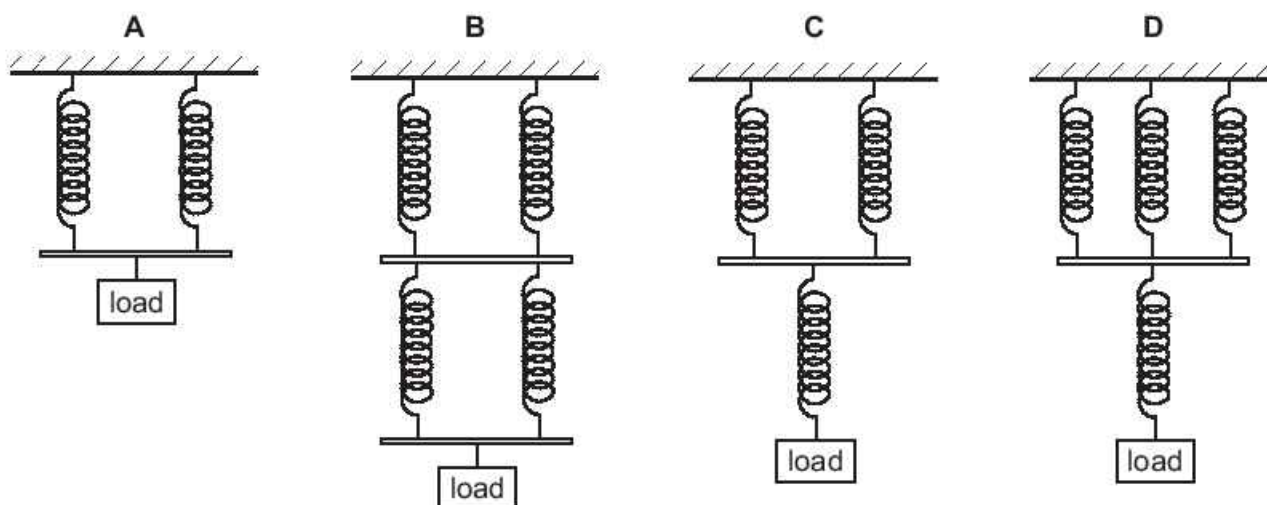
At room temperature, which can sustain the largest plastic deformation?

- A the ductile material aluminium C the brittle material glass
- B the brittle material carbon D the ductile material steel

41 9702/01/O/N/08/Q21

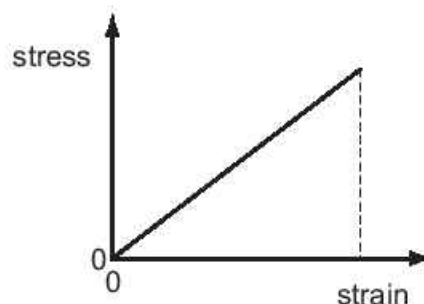
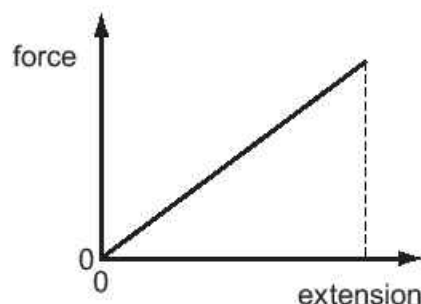
A number of similar springs, each having the same spring constant, are joined in four arrangements. The same load is applied to each.

Which arrangement gives the greatest extension?



42 9702/01/O/N/08/Q22

The graphs show how force varies with extension and stress varies with strain for the loading of a metal wire.



The Young modulus for this wire is equal to

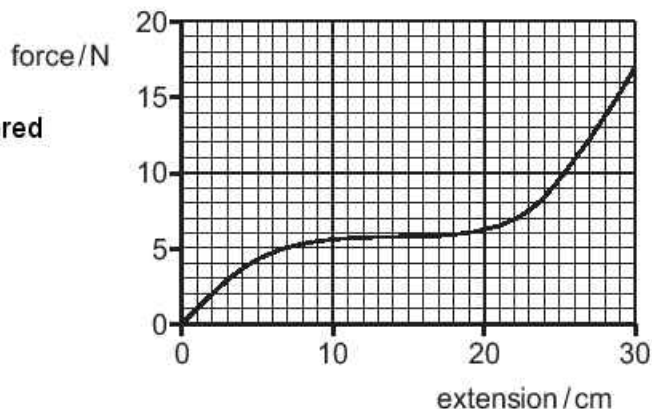
- A the gradient of the force-extension graph.
- B the area between the force-extension graph and the extension axis.
- C the gradient of the stress-strain graph.
- D the area between the stress-strain graph and the strain axis.

43 9702/01/M/J/09/Q21

A rubber band is stretched by hanging weights on it and the force-extension graph is plotted from the results.

What is the best estimate of the strain energy stored in the rubber band when it is extended 30 cm?

- A 2.0 J B 2.6 J
- C 5.1 J D 200 J



44 9702/01/M/J/09/Q20

Two steel wires P and Q have lengths l and $2l$ respectively, and cross-sectional areas A and $\frac{A}{2}$ respectively. Both wires obey Hooke's law.

What is the ratio $\frac{\text{tension in P}}{\text{tension in Q}}$ when both wires are stretched to the same extension?

- A** $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{2}{1}$ **D** $\frac{4}{1}$

45 9702/12/O/N/09/Q19

Which row best defines elastic and plastic behaviour of a material?

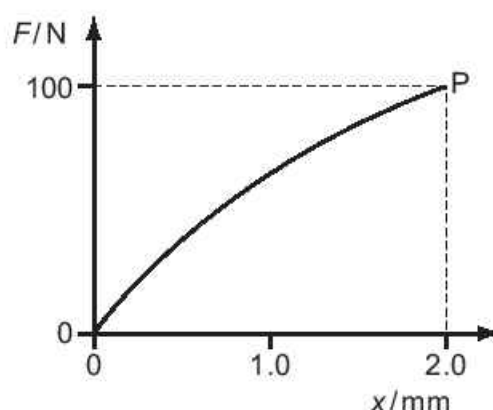
	elastic behaviour of a material	plastic behaviour of a material
A	extends only within the limit of proportionality	extends beyond the limit of proportionality
B	has a linear force-extension curve	has a horizontal force-extension curve
C	obeys Hooke's Law	extends continuously under a steady load
D	returns to its original shape and size	suffers permanent deformation

46 9702/12/O/N/09/Q20

The graph shows the non-linear force-extension curve for a wire made from a new composite material.

What could be the value of the strain energy stored in the wire when it is stretched to point P?

- A** 0.09 J **B** 0.10 J
C 0.11 J **D** 0.20 J



47 9702/12/O/N/09/Q21

A steel string on an electric guitar has the following properties.

diameter = 5.0×10^{-4} m

Young modulus = 2.0×10^{11} Pa

tension = 20 N

The string snaps, and contracts elastically.

By what percentage does a length l of a piece of the string contract?

- A** 5.1×10^{-4} % **B** 5.1×10^{-2} % **C** 1.3×10^{-4} % **D** 1.3×10^{-2} %

48 9702/11/O/N/10/Q20

A student writes some statements about solids, liquids and gases.

- 1 Solids are rigid because the molecules in a solid vibrate.
- 2 Liquids flow because the molecules in a liquid are closer than in a gas.
- 3 Gases are less dense than liquids because the molecules in a gas move randomly.

Which statements are correct?

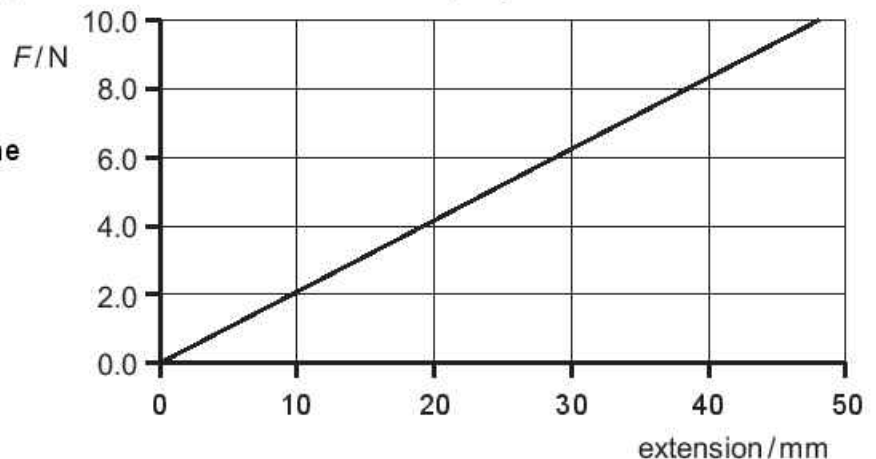
- A** 1 only **B** 1 and 3 only **C** 2 and 3 only **D** none of the above

49 9702/11/O/N/10/Q21

The graph shows how force depends on extension for a certain spring.

What is the energy stored in the spring when the extension is 30 mm?

- A 0.095 J B 0.19 J
C 0.25 J D 0.95 J



50 9702/11/O/N/10/Q22

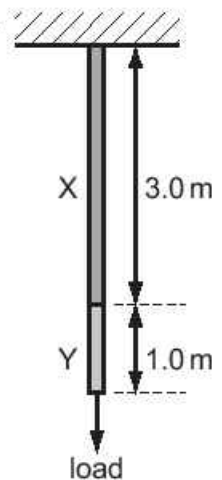
A wire consists of a 3.0 m length of metal X joined to a 1.0 m length of metal Y. The cross-sectional area of the wire is uniform.

A load hung from the wire causes metal X to stretch by 1.5 mm and metal Y to stretch by 1.0 mm.

The same load is then hung from a second wire of the same cross-section, consisting of 1.0 m of metal X and 3.0 m of metal Y.

What is the total extension of this second wire?

- A 2.5 mm B 3.5 mm C 4.8 mm D 5.0 mm



51 9702/12/O/N/10/Q19

Which group of materials contains two polymers?

- | | | | | | | | |
|---|--------|------|-------------|---|-------|---------|--------|
| A | copper | sand | polystyrene | C | nylon | sugar | rubber |
| B | glass | wood | aluminium | D | stone | diamond | steel |

52 9702/12/O/N/10/Q21

Two wires P and Q are made from the same material.

Wire P is initially twice the diameter and twice the length of wire Q. The same force, applied to each wire, causes the wires to extend elastically.

What is the ratio of the extension in P to that in Q?

- A $\frac{1}{2}$ B 1 C 2 D 4

53 9702/11/O/N/11/Q23

The Young modulus of steel is determined using a length of steel wire and is found to have the value E .

Another experiment is carried out using a wire of the same steel, but of half the length and half the diameter.

What value is obtained for the Young modulus in the second experiment?

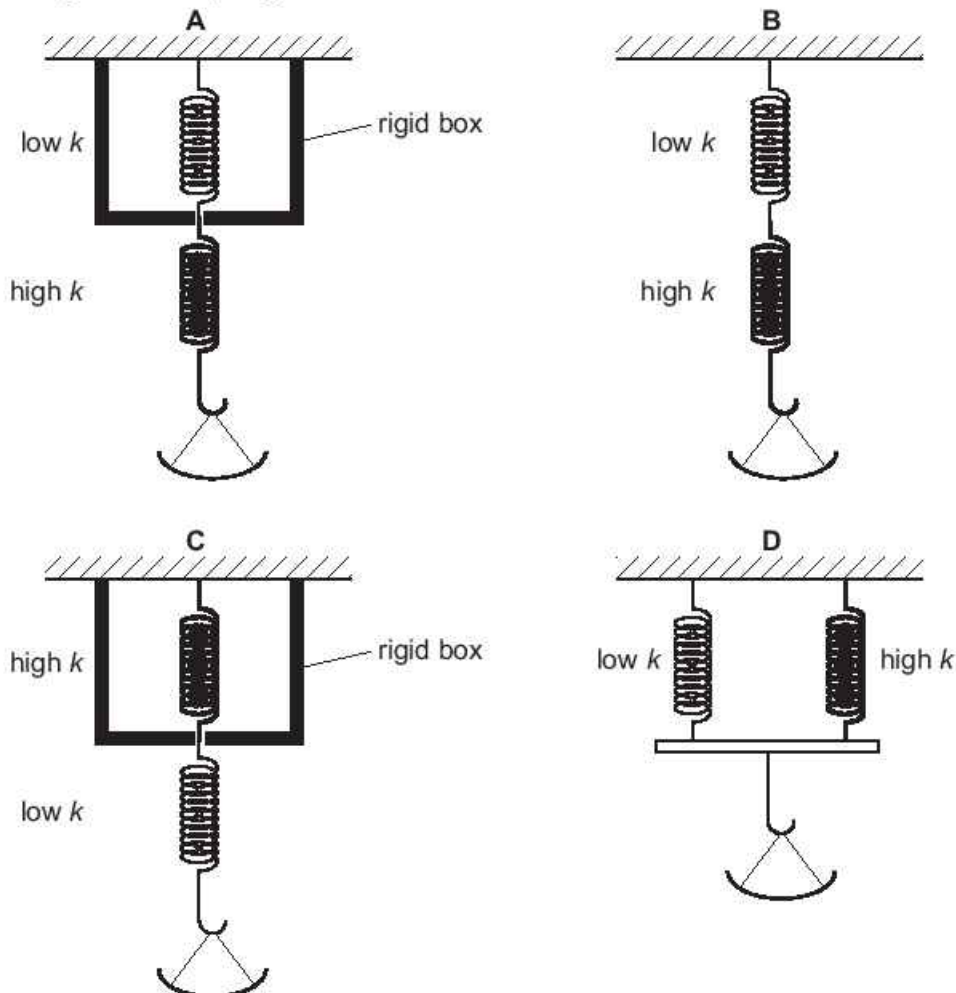
- A $\frac{1}{2}E$ B E C $2E$ D $4E$

54 9702/12/O/N/10/Q22

To determine the mass of food in a pan, a scale is used that has high sensitivity for small masses but low sensitivity for large masses.

To do this, two springs are used, each with a different spring constant k . One of the springs has a low spring constant and the other has a high spring constant.

Which arrangement of springs would be suitable?



55 9702/11/M/J/11/Q21

The Young modulus E can be determined from measurements made when a wire is stretched.

Which quantities would be measured in order to determine E ?

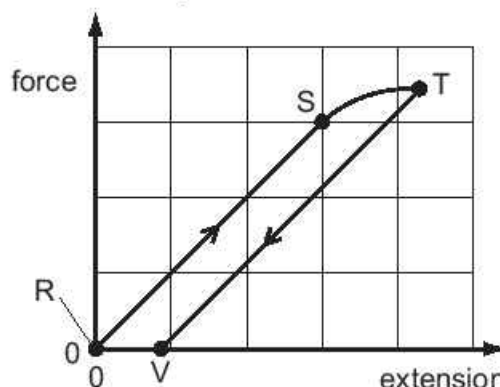
A	mass of stretching load	original length of wire	diameter of wire	extension of wire
B	mass of stretching load	new length of wire	cross-sectional area of wire	diameter of wire
C	mass of wire	original length of wire	cross-sectional area of wire	new length of wire
D	mass of wire	new length of wire	diameter of wire	extension of wire

56 9702/12/M/J/15/Q22

A long, thin metal wire is suspended from a fixed support and hangs vertically. Masses are suspended from its lower end.

The load on the lower end is increased from zero and then decreased again back to zero.

The diagram shows the force-extension graph produced.



Where on the graph would the elastic limit be found?

- A anywhere between point R and point S
- B beyond point S but before point T
- C exactly at point S
- D exactly at point T

57 9702/12/M/J/11/Q23

The behaviour of a wire under tensile stress may be described in terms of the Young modulus E of the material of the wire and of the force per unit extension k of the wire.

For a wire of length L and cross-sectional area A , what is the relation between E and k ?

- A $E = \frac{A}{kL}$
- B $E = \frac{kA}{L}$
- C $E = \frac{kL}{A}$
- D $E = \frac{L}{kA}$

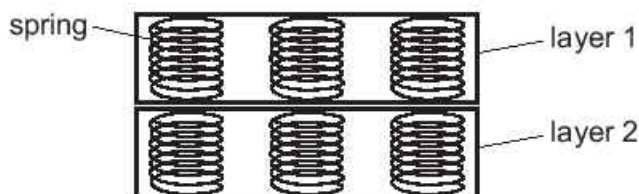
58 9702/12/M/J/11/Q24

The diagram shows the structure of part of a mattress.

The manufacturer wants to design a softer mattress (one which will compress more for the same load).

Which change will **not** have the desired effect?

- A using more layers of springs
- B using more springs per unit area
- C using springs with a smaller spring constant
- D using springs made from wire with a smaller Young modulus



59 9702/11/O/N/11/Q5

The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

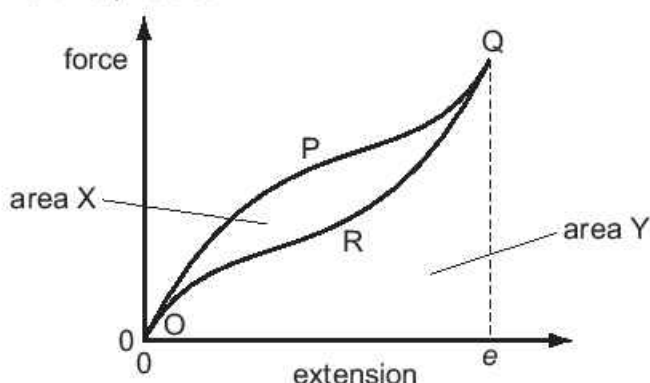
The wire is extended by a known force and the following measurements are made.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	measurement	symbol	value
A	length of wire before force applied	l	$2.043 \pm 0.002 \text{ m}$
B	diameter of wire	d	$0.54 \pm 0.02 \text{ mm}$
C	force applied	F	$19.62 \pm 0.01 \text{ N}$
D	extension of wire with force applied	x	$5.2 \pm 0.2 \text{ mm}$

60 9702/11/O/N/11/Q24

A rubber band is stretched and then relaxed to its original length. The diagram shows the force-extension graph for this process.



As the force is increased, the curve follows the path OPQ to extension e . As the force is reduced, the curve follows the path QRO to return to zero extension.

The area labelled X is between the curves OPQ and QRO. The area labelled Y is bounded by the curve QRO and the horizontal axis.

Which statement about the process is correct?

- A Area X is the energy which heats the band as it is stretched to e .
- B (Area X + area Y) is the minimum energy required to stretch the band to e .
- C Area X is the elastic potential energy stored in the band when it is stretched to e .
- D (Area Y – area X) is the net work done on the band during the process.

61 9702/11/O/N/11/Q25

When describing the behaviour of a spring, the spring constant is used.

Different loads are used to extend the spring by different amounts.

To find the spring constant, which quantities are required?

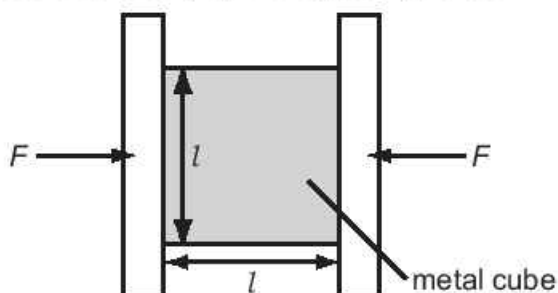
- A the elastic limit and the loads
- B the elastic limit, extensions and the length of the spring
- C the loads and the extensions of the spring
- D the loads and the length of the spring

62 9702/11/O/N/11/Q26

A metal cube of side l is placed in a vice and compressed elastically by two opposing forces F .

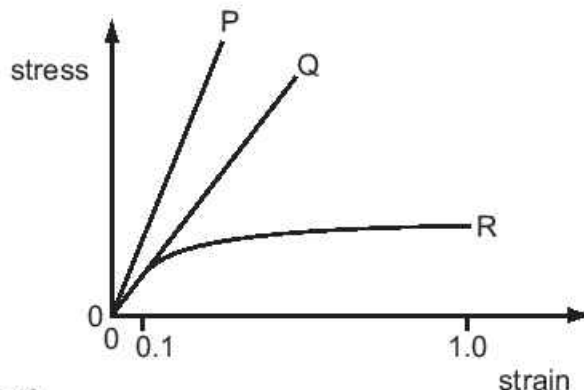
How will Δl , the amount of compression, relate to l ?

- A $\Delta l \propto \frac{1}{l^2}$
- B $\Delta l \propto \frac{1}{l}$
- C $\Delta l \propto l$
- D $\Delta l \propto l^2$



63 9702/12/O/N/11/Q21

The graph shows the relationship between stress and strain for three wires of the same linear dimensions but made from different materials.



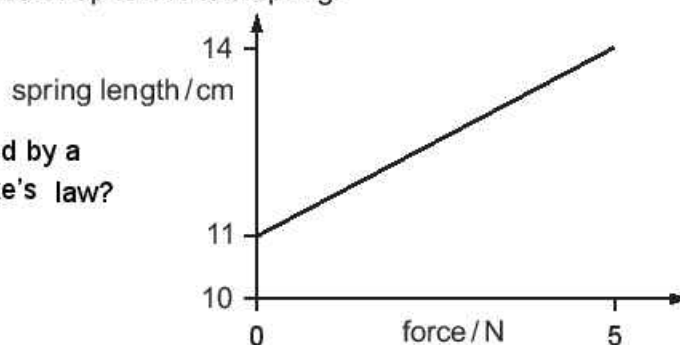
Which statements are correct?

- 1 The extension of P is approximately twice that of Q for the same stress.
- 2 The ratio of the Young modulus for P to that of Q is approximately two.
- 3 For strain less than 0.1, R obeys Hooke's law.

A 1, 2 and 3 **B** 1 and 3 only **C** 2 and 3 only **D** 2 only

64 9702/12/O/N/11/Q22

The graph shows the effect of applying a force of up to 5 N to a spring.

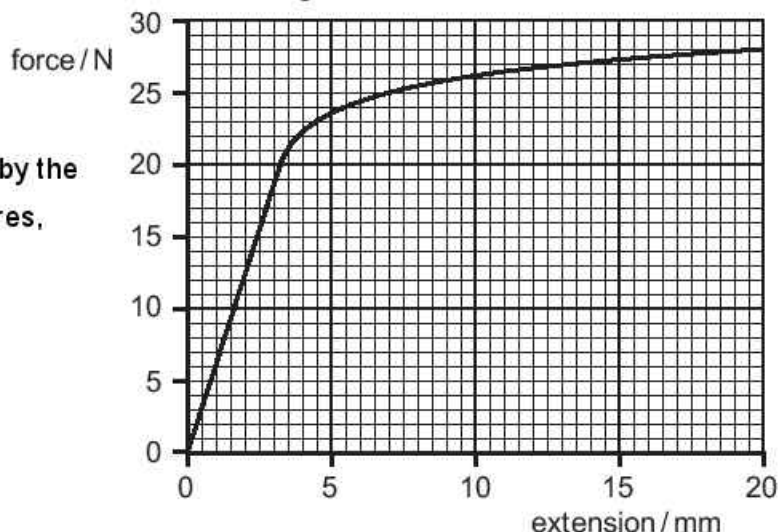


What is the total increase in length produced by a 7 N force, assuming the spring obeys Hooke's law?

- A** 4.2 cm **B** 5.6 cm
C 15.2 cm **D** 19.6 cm

65 9702/12/M/J/12/Q24

The graph is a force-extension graph for a wire that is being stretched.



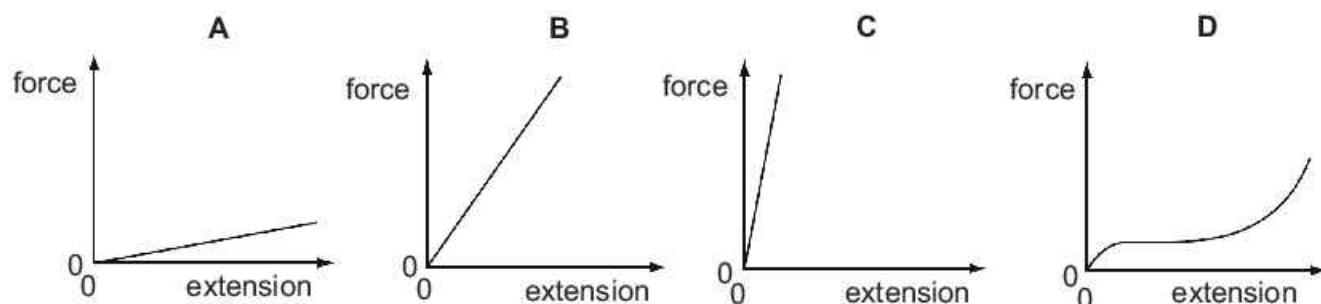
How much work needs to be done by the tensile force, to two significant figures, to cause an extension of 7.0 mm?

- A** 0.088 J **B** 0.12 J
C 0.53 J **D** 120 J

66 9702/12/O/N/11/Q23

The following force-extension graphs are drawn to the same scale.

Which graph represents the deformed object with the greatest amount of elastic potential energy?



67 9702/12/M/J/12/Q25

A wire stretches 8 mm under a load of 60 N.

A second wire of the same material, with half the diameter and a quarter of the original length of the first wire, is stretched by the same load.

Assuming that Hooke's law is obeyed, what is the extension of this wire?

- A** 1 mm **B** 4 mm **C** 8 mm **D** 16 mm

68 9702/13/M/J/12/Q20

What is represented by the gradient of a graph of force (vertical axis) against extension (horizontal axis)?

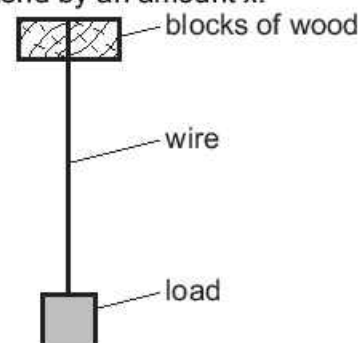
- A** elastic limit **B** spring constant **C** stress **D** Young modulus

69 9702/13/M/J/12/Q21

The diagram shows a wire of diameter D and length L that is firmly clamped at one end between two blocks of wood. A load is applied to the wire which causes it to extend by an amount x .

By how much would a wire of the same material, but of diameter $2D$ and length $3L$, extend when the same load is applied?

- A** $\frac{2}{3}x$ **B** $\frac{3}{4}x$
C $\frac{4}{3}x$ **D** $\frac{3}{2}x$



70 9702/13/M/J/12/Q22

Which property of a metal wire depends on its Young modulus?

- A** ductility **B** elastic limit **C** spring constant **D** ultimate tensile stress

71 9702/13/M/J/12/Q23

When white sugar granules are heated, they melt. When the melt is cooled quickly, a brittle solid form of toffee is produced.

How does the structure of the sugar change?

- A** amorphous to polymeric **C** crystalline to polymeric
B crystalline to amorphous **D** polymeric to amorphous

72 9702/12/O/N/12/Q24

Which row gives the correct description for the arrangement of atoms in the four types of material?

	atoms have no long-range order	atoms form giant chain-like molecules	atoms have an ordered arrangement in regions, but these ordered regions are at angles to one another	atoms are arranged in an ordered way throughout
A	crystalline	amorphous	polymeric	polycrystalline
B	polycrystalline	crystalline	amorphous	polymeric
C	polymeric	polycrystalline	crystalline	amorphous
D	amorphous	polymeric	polycrystalline	crystalline

73 9702/12/O/N/12/Q25

What leads to the conclusion that the movement of molecules is random?

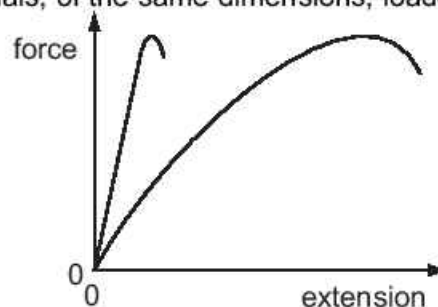
- A evaporation of water at room temperature
- B conduction of electricity in water
- C convection currents in air
- D motion of dust particles in air

74 9702/12/O/N/12/Q26

The diagram shows the force-extension graphs for two materials, of the same dimensions, loaded to fracture.

What describes the behaviour of the materials?

- A Both materials are brittle.
- B Both materials obey Hooke's law.
- C Both materials are plastic.
- D Both materials have the same ultimate tensile stress.



75 9702/12/O/N/12/Q27

Two wires, X and Y, are made from different metals and have different dimensions. The Young modulus of wire X is twice that of wire Y. The diameter of wire X is half that of wire Y.

Both wires are extended with equal strain and obey Hooke's law.

What is the ratio $\frac{\text{tension in wire X}}{\text{tension in wire Y}}$?

- A $\frac{1}{8}$
- B $\frac{1}{2}$
- C 1
- D 8

76 9702/13/O/N/12/Q24

A trolley is held at rest between two steel springs.

Each spring has an unstretched length of 0.10 m.

Spring P has spring constant 60 N m^{-1} .

Spring Q has spring constant 120 N m^{-1} .

Spring P has an extension of 0.40 m.



What is the extension of spring Q?

- A 0.10 m
- B 0.20 m
- C 0.30 m
- D 0.80 m

77 9702/13/O/N/12/Q25

A lift is supported by two steel cables, each of length 10 m and diameter 0.5 cm.

The lift drops 1 mm when a man of mass 80 kg steps into the lift.

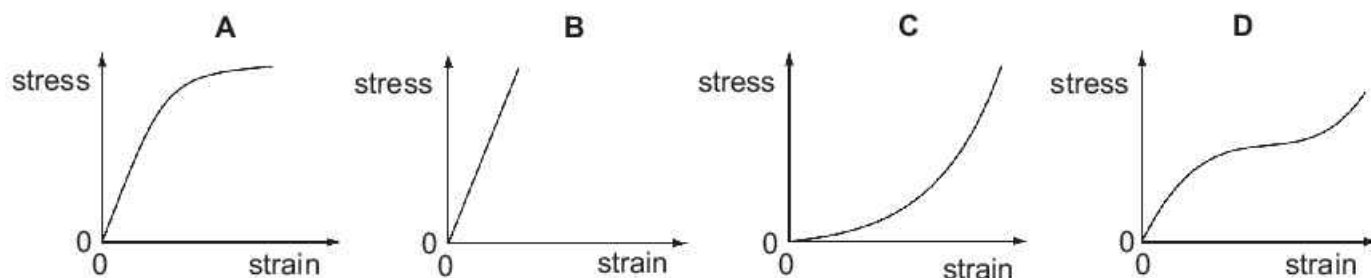
What is the best estimate of the value of the Young modulus of the steel?

- A** $2 \times 10^{10} \text{ Nm}^{-2}$ **B** $4 \times 10^{10} \text{ Nm}^{-2}$ **C** $2 \times 10^{11} \text{ Nm}^{-2}$ **D** $4 \times 10^{11} \text{ Nm}^{-2}$

78 9702/11/M/J/13/Q22

The stress-strain graphs for four different materials are shown below.

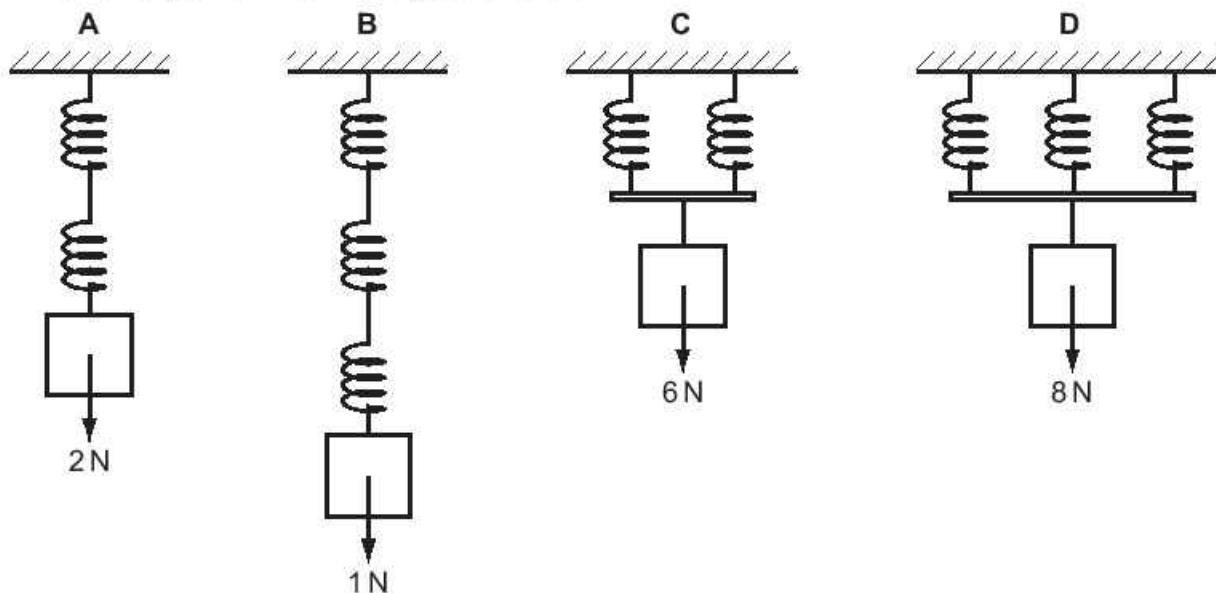
Which diagram shows the stress-strain graph for a ductile metal?



79 9702/11/M/J/13/Q23

A number of identical springs, each having the same spring constant, are joined in four arrangements. A different load is applied to each arrangement.

Which arrangement has the largest extension?



80 9702/01/O/N/06/Q23

What is the unit of the Young modulus?

- A** Nm^{-1} **B** Nm **C** Nm^{-2} **D** Nm^2

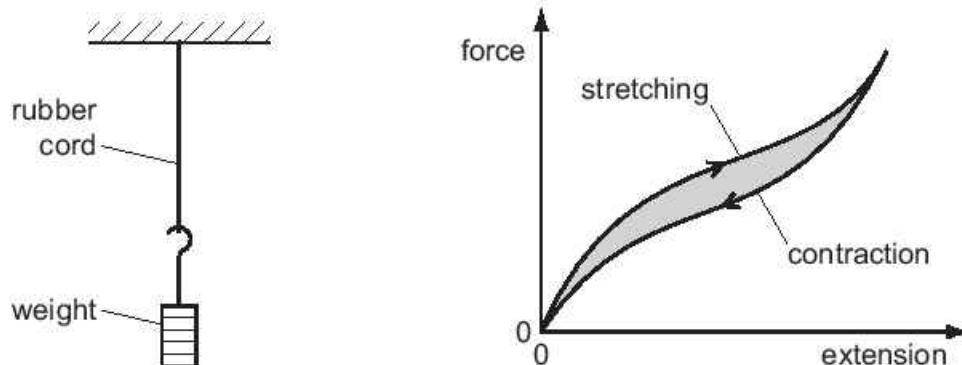
81 9702/12/O/N/13/Q21

Which two substances are normally both crystalline?

- A** copper and diamond **C** diamond and glass
B copper and glass **D** diamond and rubber

82 9702/12/M/J/13/Q22

A rubber cord hangs from a rigid support. A weight attached to its lower end is gradually increased from zero, and then gradually reduced to zero.



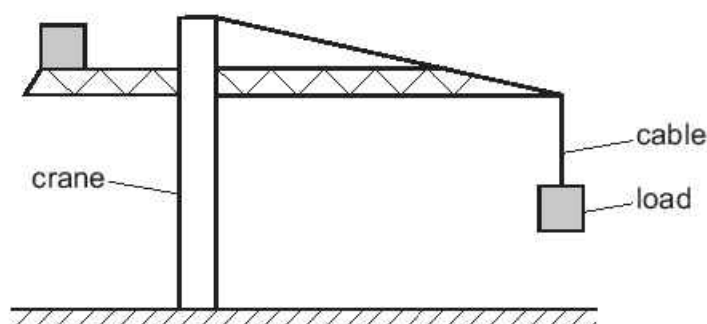
The force-extension curve for contraction is below the force-extension curve for stretching.

What does the shaded area between the curves represent?

- A the amount of elastic energy stored in the rubber
- B the amount of thermal energy dissipated in the rubber
- C the work done on the rubber cord during stretching
- D the work done by the rubber cord during contraction

83 9702/12/M/J/13/Q23

The diagram shows a large crane on a construction site lifting a cube-shaped load.



A model is made of the crane, its load and the cable supporting the load.

The material used for each part of the model is the same as that in the full-size crane, cable and load. The model is one tenth full-size in all linear dimensions.

What is the ratio $\frac{\text{stress in the cable on the full-size crane}}{\text{stress in the cable on the model crane}}$?

- A 10^0
- B 10^1
- C 10^2
- D 10^3

84 9702/13/O/N/13/Q22

A lift is supported by two steel cables each of length 20 m.

Each of the cables consists of 100 parallel steel wires, each wire of cross-sectional area $3.2 \times 10^{-6} \text{ m}^2$. The Young modulus of steel is $2.1 \times 10^{11} \text{ N m}^{-2}$.

Which distance does the lift move downward when a man of mass 70 kg steps into it?

- A 0.010 mm
- B 0.020 mm
- C 0.10 mm
- D 0.20 mm

85 9702/01/O/N/08/Q19

Which properties best describe modelling clay?

- A brittle and ductile
- B ductile and elastic
- C elastic and plastic
- D plastic and ductile

86 9702/12/O/N/13/Q24

A steel spring has a spring constant of 150 N m^{-1} . When a 25 N weight is hung from the spring, it has a stretched length of 55 cm.

What was the original length of the spring?

- A 0.38 m
- B 0.49 m
- C 0.61 m
- D 0.72 m

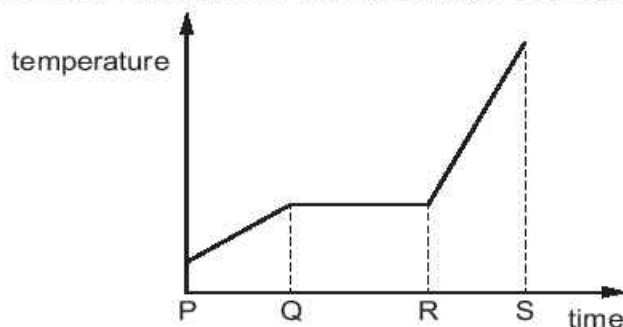
87 9702/13/O/N/13/Q23

What is equal to the Young modulus of a material that is extended elastically within the limit of proportionality?

- A area under the force-extension graph
- B area under the stress-strain graph
- C gradient of the force-extension graph
- D gradient of the stress-strain graph

88 9702/11/M/J/14/Q19

A crystalline solid is heated at a constant rate and the change of temperature with time is shown in the graph below.



Which statement about the particles in the material is correct?

- A In the time from P to Q, the particles are arranged randomly.
- B In the time from Q to R, some particles are arranged regularly and some particles are arranged randomly.
- C In the time from R to S, the particles are widely spaced.
- D The arrangement of the particles is the same in the time from P to S.

89 9702/13/M/J/14/Q18

A ball drops onto a horizontal surface and bounces elastically.

What happens to the kinetic energy of the ball during the very short time that it is in contact with the surface?

- A Most of the kinetic energy is lost as heat and sound energy.
- B The kinetic energy decreases to zero and then returns to its original value.
- C The kinetic energy remains constant because it is an elastic collision.
- D The kinetic energy remains constant in magnitude but changes direction.

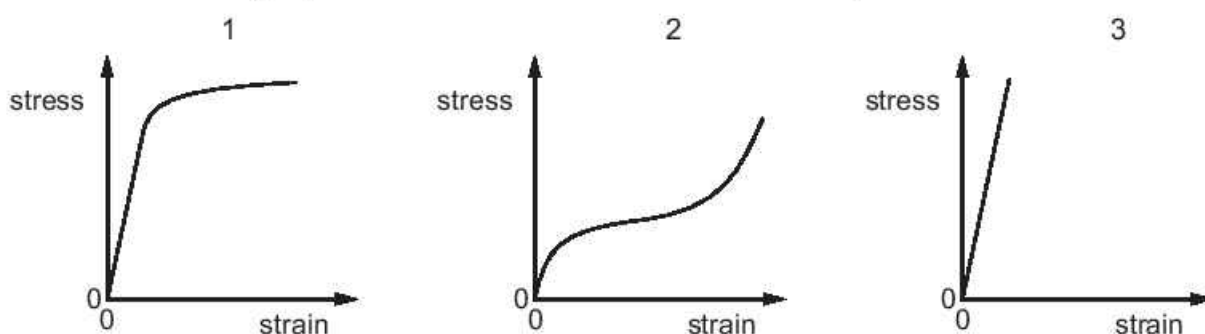
90 9702/13/M/J/14/Q20

Which process does **not** require energy to be supplied?

- A boiling
- B evaporation
- C freezing
- D melting

91 9702/11/M/J/14/Q20

The stress-strain graphs for three different materials are shown, not drawn to the same scales.



The three materials are copper, rubber and glass.
Which materials are represented by the graphs?

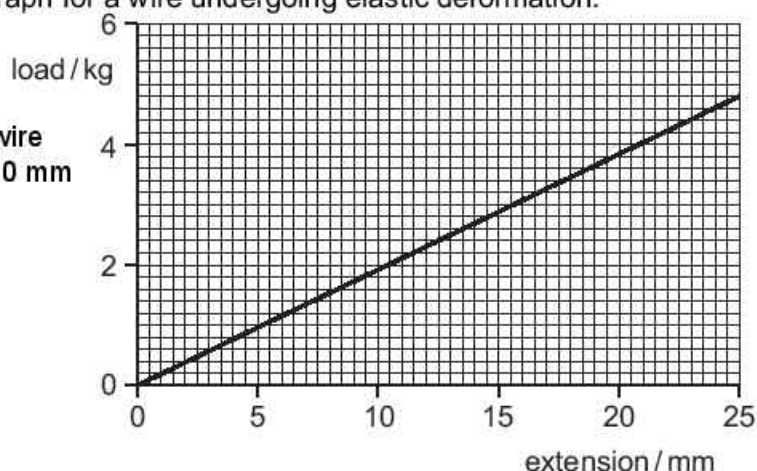
	1	2	3
A	copper	glass	rubber
B	copper	rubber	glass
C	glass	copper	rubber
D	glass	rubber	copper

92 9702/11/M/J/14/Q21

The graph is a load-extension graph for a wire undergoing elastic deformation.

How much work is done on the wire to increase the extension from 10 mm to 20 mm?

- A 0.028 J B 0.184 J
C 0.28 J D 0.37 J

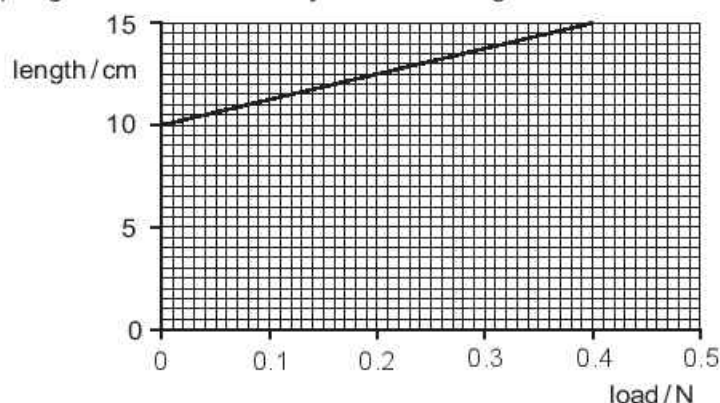


93 9702/12/M/J/14/Q20

The graph shows the length of a spring as it is stretched by an increasing load.

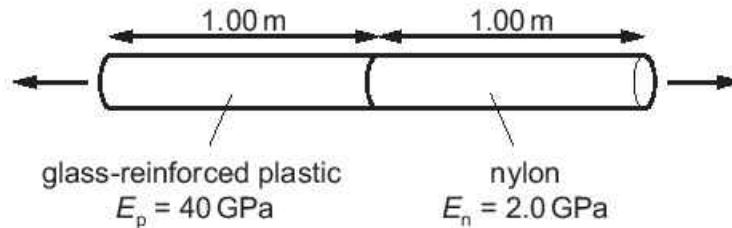
What is the spring constant?

- A 8.0 N m^{-1}
B 2.7 N m^{-1}
C 0.13 N m^{-1}
D 0.080 N m^{-1}



94 9702/12/M/J/14/Q21

A composite rod is made by attaching a glass-reinforced plastic rod and a nylon rod end to end, as shown.



The rods have the same cross-sectional area and each rod is 1.00 m in length. The Young modulus E_p of the plastic is 40 GPa and the Young modulus E_n of the nylon is 2.0 GPa.

The composite rod will break when its total extension reaches 3.0 mm.

What is the greatest tensile stress that can be applied to the composite rod before it breaks?

- A $7.1 \times 10^{-14}\text{ Pa}$ B $7.1 \times 10^{-2}\text{ Pa}$ C $5.7 \times 10^6\text{ Pa}$ D $5.7 \times 10^9\text{ Pa}$

95 9702/13/M/J/14/Q19

When the water in a pond freezes, it changes from a liquid to a solid. When this occurs, it changes volume and exchanges energy with the surroundings.

Which row is correct?

	change of volume	energy exchange
A	contracts	gives out energy to the surroundings
B	contracts	takes in energy from the surroundings
C	expands	gives out energy to the surroundings
D	expands	takes in energy from the surroundings

96 9702/13/M/J/14/Q23

An elastic material with a Young modulus E is subjected to a tensile stress S . Hooke's Law is obeyed.

What is the expression for the elastic energy stored per unit volume of the material?

- A $\frac{S^2}{2E}$ B $\frac{S^2}{E}$ C $\frac{E}{2S^2}$ D $\frac{2E}{S^2}$

97 9702/12/O/N/14/Q18

Which statement about molecules in a gas is correct?

- A In Brownian motion experiments, the molecules can be seen moving randomly in all directions.
 B The pressure exerted by a gas is caused by molecules bouncing against each other and changing kinetic energy.
 C The pressure exerted by a gas is caused by molecules rebounding from the walls of a container and changing momentum.
 D When the average speed of the molecules in a closed container increases, the density must also increase.

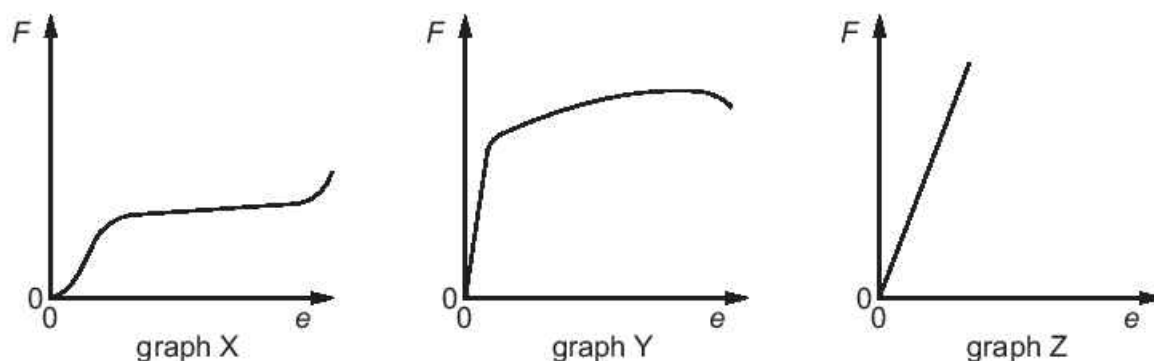
98 9702/12/O/N/14/Q17

What is the correct name for a material containing long-chain molecules that are tangled and coiled?

- A amorphous metal C crystalline metal
 B amorphous polymer D crystalline polymer

99 9702/13/M/J/14/Q24

Cylindrical samples of steel, glass and rubber are each subjected to a gradually increasing tensile force F . The extensions e are measured and graphs are plotted as shown below.



Which row correctly relates the graphs to the materials?

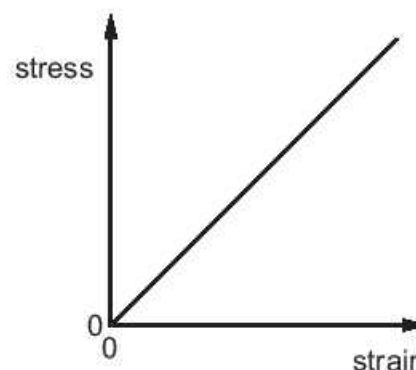
	steel	glass	rubber
A	X	Y	Z
B	X	Z	Y
C	Y	X	Z
D	Y	Z	X

100 9702/11/O/N/14/Q20

The stress-strain graph for a glass rod, up to the point at which it breaks, is shown below.

Which statement about the glass rod is correct?

- A** Hooke's law is obeyed for all values of stress up to the breaking point.
- B** The glass is ductile.
- C** The glass shows plastic deformation.
- D** When the cross-sectional area of the rod is doubled, the ultimate tensile stress of the rod is halved.

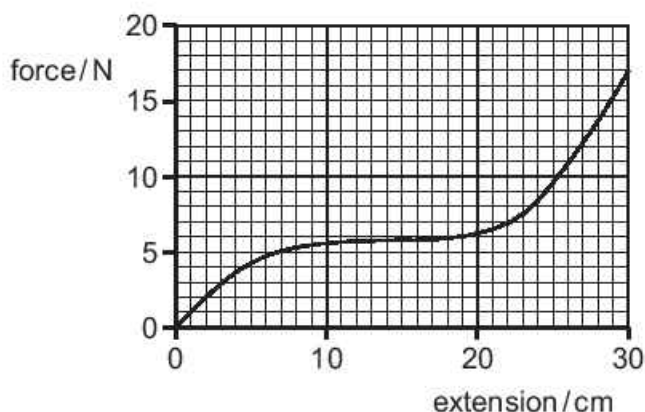


101 9702/11/O/N/14/Q21

A rubber band is stretched by hanging weights on it and the force-extension graph is plotted from the results.

What is the best estimate of the strain energy stored in the rubber band when it is extended 30 cm?

- A** 1.8 J **B** 2.6 J
- C** 5.1 J **D** 200 J



102 9702/13/O/N/14/Q21

A student is studying Brownian motion.

Using a microscope, she observes particles of smoke in a glass container, illuminated by a strong light. The particles of smoke have a zig-zag path, constantly changing speed and direction.

What happens to the smoke particles if the air in the container is heated?

- A The smoke particles become easier to see.
- B The smoke particles change direction more frequently.
- C The smoke particles increase in volume.
- D The smoke particles move further apart.

103 9702/13/O/N/14/Q23

What is meant by the ultimate tensile stress of a material?

- A the maximum force that can be applied to a bar of the material before it bends
- B the maximum inter-atomic force before the atomic bonds of the material break
- C the maximum stretching force per unit cross-sectional area before the material breaks
- D the maximum tensile force in a wire of the material before it breaks

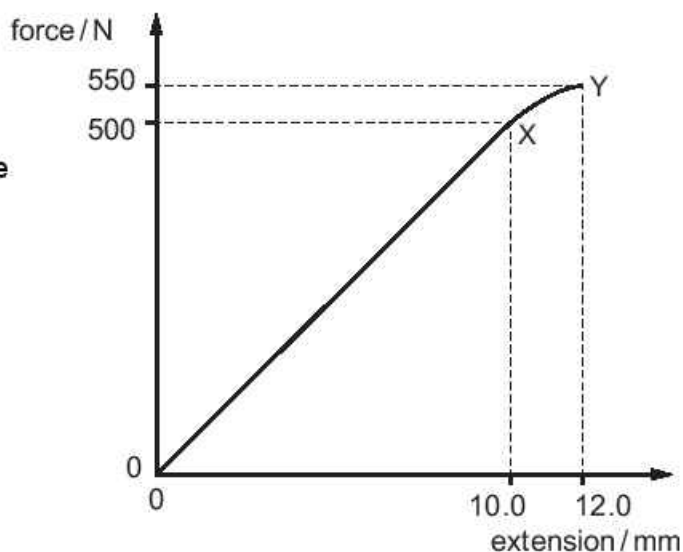
104 9702/13/O/N/14/Q24

The graph shows the behaviour of a sample of a metal when it is stretched until it starts to undergo plastic deformation.

What is the total work done in stretching the sample from zero to 12.0 mm extension?

Simplify the calculation by treating the curve XY as a straight line.

- A 3.30 J B 3.55 J
- C 3.60 J D 6.60 J

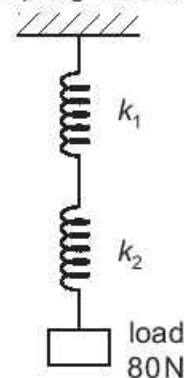


105 9702/13/O/N/14/Q25

Two springs, one with spring constant $k_1 = 4 \text{ kN m}^{-1}$ and the other with spring constant $k_2 = 2 \text{ kN m}^{-1}$, are connected as shown.

What is the total extension of the springs when supporting a load of 80 N?

- A 1.3 cm B 4 cm C 6 cm D 60 cm



106 9702/11/M/J/15/Q20

Descriptions of three different types of material are listed.

- 1 a polycrystalline material made up of large numbers of small crystals
- 2 an amorphous material with little or no ordered arrangement of molecules
- 3 a polymeric material consisting of long chains of molecules

Which row correctly matches the descriptions to nylon, copper and glass?

	1	2	3
A	copper	glass	nylon
B	copper	nylon	glass
C	glass	nylon	copper
D	nylon	copper	glass

107 9702/11/M/J/15/Q21

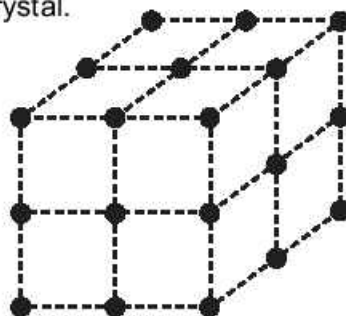
The diagram shows the arrangement of atoms in a particular crystal.

The mass of each atom is $3.5 \times 10^{-25} \text{ kg}$.

The density of the crystal is $9.2 \times 10^3 \text{ kg m}^{-3}$.

What is the shortest distance between the centres of two adjacent atoms?

- A $3.8 \times 10^{-29} \text{ m}$ B $6.2 \times 10^{-15} \text{ m}$
 C $3.4 \times 10^{-10} \text{ m}$ D $3.0 \times 10^{-9} \text{ m}$



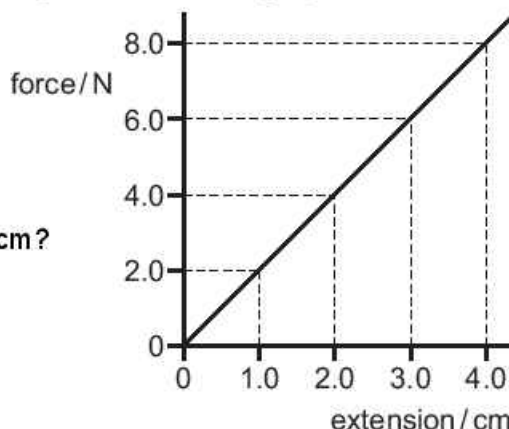
108 9702/11/M/J/15/Q23

The variation with applied force of the extension of a spring is shown in the graph.

When there is no force applied to the spring, it has a length of 1.0 cm.

What is the increase in the strain energy stored in the spring when its length is increased from 2.0 cm to 3.0 cm?

- A 0.020 J B 0.030 J
 C 0.040 J D 0.050 J



109 9702/12/M/J/15/Q21

To determine the Young modulus of a wire, several measurements are taken.

In which row can the measurement **not** be taken **directly** with the stated apparatus?

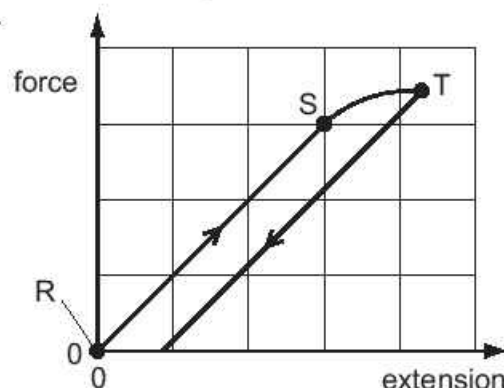
	measurement	apparatus
A	area of cross-section of wire	micrometer screw gauge
B	extension of wire	vernier scale
C	mass of load applied to wire	electronic balance
D	original length of wire	metre rule

110 9702/12/M/J/15/Q22

A long, thin metal wire is suspended from a fixed support and hangs vertically. Masses are suspended from its lower end.

The load on the lower end is increased from zero and then decreased again back to zero.

The diagram shows the force-extension graph produced.

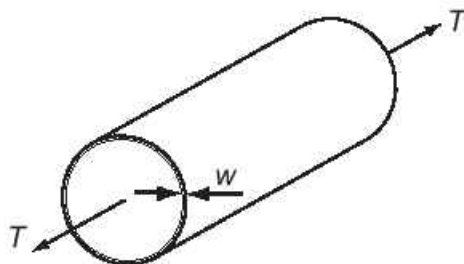


Where on the graph would the elastic limit be found?

- A anywhere between point R and point S
- B just beyond point S
- C exactly at point S
- D exactly at point T

111 9702/12/M/J/15/Q23

The diagram represents a steel tube with wall thickness w which is small in comparison with the diameter of the tube.



The tube is under tension, caused by a force T , parallel to the axis of the tube. To reduce the stress in the material of the tube, it is proposed to thicken the wall.

The tube diameter and the tension being constant, which wall thickness gives half the stress?

- A $\frac{w}{2}$
- B $\sqrt{2}w$
- C $2w$
- D $4w$

112 9702/13/M/J/15/Q20

When ice melts, it contracts.

Which row is correct for ice turning into water?

	distance between molecules	density
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

113 9702/13/M/J/15/Q22

A steel bar of circular cross-section is under tension T , as shown.

The diameter of the wide portion is double the diameter of the narrow portion.



What is the value of $\frac{\text{stress in the wide portion}}{\text{stress in the narrow portion}}$?

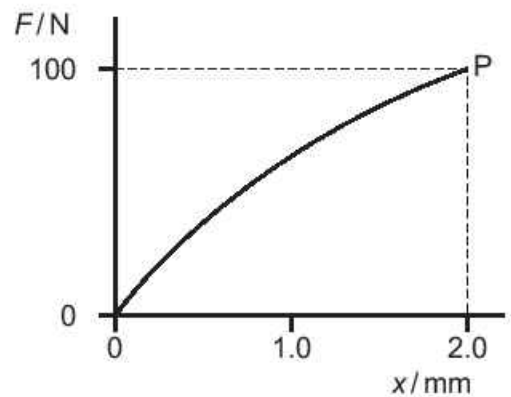
- A 0.25 B 0.50 C 2.0 D 4.0

114 9702/13/M/J/15/Q23

The graph shows the non-linear force-extension curve for a wire made from a new composite material.

What could be the value of the strain energy stored in the wire when it is stretched elastically to point P?

- A 0.09 J B 0.10 J
C 0.11 J D 0.20 J



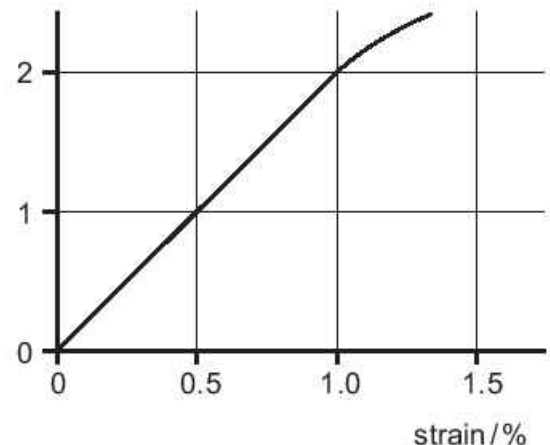
115 9702/13/M/J/15/Q24

The diagram shows the stress-strain graph for bone.

What is the Young modulus of bone?

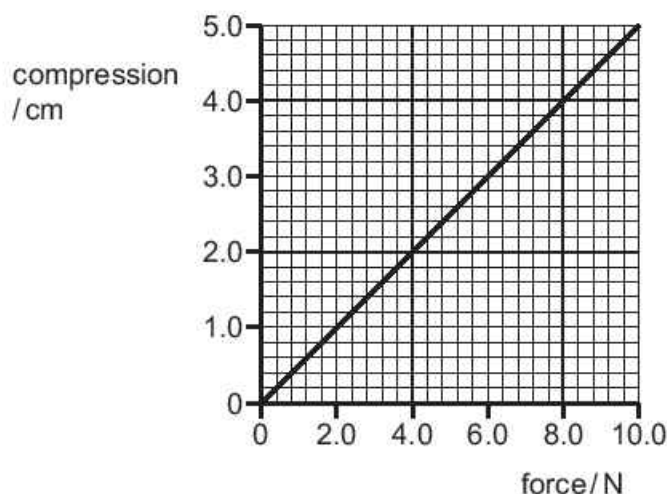
- A $1 \times 10^6 \text{ Nm}^{-2}$ B $2 \times 10^6 \text{ Nm}^{-2}$
C $1 \times 10^8 \text{ Nm}^{-2}$ D $2 \times 10^8 \text{ Nm}^{-2}$

stress
 $/ 10^6 \text{ Nm}^{-2}$

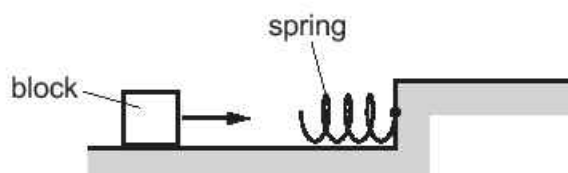


116 9702/11/M/J/17/Q21

The variation of the compression of a spring with the force applied to it is shown in the graph.



A block slides along a horizontal frictionless surface towards the spring, as shown.



The block is brought to rest by the spring. When the spring reaches a compression of 4.0 cm, all of the kinetic energy of the block is transferred to the elastic potential energy of the spring.

What is the kinetic energy of the block when it first makes contact with the spring?

- A 0.16 J B 0.32 J C 16 J D 32 J

117 9702/11/M/J/17/Q20

A wire of diameter d and length l hangs vertically from a fixed point. The wire is extended by hanging a mass M on its end. The Young modulus of the wire is E . The acceleration of free fall is g .

Which equation is used to determine the extension x of the wire?

- A $x = \frac{Ml}{\pi d^2 E}$ B $x = \frac{Mgl}{\pi d^2 E}$ C $x = \frac{4Mgl}{\pi d E}$ D $x = \frac{4Mgl}{\pi d^2 E}$

118 9702/12/M/J/17/Q20

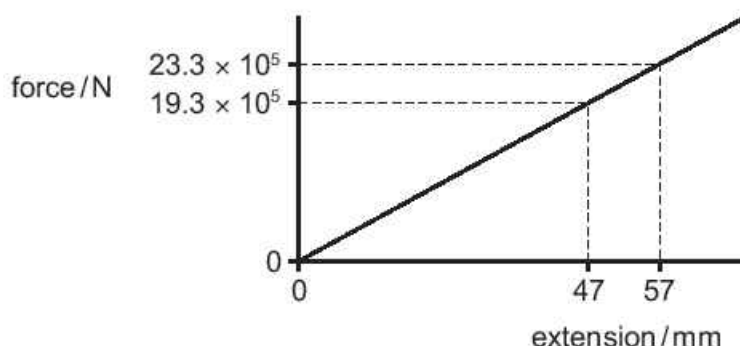
What are the units of stress, strain and the Young modulus?

	stress	strain	Young modulus
A	newton	metre	pascal
B	newton	no unit	newton
C	pascal	metre	newton
D	pascal	no unit	pascal

119 9702/12/M/J/17/Q19

A cable on a suspension bridge supports a weight of $19.3 \times 10^5 \text{ N}$. This weight causes the cable to stretch by 47 mm.

A lorry crossing the bridge then increases the force on the cable to $23.3 \times 10^5 \text{ N}$. The force-extension graph for the cable is shown.

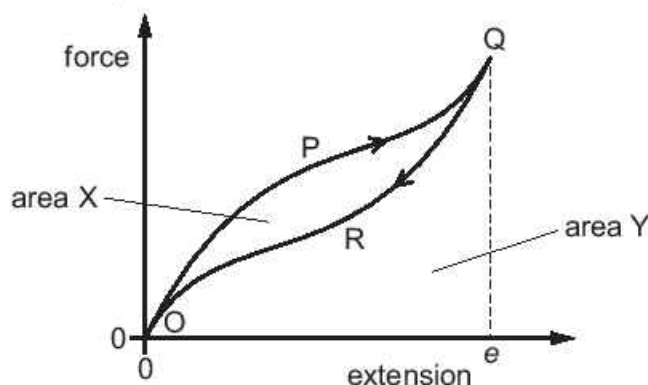


What is the **increase** in strain energy in the cable when the lorry is crossing the bridge?

- A 21 kJ B 23 kJ C 45 kJ D 66 kJ

120 9702/12/M/J/17/Q21

A rubber band is stretched and then relaxed to its original length. The diagram shows the force-extension graph for this process.



As the force is increased, the curve follows the path OPQ to extension e . As the force is reduced, the curve follows the path QRO to return to zero extension.

The area labelled X is between the curves OPQ and QRO. The area labelled Y is bounded by the curve QRO and the horizontal axis.

Which statement about the process is correct?

- A Area X is the energy which heats the band as it is stretched to extension e .
 B (Area X + area Y) is the minimum energy required to stretch the band to extension e .
 C Area X is the elastic potential energy stored in the band when it is stretched to extension e .
 D (Area Y – area X) is the net work done on the band during the process.

121 9702/11/O/N/17/Q20

A spring is loaded with weights. When the weights are removed, the spring returns to its original length.

The spring is then loaded with heavier weights. When the weights are removed, the spring is longer than it was originally.

Which types of deformation are shown by this experiment?

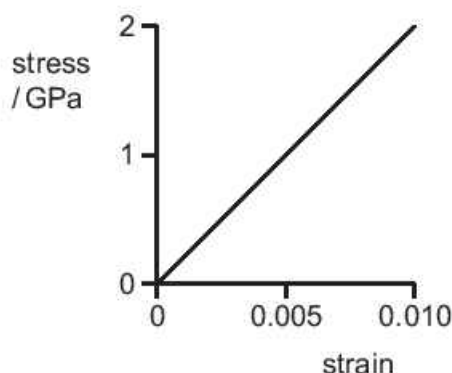
- A** both elastic and plastic deformation **C** neither elastic nor plastic deformation
B elastic deformation only **D** plastic deformation only

122 9702/11/O/N/17/Q21

The stress-strain graph for a metal is shown.

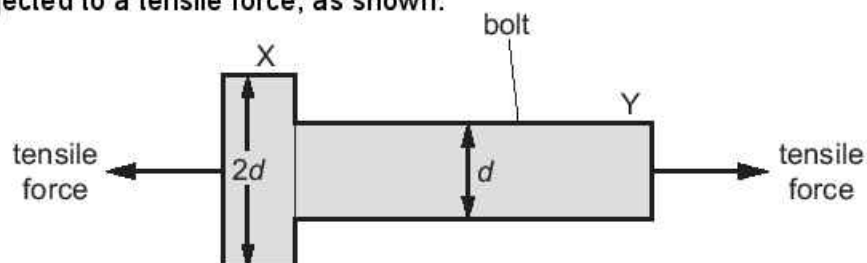
What is the strain energy per unit volume of a rod made from this metal when the strain of the rod is 0.010?

- A** 10 kJ m^{-3}
B 100 kJ m^{-3}
C 1.0 MJ m^{-3}
D 10 MJ m^{-3}



123 9702/12/O/N/17/Q20

A bolt is subjected to a tensile force, as shown.



The bolt has a circular cross-section. At end X the diameter is $2d$. At end Y the diameter is d .

What is the ratio $\frac{\text{stress at Y}}{\text{stress at X}}$?

- A** 0.25 **B** 0.50 **C** 2.0 **D** 4.0

124 9702/12/O/N/17/Q21

A rectangular block of steel supporting a very large component of a bridge has a height of 15 cm and a cross-section of $20 \text{ cm} \times 12 \text{ cm}$. It is designed to compress 1 mm when under maximum, evenly distributed, load.

The Young modulus of steel is $2.0 \times 10^{11} \text{ N m}^{-2}$.

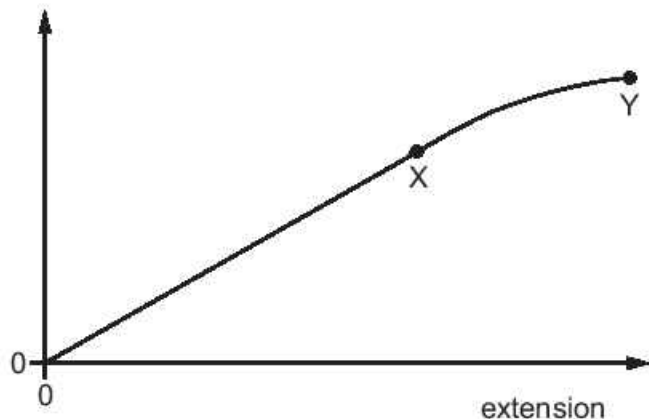
What is the maximum load it can support?

- A** 32 MN **B** 56 GN **C** 720 GN **D** 32 TN

125 9702/13/O/N/17/Q20

A sample of metal is subjected to a force which increases to a maximum value and then decreases back to zero. A force-extension graph for the sample is shown.

When the sample contracts, it follows the same force-extension curve as when it was being stretched. What is the behaviour of the metal between X and Y?



- A both elastic and plastic
- B not elastic and not plastic
- C plastic but not elastic
- D elastic but not plastic

126 9702/13/O/N/17/Q21

A load is hung from the end of a metal wire. The load is increased and the wire stretches elastically. The table shows the length of the wire for different loads.

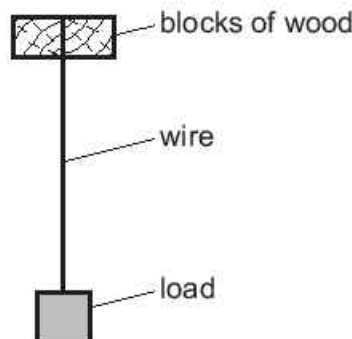
load / kN	length / mm
0	500.0
1.0	502.0
2.0	504.0
3.0	506.0
4.0	508.0

When the load is 4.0 kN, what is the strain energy stored in the wire?

- A 16 J
- B 32 J
- C 1.0 kJ
- D 2.0 kJ

127 9702/11/M/J/18/Q18

The diagram shows a wire of diameter D and length L that is firmly clamped at one end between two blocks of wood. A load is applied to the wire which extends its length by x .



A second wire is made of the same material, but of diameter $2D$ and length $3L$. Both wires obey Hooke's law.

What is the extension of the second wire when the same load is applied?

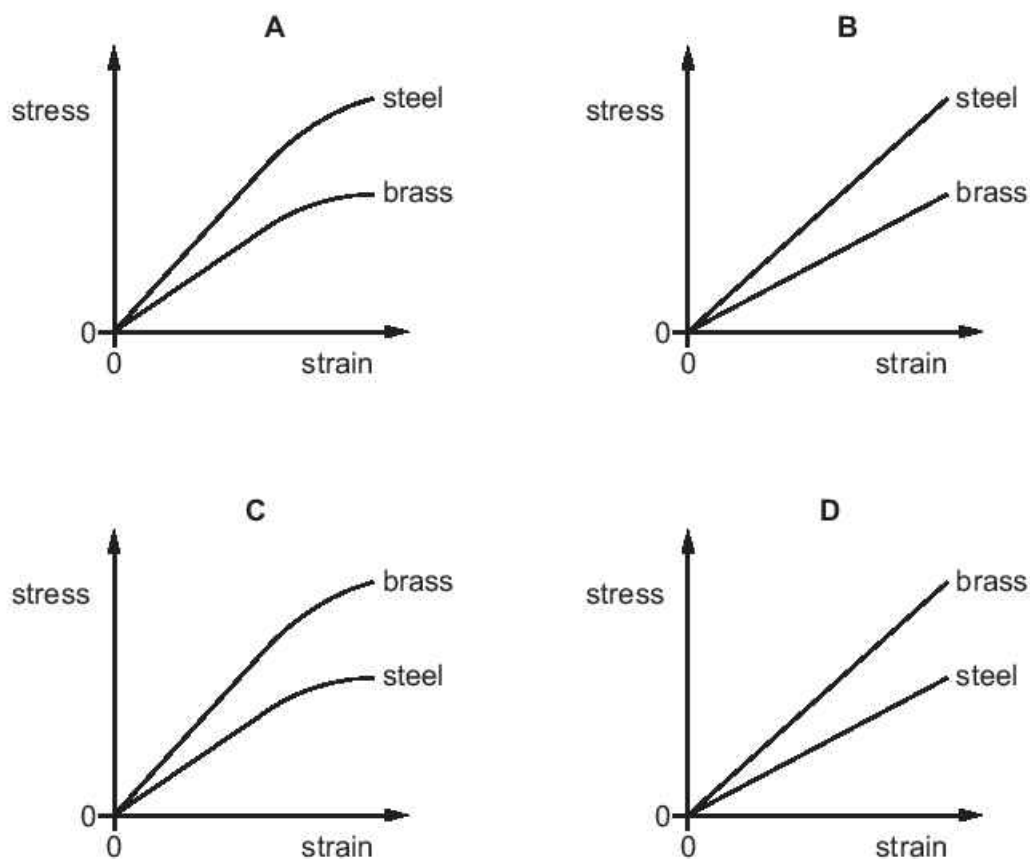
- A $\frac{2}{3}x$
- B $\frac{3}{4}x$
- C $\frac{4}{3}x$
- D $\frac{3}{2}x$

128 9702/11/M/J/18/Q19

Two wires, one made of brass and the other of steel, are stretched in an experiment. Both wires obey Hooke's law during this experiment.

The Young modulus for brass is less than the Young modulus for steel.

Which graph shows how the stress varies with strain for both wires in this experiment?



129 9702/12/M/J/18/Q20

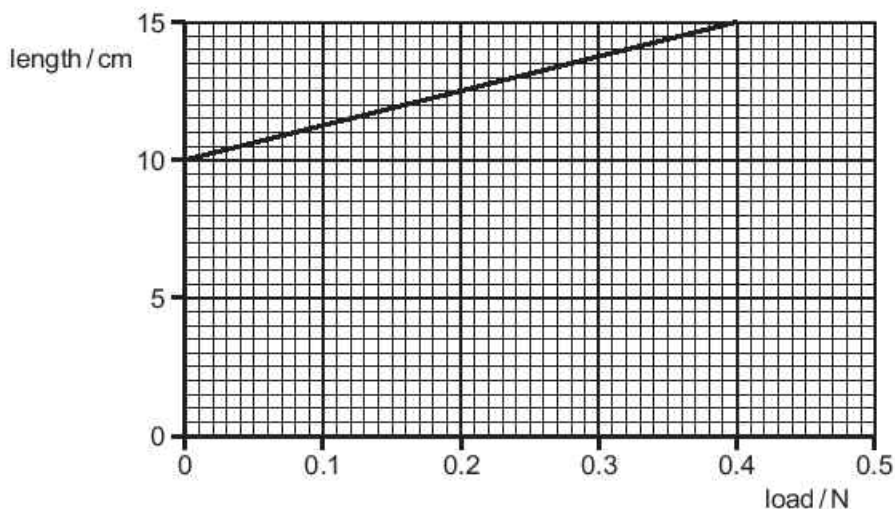
An elastic material with Young modulus E is subjected to a tensile stress S . Hooke's law is obeyed.

What is the expression for the elastic energy stored per unit volume of the material?

- A** $\frac{E}{2S^2}$
 B $\frac{2E}{S^2}$
 C $\frac{S^2}{E}$
 D $\frac{S^2}{2E}$

130 9702/12/M/J/18/Q21

The graph shows the length of a spring as it is stretched by an increasing load.



What is the spring constant of the spring?

- A** 0.080 Nm^{-1} **B** 0.13 Nm^{-1} **C** 2.7 Nm^{-1} **D** 8.0 Nm^{-1}

131 9702/13/M/J/18/Q18

Data for a steel wire on an electric guitar are listed.

diameter = $5.0 \times 10^{-4} \text{ m}$

Young modulus = $2.0 \times 10^{11} \text{ Pa}$

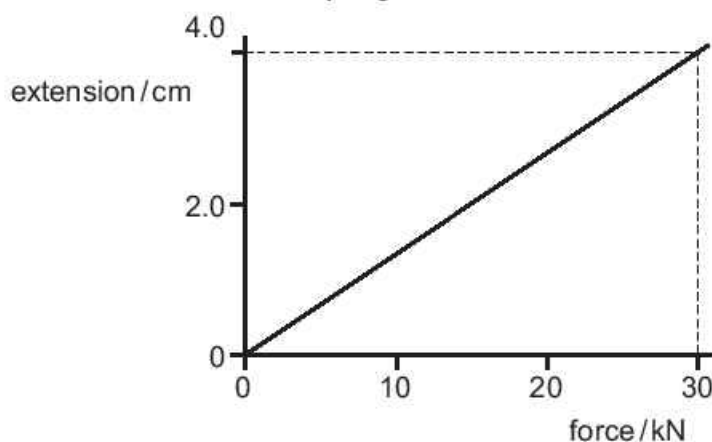
tension = 20 N

The wire snaps and contracts elastically. Assume the wire obeys Hooke's law. By what percentage does the length l of a piece of the wire contract?

- A** $1.3 \times 10^{-4} \%$ **B** $5.1 \times 10^{-4} \%$ **C** $1.3 \times 10^{-2} \%$ **D** $5.1 \times 10^{-2} \%$

132 9702/13/M/J/18/Q19

The graph shows how the extension of a spring varies with the force used to stretch it.



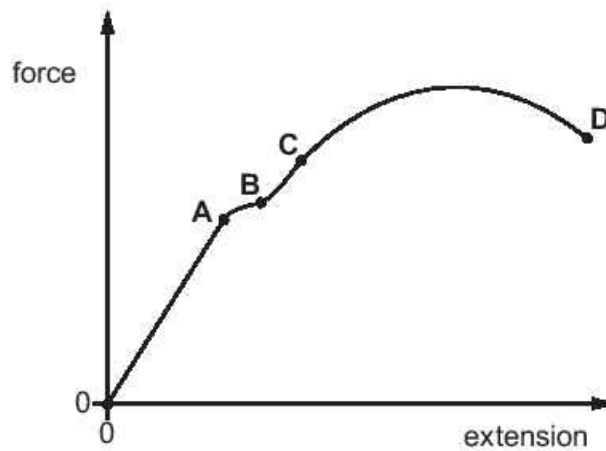
What is the strain energy in the spring when the extension is 4.0 cm ?

- A** 60 J **B** 120 J **C** 600 J **D** 1200 J

133 9702/12/F/M/18/Q19

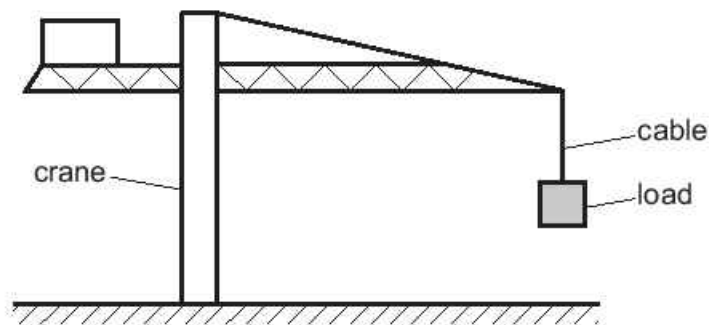
The force-extension graph of a metal wire is shown.

At which point on the graph does the metal wire stop obeying Hooke's law?



134 9702/12/F/M/18/Q20

The diagram shows a large crane on a construction site lifting a cube-shaped load at a constant speed.



A model is made of the crane, its load and the cable supporting the load.

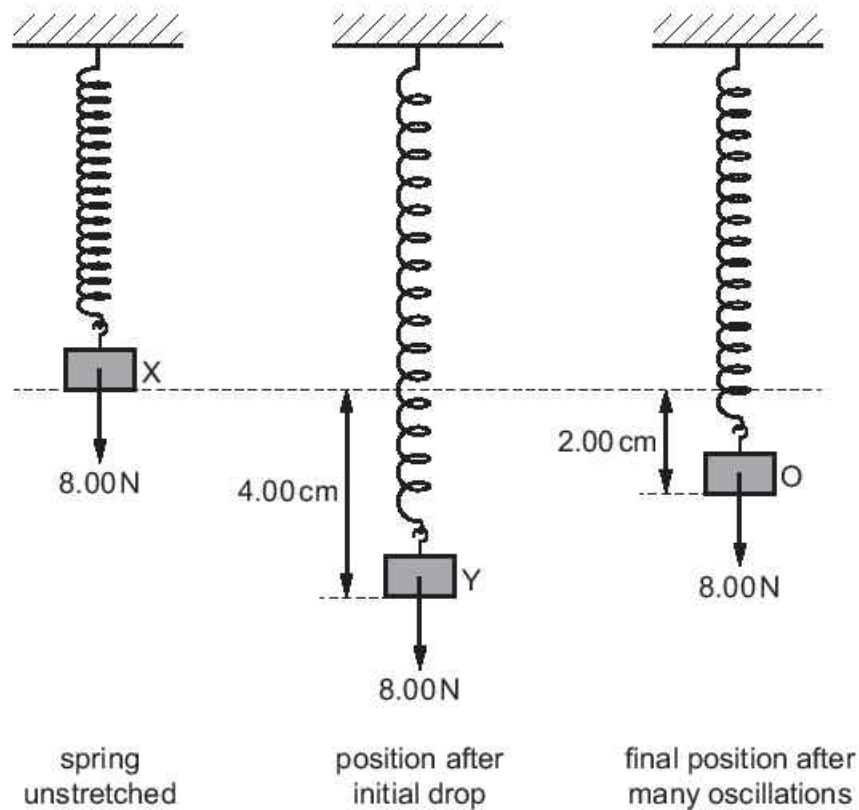
The material used for each part of the model is the same as that in the full-size crane, cable and load. The model is one tenth full-size in all linear dimensions.

What is the ratio $\frac{\text{stress in the cable on the full-size crane}}{\text{stress in the cable on the model crane}}$?

- A** 0.1 **B** 1 **C** 10 **D** 100

135 9702/11/O/N/18/Q16

An 8.00 N weight is attached to the lower end of a spring which is fixed at its upper end. The weight is initially held at rest at position X and the spring is unstretched. The weight is then released and falls to position Y, which is 4.00 cm below X. The weight oscillates and then eventually comes to rest at O, which is 2.00 cm below X.

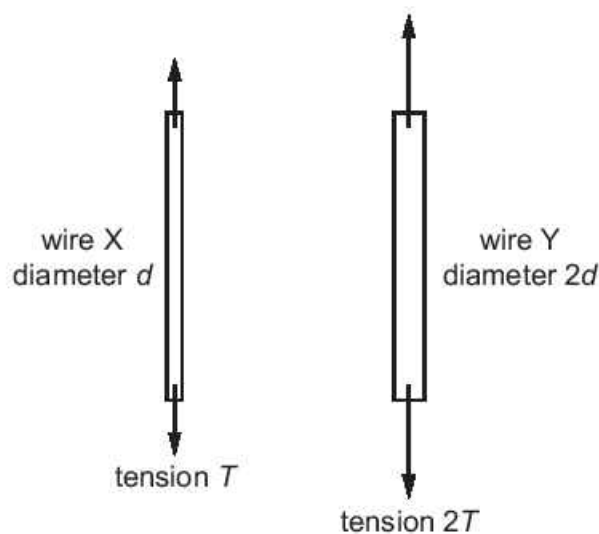


How much energy is lost from the system?

- A** 0.04 J **B** 0.08 J **C** 0.16 J **D** 0.32 J

136 9702/11/O/N/18/Q18

Two wires X and Y are made from the same material. Wire Y has twice the diameter and experiences twice the tension of wire X. The wires obey Hooke's law and have the same original length.



Wire X has extension e .

What is the extension of wire Y?

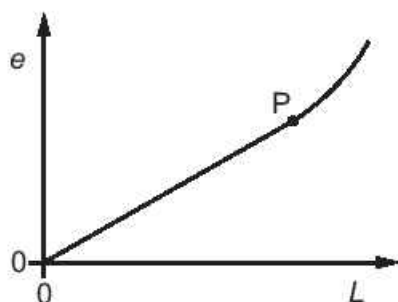
- A $\frac{e}{4}$ B $\frac{e}{2}$ C e D $2e$

137 9702/11/O/N/18/Q19

What is represented by the gradient of a graph of force (vertical axis) against extension (horizontal axis) for a wire obeying Hooke's law?

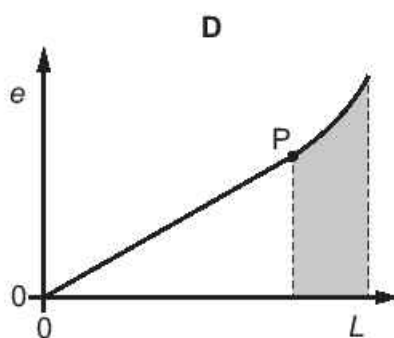
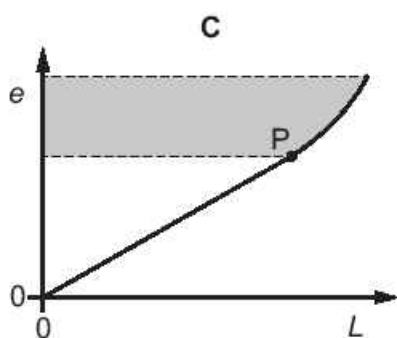
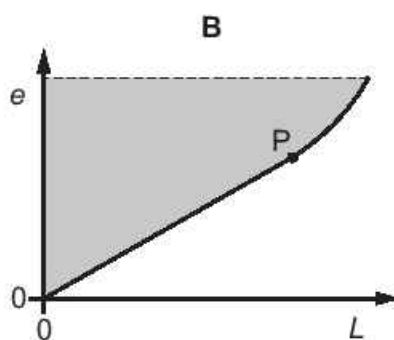
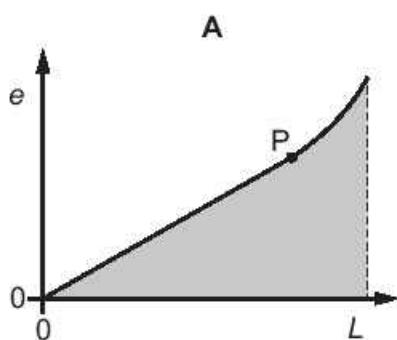
- A elastic limit
B spring constant
C stress
D Young modulus

Forces are applied to the ends of a rod so that its length increases. The variation with load L of the extension e of the rod is shown.



The point P is the elastic limit.

Which shaded area represents the work done during the plastic deformation of the rod?



139 9702/12/O/N/18/Q20

A metal cylinder is able to withstand a compressive force of 4.0 kN without deforming plastically.



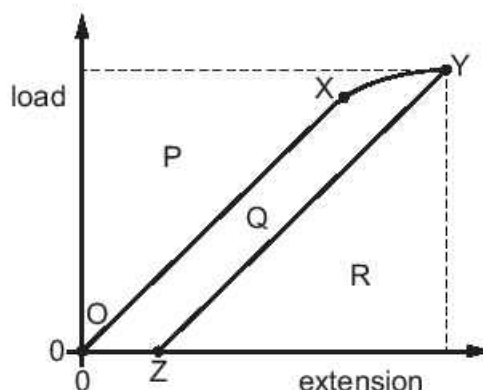
The cylinder has cross-sectional area A and would be at its elastic limit when a stress σ is applied.

What is a possible pair of values for A and σ ?

	A/m^2	σ/MPa
A	1.5×10^{-5}	50
B	1.5×10^{-5}	80
C	7.5×10^{-5}	50
D	7.5×10^{-5}	80

140 9702/12/O/N/18/Q21

A wire has both elastic and plastic properties. When it is slowly loaded, its extension varies with load as shown by line OXY. The removal of the load is represented by line YZ. This creates areas P, Q and R on the graph.

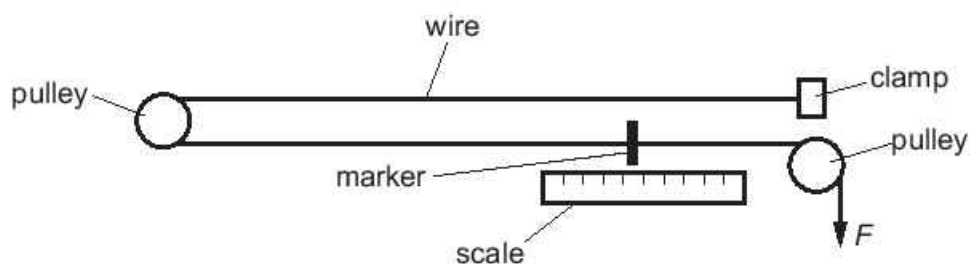


Which area represents the maximum elastic potential energy stored in the wire?

- A** P **B** Q **C** Q + R **D** R

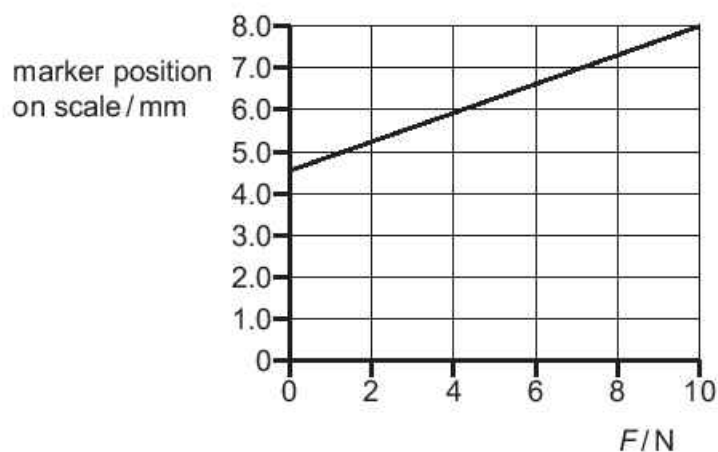
141 9702/13/O/N/18/Q19

In an experiment to measure the Young modulus of a metal, a wire of the metal of diameter 0.25 mm is clamped, as shown.



The wire passes from a clamp, around a frictionless pulley, and then to a second frictionless pulley where loads F are applied to it. A marker is attached to the wire so that the total length of wire between the clamp and the marker is initially 3.70 m. A scale is fixed near to this marker.

The graph shows how the reading on the scale varies with F .

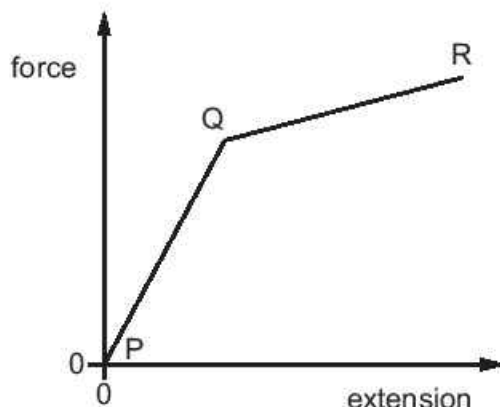


What is the Young modulus of the metal?

- A $5.5 \times 10^{10} \text{ Pa}$
- B $9.4 \times 10^{10} \text{ Pa}$
- C $1.6 \times 10^{11} \text{ Pa}$
- D $2.2 \times 10^{11} \text{ Pa}$

142 9702/13/O/N/18/Q20

A scientist is investigating the properties of a new material. She plots a force-extension graph for the material up to its breaking point.

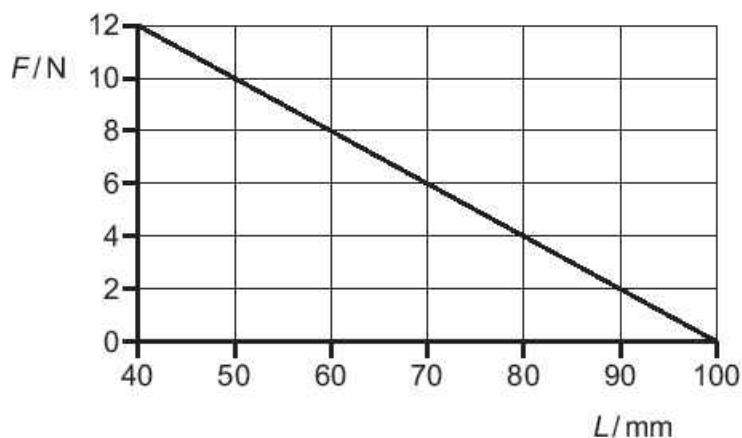


Which statement **must** be correct?

- A The area under the graph from P to R is the strain energy stored in the material.
- B The area under the graph from P to R is the work done in stretching the material.
- C The material stretches elastically from Q to R.
- D The material stretches plastically from P to Q.

143 9702/11/M/J/19/Q20

A spring of original length 100 mm is compressed by a force. The graph shows the variation of the compressing force F with the length L of the spring.

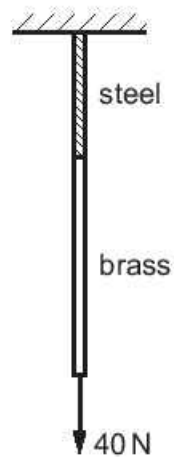


What is the energy stored in the spring when the length is 70 mm?

- A 0.090 J
- B 0.21 J
- C 0.27 J
- D 0.63 J

144 9702/11/M/J/19/Q21

A 0.80 m length of steel wire and a 1.4 m length of brass wire are joined together. The combined wires are suspended from a fixed support and a force of 40 N is applied, as shown.



The Young modulus of steel is 2.0×10^{11} Pa.

The Young modulus of brass is 1.0×10^{11} Pa.

Each wire has a cross-sectional area of $2.4 \times 10^{-6} \text{ m}^2$.

The wires obey Hooke's law.

What is the total extension? Ignore the weights of the wires.

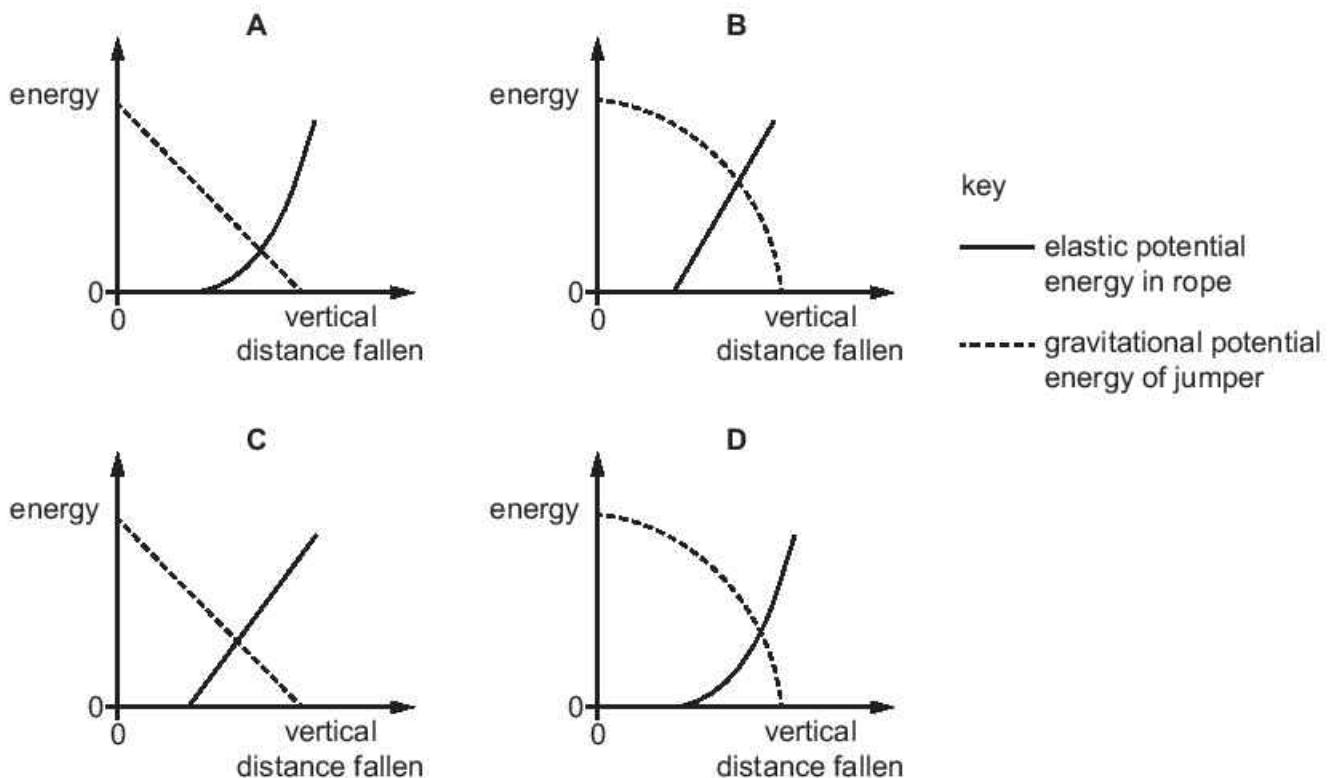
- A** $1.7 \times 10^{-4} \text{ m}$ **B** $3.0 \times 10^{-4} \text{ m}$ **C** $3.9 \times 10^{-4} \text{ m}$ **D** $9.0 \times 10^{-4} \text{ m}$

145 9702/12/M/J/19/Q18

A bungee jumper jumps off a high bridge, when attached to it by a long elastic rope which obeys Hooke's law.

The gravitational potential energy of the jumper is measured relative to the lowest point reached by the jumper.

Which graph shows the variation of the gravitational potential energy of the jumper, and the elastic potential energy in the rope, with the vertical distance fallen from the top of the bridge?



146 9702/12/M/J/19/Q20

A wire X is stretched by a force and gains elastic potential energy E .

The same force is applied to wire Y of the same material, with the same initial length but twice the diameter of wire X. Both wires obey Hooke's law.

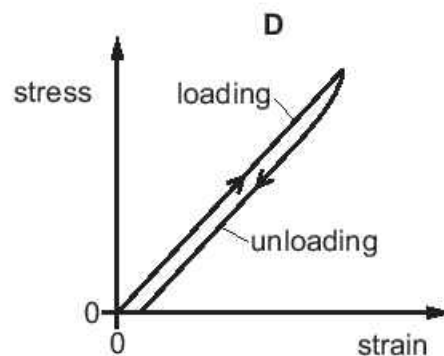
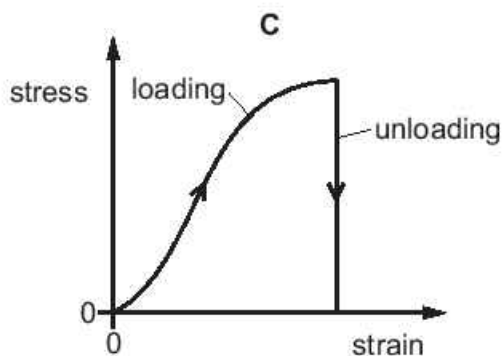
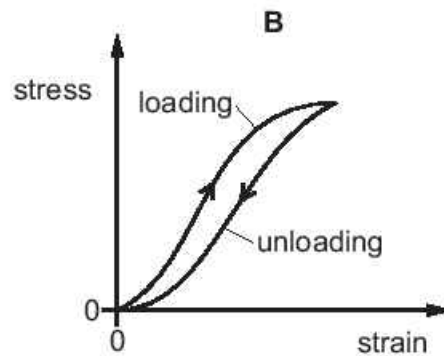
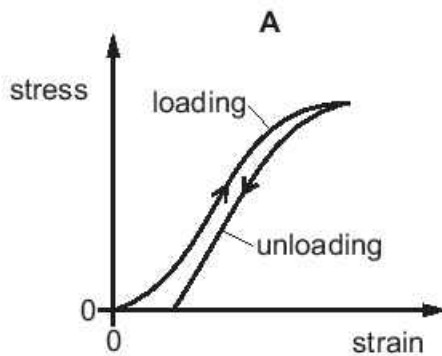
What is the gain in elastic potential energy of wire Y?

- A** $0.25E$ **B** $0.5E$ **C** $2E$ **D** $4E$

147 9702/12/M/J/19/Q22

The stress-strain graphs for loading and unloading four different materials are shown.

Which material exhibits purely elastic behaviour?



148 9702/13/M/J/19/Q19

Four solid steel rods, each of length 2.0 m and cross-sectional area 250 mm^2 , equally support an object weighing 10 kN. The weight of the object causes the rods to contract by 0.10 mm. The rods obey Hooke's law.

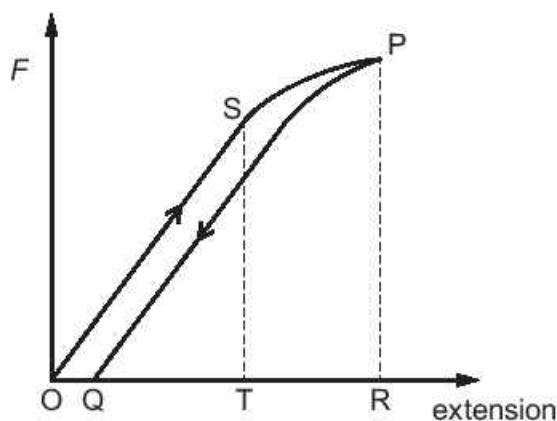
What is the Young modulus of steel?

- A** $2.0 \times 10^8 \text{ Nm}^{-2}$
- B** $2.0 \times 10^{11} \text{ Nm}^{-2}$
- C** $8.0 \times 10^8 \text{ Nm}^{-2}$
- D** $8.0 \times 10^{11} \text{ Nm}^{-2}$

149 9702/13/M/J/19/Q20

A wire is attached at one end to a fixed point. A tensile force F is applied to the other end of the wire, causing it to extend. This is shown on the graph by the line OSP.

The force F is then gradually reduced to zero and the wire contracts. This is shown on the graph by the line PQ.

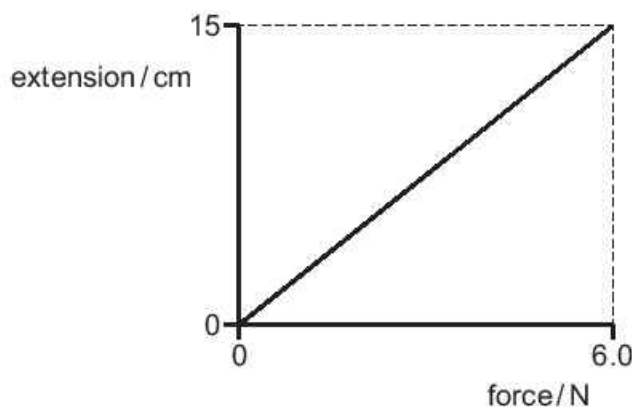


Which area on the graph represents the work done by the wire as it contracts?

- A** OSTO **B** OSPRO **C** QPRQ **D** OSPQO

150 9702/11/O/N/19/Q19

An extension–force graph for a spring is shown.

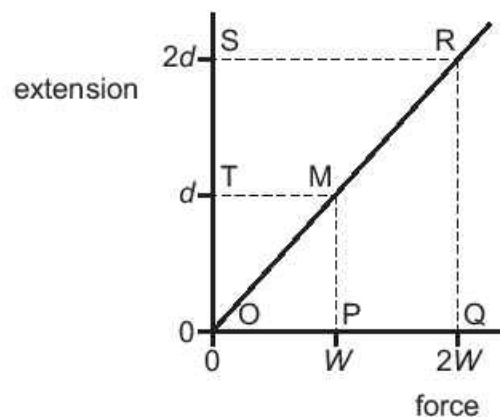


What is the spring constant of the spring?

- A** 0.025 Nm^{-1} **B** 0.40 Nm^{-1} **C** 2.5 Nm^{-1} **D** 40 Nm^{-1}

151 9702/11/O/N/19/Q20

A wire is extended by a force. The graph shows how the extension of the wire varies with the force applied.



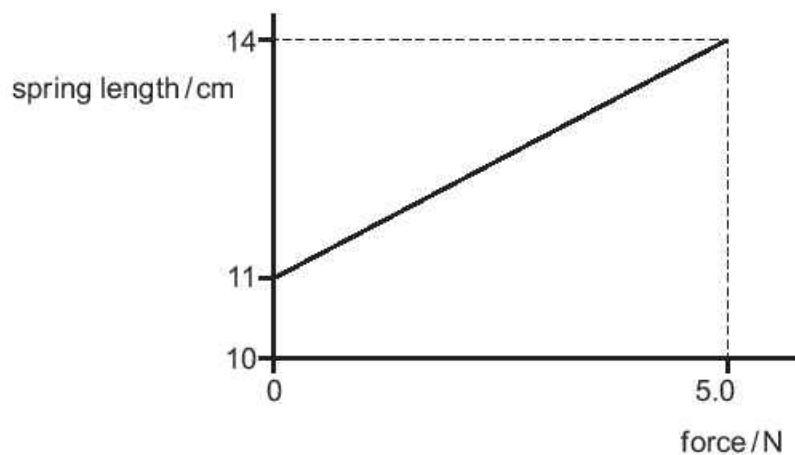
Initially a force W gives an extension d . The force is then increased to $2W$. This increases the extension to $2d$.

Which area of the graph represents the work done by the force when the force increases from W to $2W$?

- A** ORQ **B** OQRS **C** ORS **D** TMRS

152 9702/12/O/N/19/Q18

The graph shows the effect of applying a force of up to 5.0 N to a spring.



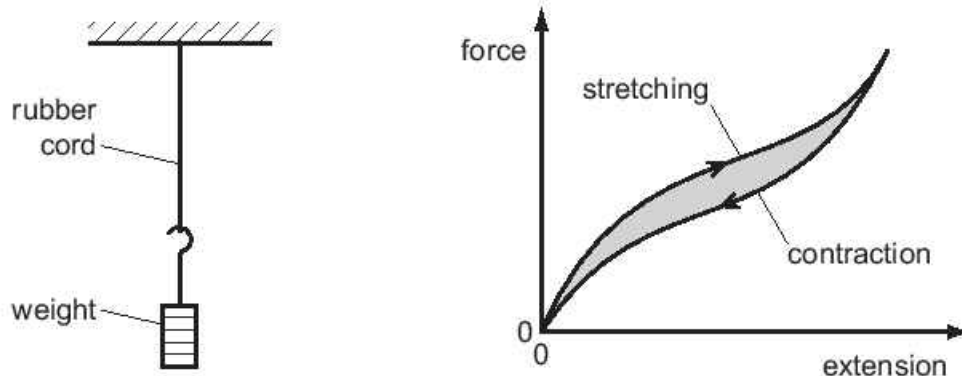
The spring obeys Hooke's law for forces up to 7.0 N.

What is the total extension of the spring produced by a 7.0 N force?

- A** 4.2 cm **B** 5.6 cm **C** 15 cm **D** 20 cm

153 9702/12/O/N/19/Q19

A rubber cord hangs from a rigid support. A weight attached to its lower end is gradually increased from zero, and then gradually reduced to zero.



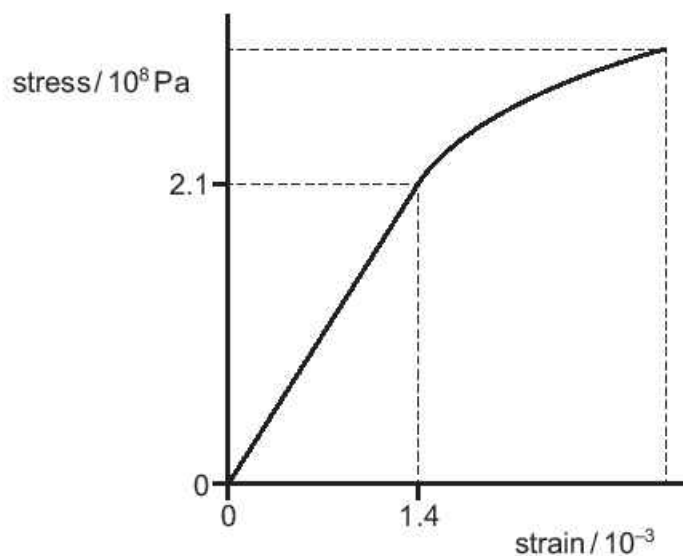
The force–extension curve for contraction is below the force–extension curve for stretching.

What does the shaded area between the curves represent?

- A the elastic potential energy stored in the rubber cord
- B the thermal energy dissipated in the rubber cord
- C the work done on the rubber cord during stretching
- D the work done by the rubber cord during contraction

154 9702/13/O/N/19/Q19

The stress–strain graph for a wire is shown.

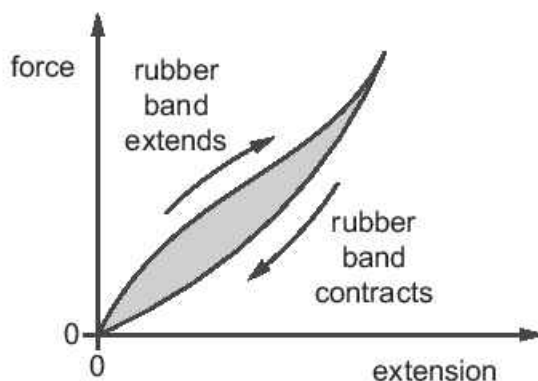


What is the Young modulus of the material of the wire?

- | | |
|------------------------------------|-----------------------------------|
| A $6.7 \times 10^{-12} \text{ Pa}$ | C $1.5 \times 10^8 \text{ Pa}$ |
| B $6.7 \times 10^{-9} \text{ Pa}$ | D $1.5 \times 10^{11} \text{ Pa}$ |

155 9702/13/O/N/19/Q20

The diagram shows a force–extension graph for a rubber band as the band is extended and then the stretching force is decreased to zero.

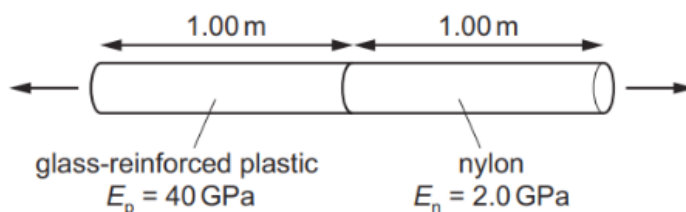


What can be deduced from the graph?

- A The rubber band does not return to its original length when the force is decreased to zero.
- B The rubber band obeys Hooke's law for the extensions shown.
- C The rubber band remains elastic for the extensions shown.
- D The shaded area represents the work done in extending the rubber band.

156 9702/12/F/M/20/Q19

A composite rod is made by attaching a glass-reinforced plastic rod and a nylon rod end to end, as shown.



The rods have the same cross-sectional area and each rod is 1.00 m in length. The Young modulus E_p of the plastic is 40 GPa and the Young modulus E_n of the nylon is 2.0 GPa.

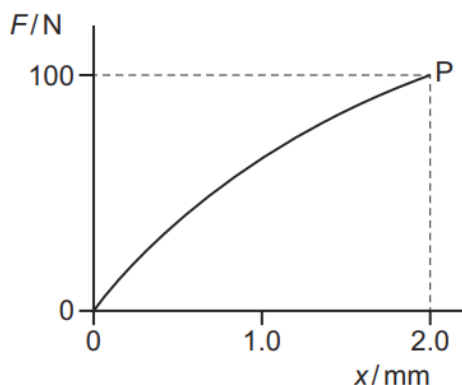
The composite rod will break when its total extension reaches 3.0 mm.

What is the greatest tensile stress that can be applied to the composite rod before it breaks?

- A $2.9 \times 10^6 \text{ Pa}$
- B $5.7 \times 10^6 \text{ Pa}$
- C $2.9 \times 10^9 \text{ Pa}$
- D $5.7 \times 10^9 \text{ Pa}$

157. 9702/12/F/M//20/Q20

The graph shows the non-linear force–extension curve for a wire made from a new composite material.



What is the best estimate of the work done in stretching the wire to point P?

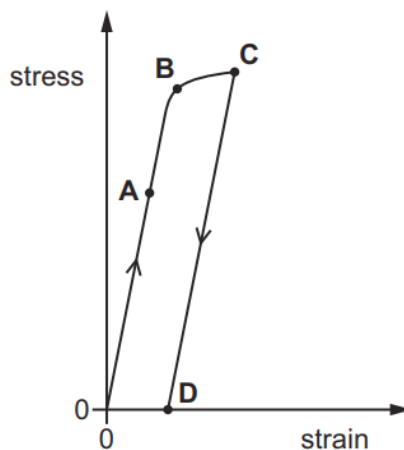
- A** 0.09 J **B** 0.10 J **C** 0.11 J **D** 0.20 J

158. 9702/11/M/J/20/Q20

A tensile force is used to extend a sample of a material. The force is then removed.

The variation with strain of the applied stress is shown on the graph.

Which point on the graph could represent the elastic limit for the material?



159. 9702/11/M/J/20/Q21

A tensile force is applied to an unstretched rubber band, causing it to stretch. The tensile force is then removed.

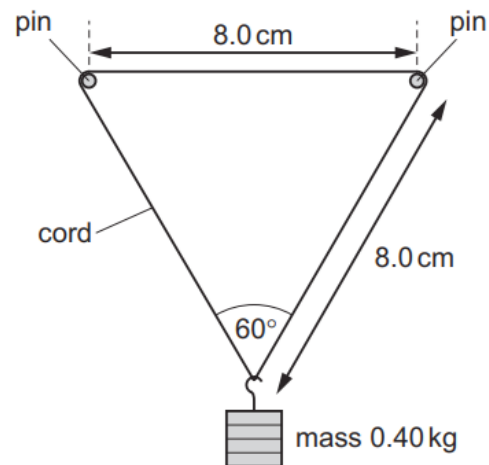
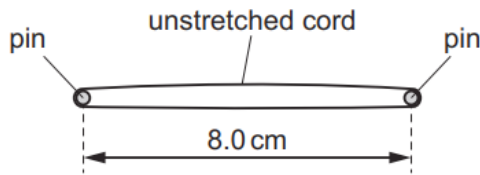
Which statement about the rubber band **must** be correct?

- A** If the rubber band stretches elastically and plastically, all the work done by the force is converted to thermal energy in the rubber.
- B** If the rubber band stretches elastically, it obeys Hooke's law.
- C** If the rubber band stretches elastically, the gradient of the force–extension graph represents the work done by the force.
- D** If the rubber band stretches plastically, the rubber band will be longer after the force is removed than it was before the force is applied.

160. 9702/12/M/J/20/Q18

An elastic cord of unstretched total length 16.0 cm and cross-sectional area $2.0 \times 10^{-6}\text{ m}^2$ is held horizontally by two smooth pins a distance 8.0 cm apart.

The cord obeys Hooke's law. A load of mass 0.40 kg is suspended centrally on the cord. The angle between the two sides of the cord supporting the load is 60° .



What is the Young modulus of the cord material?

- A** $5.7 \times 10^5\text{ Pa}$ **B** $1.1 \times 10^6\text{ Pa}$ **C** $2.3 \times 10^6\text{ Pa}$ **D** $3.9 \times 10^6\text{ Pa}$

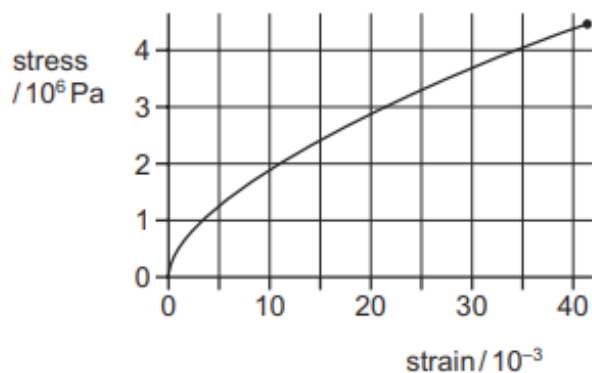
161. 9702/12/M/J/20/Q20

Which statement describes what is meant by the plastic deformation of a material?

- A** It always obeys Hooke's law.
B It does not return to its original length when the extending force is removed.
C It never obeys Hooke's law.
D It returns to its original length when the extending force is removed.

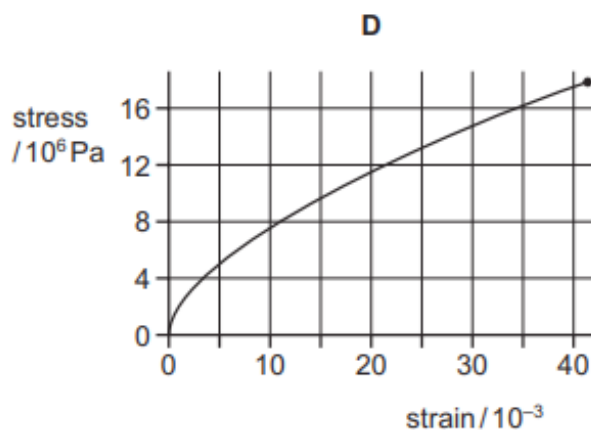
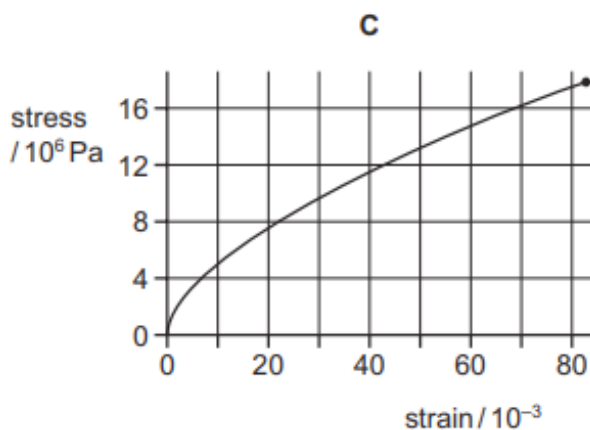
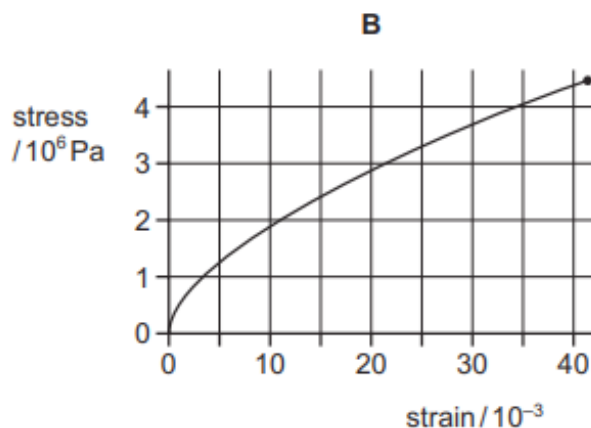
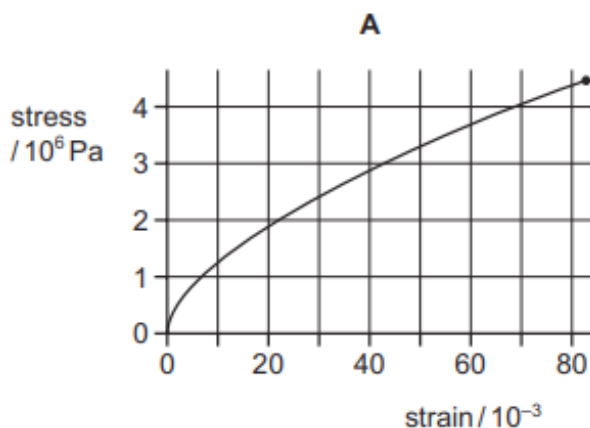
162. 9702/12/M/J/20/Q19

A student is investigating the mechanical properties of a metal. He applies different loads to a long thin wire up to its breaking point, and measures the extension of the wire for each load. He then plots a graph of stress against strain.



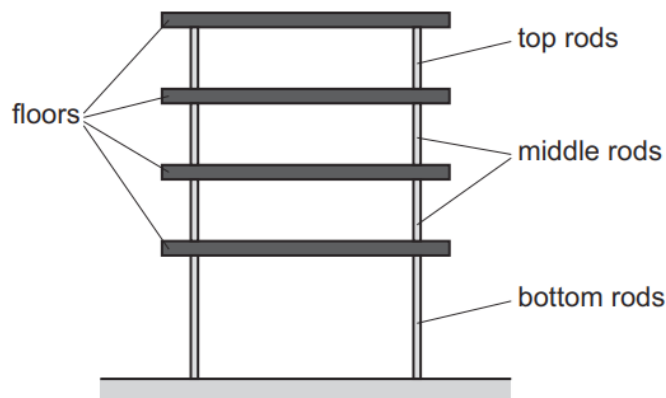
The student repeats the experiment with a wire made from the same metal, with twice the original length and half the diameter.

Which graph is obtained?



163. 9702/13/M/J/20/Q20

The diagram shows a simplified model of a building with four identical heavy floors.



The spacing of the bottom floor from the ground is twice that of the spacing between the floors. Between each floor are equal numbers of vertical steel supporting rods of negligible mass compared with the floors. The rods are of different diameters so that the stress in each rod is the same.

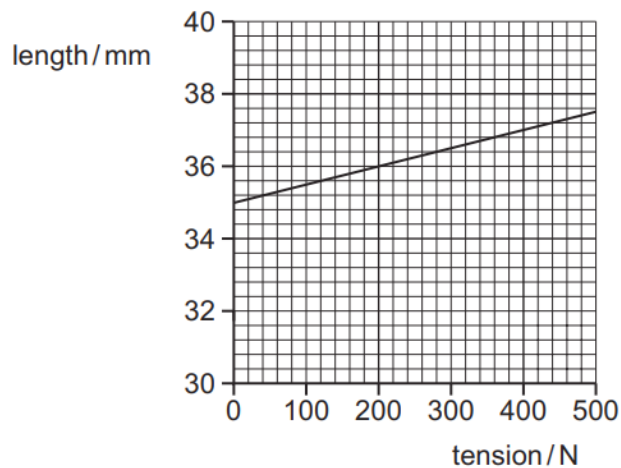
What is the ratio $\frac{\text{diameter of bottom rods}}{\text{diameter of top rods}}$?

- A** 2 **B** 4 **C** 8 **D** 16

164. 9702/13/M/J/20/Q21

The Achilles tendon in a rabbit's leg is stretched when the rabbit jumps.

The graph shows the variation with tension of the length of the tendon.

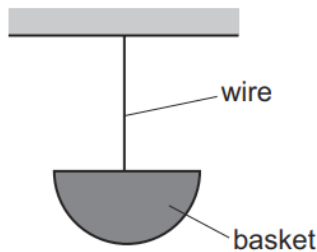


What is the strain energy in the tendon when the tension is 400 N?

- A** 0.40 J **B** 0.80 J **C** 2.4 J **D** 7.4 J

165. 9702/11/O/N/20/Q19

A wire of circular cross-section, which obeys Hooke's law, is used to suspend a basket as shown.



The Young modulus for the material of the wire is 2.5×10^{11} Pa.

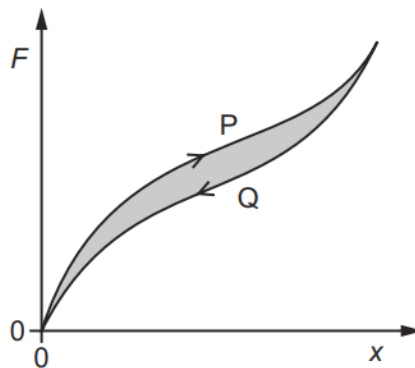
When a weight of 34 N is added to the basket, the strain in the wire increases by 6.0×10^{-5} .

What is the radius of the wire?

- A** 7.2×10^{-7} m **B** 2.3×10^{-6} m **C** 8.5×10^{-4} m **D** 1.7×10^{-3} m

166. 9702/11/O/N/20/Q20

An unstretched rubber cord is stretched by a force. The force F is plotted against the extension x . F is slowly increased from zero, causing the cord to extend along path P. F is then reduced back to zero along path Q.



What is represented by the shaded area?

- A** the elastic energy stored in the rubber cord
B the energy that causes plastic deformation
C the energy dissipated as heat
D the work done to extend the rubber cord

167. 9702/12/O/N/20/Q20

A mass of 60.0 g is suspended from a spring and the distance from the bottom of the spring to the floor is measured to be 16.4 cm.

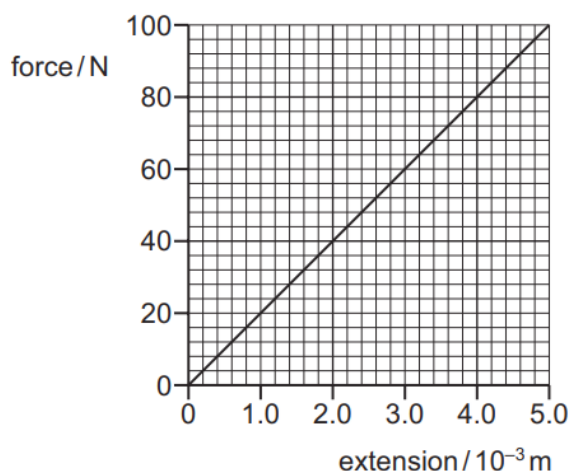
The mass is replaced with a 100.0 g mass and the distance from the bottom of the spring to the floor is now measured to be 12.6 cm. The spring obeys Hooke's law.

What is the spring constant of the spring?

- A** 1.05 N m^{-1} **B** 1.35 N m^{-1} **C** 10.3 N m^{-1} **D** 103 N m^{-1}

168. 9702/12/O/N/20/Q21

The graph shows the force–extension graph for a wire.



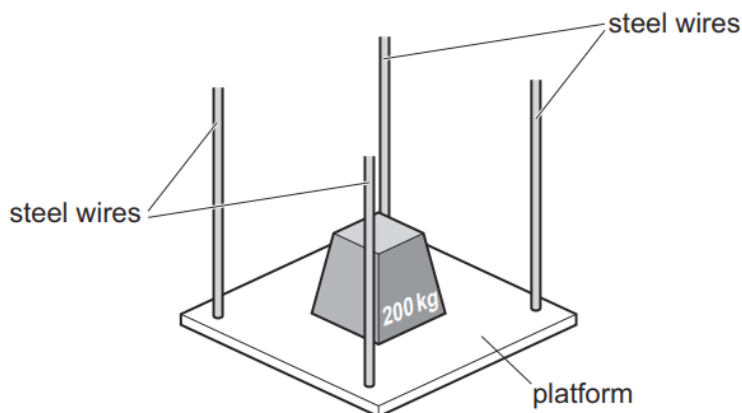
The wire is already extended by a force of 60 N.

How much work is done to increase the extension of the wire by 2.0 mm?

- A** 0.040 J **B** 0.090 J **C** 0.16 J **D** 0.25 J

169. 9702/13/O/N/20/Q20

A platform is suspended by four steel wires. Each wire is 5.0 m long and has a diameter of 3.0 mm. The Young modulus of steel is $2.1 \times 10^{11} \text{ Pa}$.



The wires obey Hooke's law when a load of mass 200 kg is placed on the platform.

How far will the platform descend because of the extension of the wires?

- A** $1.7 \times 10^{-4} \text{ m}$ **B** $4.1 \times 10^{-4} \text{ m}$ **C** $1.7 \times 10^{-3} \text{ m}$ **D** $6.6 \times 10^{-3} \text{ m}$

170. 9702/13/O/N/20/Q21

A tensile force of 7.00 MN is applied to a sample of steel. This causes the sample to extend by 5.00 mm in the direction of the force. The sample obeys Hooke's law.

What is the work done to extend the sample?

- A** 17.5 J **B** 35.0 J **C** 17.5 kJ **D** 35.0 kJ

171. 9702/12/F/M/21/Q19

Which expression is equal to the stress on a wire?

- A** $\frac{\text{extension}}{\text{original length}}$
- B** $\frac{\text{force}}{\text{cross-sectional area}}$
- C** $\frac{\text{force}}{\text{extension}}$
- D** $\frac{\text{Young modulus}}{\text{original length}}$

172. 9702/12/F/M/21/Q20

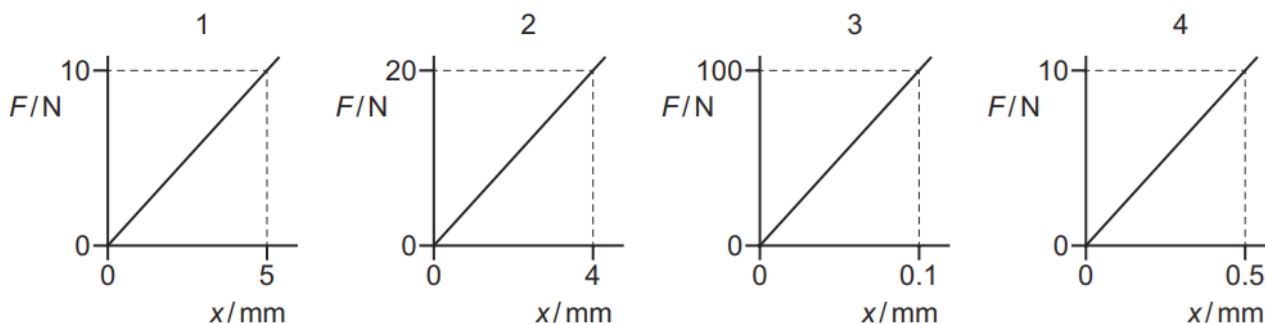
A wire is stretched by applying increasing values of force F . For each value of force applied, the extension x is recorded. A force–extension graph is plotted from the data obtained.

Which statement about the area under the graph **must** be correct?

- A** It can be calculated as $\frac{1}{2}Fx$.
- B** It is the elastic potential energy stored in the stretched sample.
- C** It is the work done in stretching the sample.
- D** It would be the same for any wire of the same material.

173. 9702/11/M/J/21/O19

The spring constants of four springs are determined by plotting the following graphs of force F against extension x .

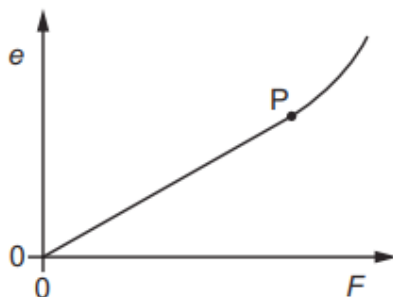


Which order of the graphs shows **decreasing** spring constants?

- A** $2 \rightarrow 1 \rightarrow 3 \rightarrow 4$
- B** $3 \rightarrow 4 \rightarrow 2 \rightarrow 1$
- C** $4 \rightarrow 2 \rightarrow 1 \rightarrow 3$
- D** $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

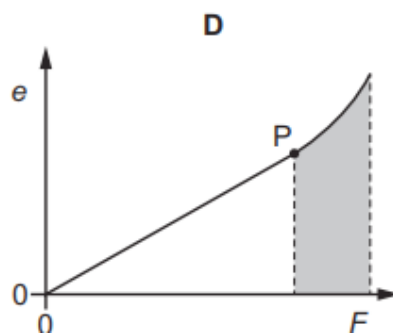
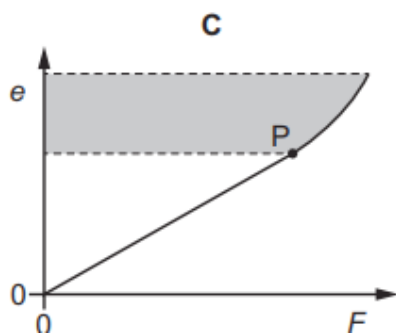
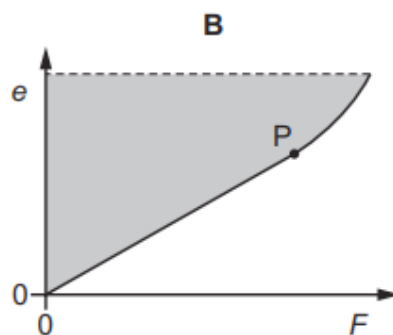
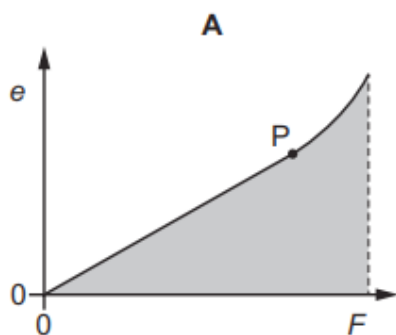
174. 9702/11/M/J/21/Q20

Forces are applied to the ends of a rod so that its length increases. The variation with force F of the extension e of the rod is shown.



The point P is the elastic limit.

Which shaded area represents the work done during the plastic deformation of the rod?



175. 9702/12/M/J/21/Q19

A steel bar of circular cross-section is under tension T , as shown.

The diameter of the wide portion is double the diameter of the narrow portion.



What is the value of $\frac{\text{stress in the wide portion}}{\text{stress in the narrow portion}}$?

A 0.25

B 0.50

C 2.0

D 4.0

176. 9702/12/M/J/21/Q20

Two guitar strings are stretched by tensile forces.

String X is stretched by a tensile force F that causes an extension x .

String Y is stretched by a tensile force $2F$ that causes an extension $2x$.

The strings obey Hooke's law.

What is the ratio $\frac{\text{strain energy in stretched string X}}{\text{strain energy in stretched string Y}}$?

A 4

B 2

C $\frac{1}{2}$

D $\frac{1}{4}$

177. 9702/13/M/J/21/Q16

A spring is initially neither compressed nor extended.

A force can be applied to this spring so that it is either compressed to a shorter length or extended to a longer length.

What is the change in the elastic potential energy in the spring when it is extended and when it is compressed?

	change in the elastic potential energy	
	spring is extended	spring is compressed
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

178. 9702/13/M/J/21/Q20

The stress σ in a material is given by the equation shown.

$$\sigma = \frac{F}{A}$$

The strain ε in the same material is given by the equation shown.

$$\varepsilon = \frac{x}{L}$$

Which expression gives the Young modulus of the material?

A $\frac{\varepsilon}{\sigma}$

B $\frac{Fx}{AL}$

C $\frac{\sigma x}{L}$

D $\frac{F}{A\varepsilon}$

179. 9702/13/M/J/21/Q21

What is an example of plastic deformation?

- A** A rubber ball is momentarily compressed every time it hits the ground.
- B** A spoon stirring some coffee in a ceramic mug hits its surface and makes a clinking sound.
- C** A toolbox is left on a horizontal plank. When the toolbox is removed, the plank is no longer straight.
- D** The spring in some bathroom weighing scales is compressed by a person standing on the scales.

180. 9702/11/O/N/21/Q19

A metal wire, of cross-sectional area A and unstretched length l , is subjected to stress σ . As a result it has strain ε .

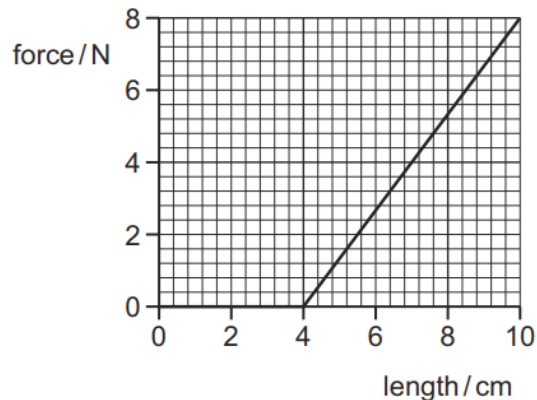
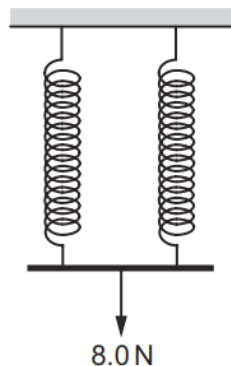
Which expression gives the Young modulus of the metal?

- A** $\frac{\varepsilon}{\sigma}$ **B** $\frac{\varepsilon A}{\sigma l}$ **C** $\frac{\sigma}{\varepsilon}$ **D** $\frac{\sigma l}{\varepsilon A}$

181. 9702/11/O/N/21/Q20

Two identical springs are connected in parallel.

A weight of 8.0 N is hung from the combination, as shown.



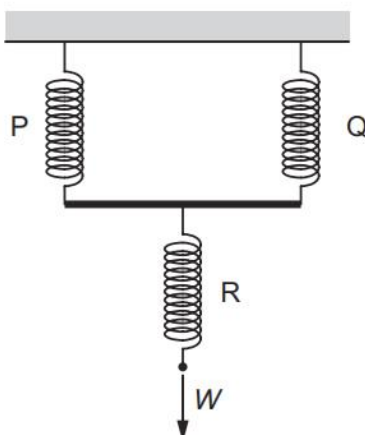
The graph shows the variation with length of the force applied to **one** of the springs.

What is the strain energy in **one** of the springs?

- A** 0.060 J **B** 0.12 J **C** 0.14 J **D** 0.24 J

182. 9702/12/O/N/21/Q21

Three springs are arranged vertically as shown.



Springs P and Q are identical and each has spring constant k . Spring R has spring constant $3k$.

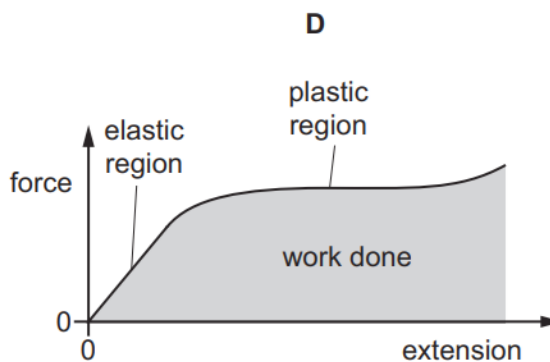
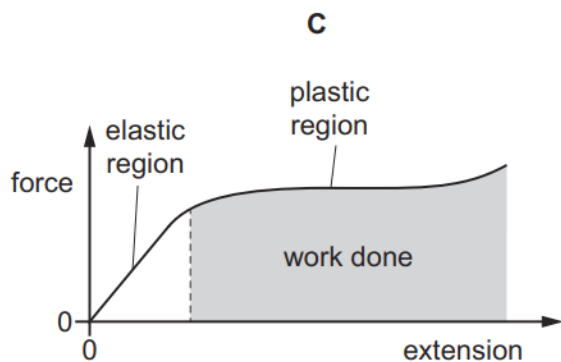
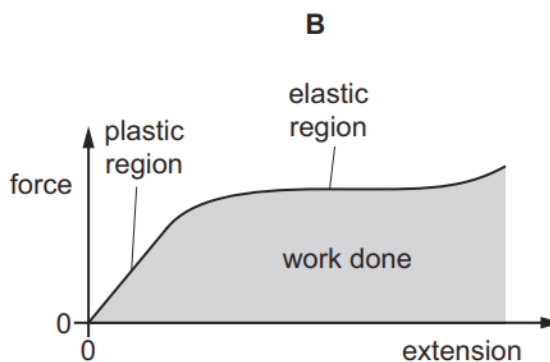
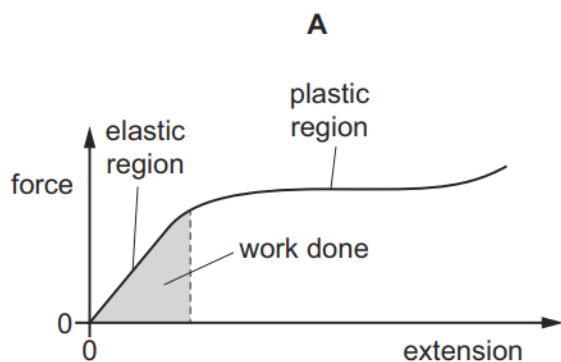
What is the increase in the overall length of the arrangement when a force W is applied as shown?

- A** $\frac{5W}{6k}$ **B** $\frac{4W}{3k}$ **C** $\frac{7}{2}kW$ **D** $4kW$

183. 9702/13/O/N/21/Q21

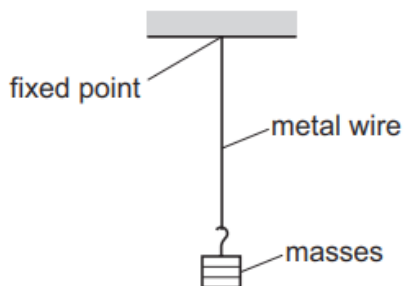
A metal wire is stretched to breaking point and the force—extension graph is plotted.

Which graph is correctly labelled with the elastic region, the plastic region and the area representing the work done to stretch the wire until it breaks?

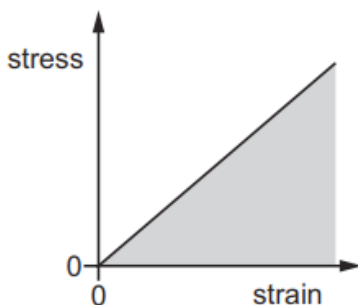


184. 9702/12/O/N/21/Q22

A length of metal wire is attached to a fixed point and hangs vertically. Masses are then suspended from the wire. Assume that the cross-sectional area of the wire remains constant.



A stress–strain graph for the wire is plotted, as shown.

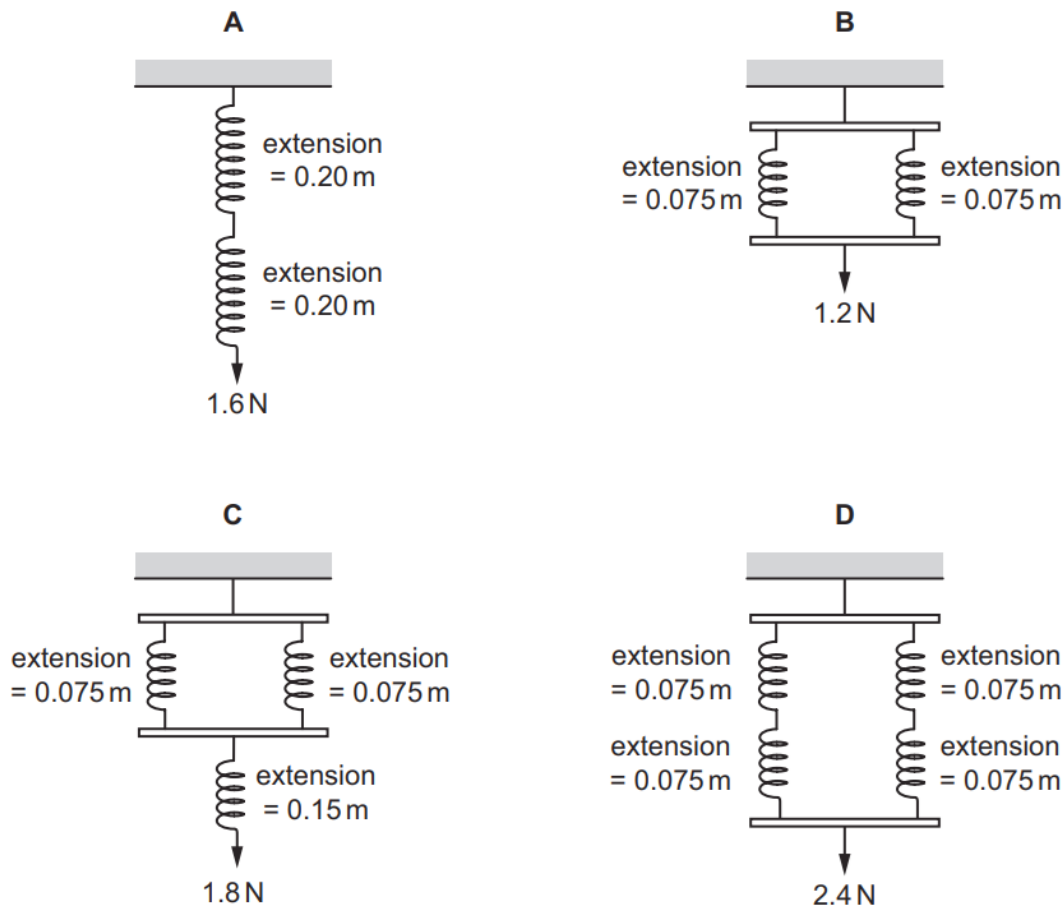


What is represented by the shaded area under the graph?

- A strain energy in the wire
- B $\frac{\text{strain energy in the wire}}{\text{cross-sectional area of the wire}}$
- C $\frac{\text{strain energy in the wire}}{\text{original length of the wire}}$
- D $\frac{\text{strain energy in the wire}}{\text{original volume of the wire}}$

185. 9702/13/O/N/21/Q20

Which spring combination has an overall spring constant of 8.0 N m^{-1} ?



186. 9702/12/F/M/22/Q18

A metal wire is stretched. The wire obeys Hooke's law.

Which quantity has a value that does **not** change?

- A** extension
- B** strain
- C** stress
- D** Young modulus

187. 9702/12/F/M/22/Q19

An object is stretched until it reaches the elastic limit.

Which statement must describe the stress on the object when it is at the elastic limit?

- A** It is the maximum stress for which the object obeys Hooke's law.
- B** It is the maximum stress that can be applied to the object before it has elastic deformation.
- C** It is the maximum stress that can be applied to the object before it has plastic deformation.
- D** It is the maximum stress the object can withstand before it breaks.

188. 9702/11/M/J/22/Q19

Two wires, P and Q, are made from the same metal and hang vertically from a steel girder.

Wire Q has half the length and twice the diameter of wire P.

Identical masses are attached to the bottom of each wire. Both wires obey Hooke's law as they are stretched by the weight of the masses.

What is the ratio $\frac{\text{extension of wire P}}{\text{extension of wire Q}}$?

A $\frac{8}{1}$

B $\frac{4}{1}$

C $\frac{1}{1}$

D $\frac{1}{2}$

189. 9702/11/M/J/22/Q20

Which statement about elastic and plastic deformation **must** be correct?

- A** Elastic deformation and plastic deformation are proportional to the applied force.
- B** Elastic deformation and plastic deformation cause no change in volume.
- C** Elastic deformation causes heating of the material but plastic deformation does not.
- D** Elastic deformation is reversible but plastic deformation is not.

190. 9702/12/M/J/22/Q19

A metal wire obeys Hooke's law and has a Young modulus of 2.0×10^{11} Pa. The wire has an original length of 1.6 m and a diameter of 0.48×10^{-3} m.

What is the spring constant of the wire?

A $7.2 \times 10^3 \text{ N m}^{-1}$

B $2.3 \times 10^4 \text{ N m}^{-1}$

C $2.9 \times 10^4 \text{ N m}^{-1}$

D $9.0 \times 10^4 \text{ N m}^{-1}$

191. 9702/12/M/J/22/Q20

A wire is being stretched by a tensile force.

Which statement about the elastic limit **must** be correct?

- A** The deformation is plastic after the elastic limit has been reached.
- B** The deformation is plastic until the elastic limit is reached.
- C** The extension is proportional to the tensile force after the elastic limit has been reached.
- D** The extension is proportional to the tensile force until the elastic limit is reached.

192. 9702/13/M/J/22/Q20

A steel wire has a length of 300 cm and a cross-sectional area of 0.50 mm^2 . The Young modulus of steel is $2.0 \times 10^{11} \text{ Pa}$.

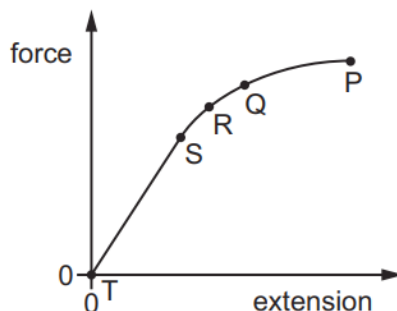
One end of the wire is attached to a fixed point. A load of 10 N is hung from the other end. The wire obeys Hooke's law.

What is the extension of the wire?

- A $3.0 \times 10^{-7} \text{ m}$
- B $3.0 \times 10^{-5} \text{ m}$
- C $3.0 \times 10^{-4} \text{ m}$
- D $3.0 \times 10^{-2} \text{ m}$

193. 9702/13/M/J/22/Q21

The extension of a copper wire is measured for different forces applied to the wire. A graph is plotted to show the variation of the force on the wire against extension. The maximum force is applied at point P.



Which statement **must** be correct?

- A Point R is the limit of proportionality.
- B The elastic potential energy of the wire at point S is given by the area under the graph between points T and S.
- C There is no plastic deformation between points Q and P.
- D The wire obeys Hooke's law up to a point between R and Q.

194. 9702/11/O/N/22/O20

An unstretched spring has a length of 2.0 cm.

The spring is then stretched within its limit of proportionality by a tensile force of 1.5 N so that the elastic potential energy stored in the spring is 0.045 J.

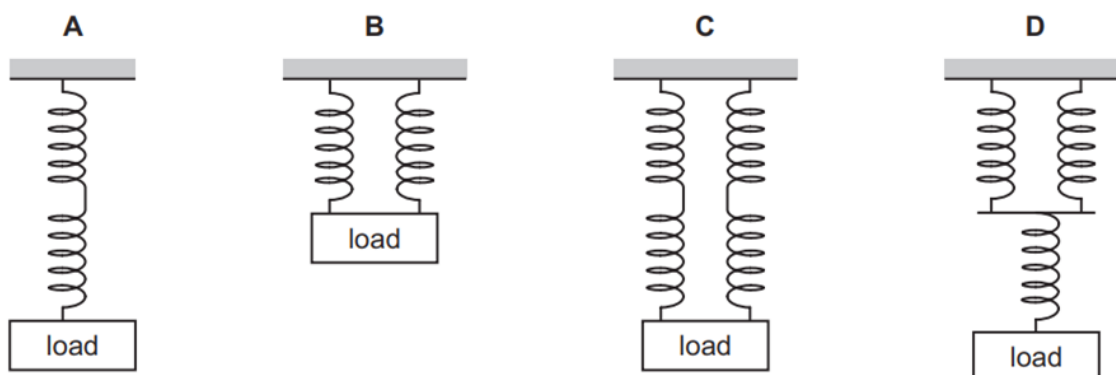
What is the stretched length of the spring?

- A 3.0 cm
- B 5.0 cm
- C 6.0 cm
- D 8.0 cm

195. 9702/11/O/N/22/Q21

Identical springs are joined in four arrangements.

Which arrangement has the same spring constant as a single spring?



196. 9702/12/O/N/22/Q20

A known tensile force acts on a metal wire. The wire does not exceed its limit of proportionality.

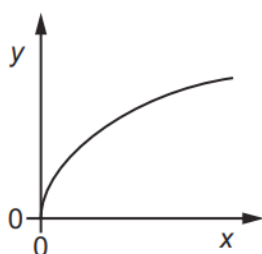
Which two measurements enable the strain of the wire to be calculated?

- A** the unstretched length of the wire and the cross-sectional area of the wire
- B** the unstretched length of the wire and the extension of the wire
- C** the Young modulus of the metal and the extension of the wire
- D** the Young modulus of the metal and the unstretched length of the wire

197. 9702/12/O/N/22/Q21

A wire is extended by different forces. The wire obeys Hooke's law.

A graph is plotted to show the variation of a quantity y with a quantity x .



What could x and y represent?

	x	y
A	elastic potential energy	extension
B	extension	force
C	force	extension
D	extension	elastic potential energy

198. 9702/13/O/N/22/Q19

A copper wire of length 3.6 m and diameter 1.22 mm is stretched by a force of 37 N. The wire obeys Hooke's law. The Young modulus of copper is 1.17×10^{11} Pa.

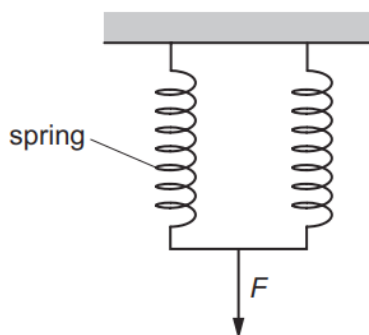
Which extension is caused by this force?

- A** 0.24 mm **B** 0.76 mm **C** 0.97 mm **D** 3.1 mm

199. 9702/13/O/N/22/Q20

A spring has spring constant k . The spring obeys Hooke's law and experiences extension x when a force F is applied to it. The resulting elastic potential energy of the spring is E_P .

The diagram shows two of these springs joined together in parallel and hanging from a fixed beam.



What is the extension and total elastic potential energy of this arrangement when the same force F is applied?

	extension	total elastic potential energy
A	$\frac{1}{2}x$	$\frac{1}{2}E_P$
B	$\frac{1}{2}x$	$\frac{1}{4}E_P$
C	x	$\frac{1}{2}E_P$
D	x	$\frac{1}{4}E_P$

200. 9702/12/F/M/23/Q18

A spring has an unstretched length of 4.50 cm. The spring is fixed at one end and a force of 35.0 N is applied to the other end so that the spring extends.

The spring obeys Hooke's law and has a spring constant of 420 N m^{-1} .

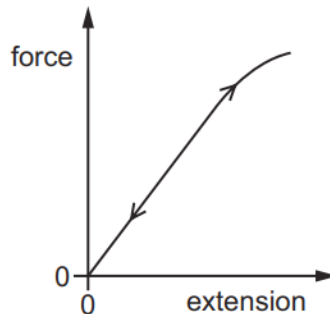
What is the strain of the extended spring?

- A** 0.019 **B** 0.083 **C** 1.85 **D** 2.67

201. 9702/12/F/M/23/Q19

A wire is fixed at one end and extended by a force that is applied to the other end. The force is slowly increased from zero and then slowly decreased back to zero.

The force–extension graph for the wire is shown.



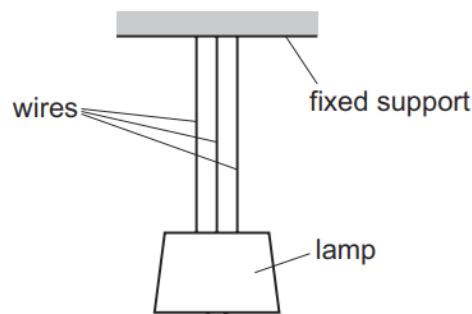
The graph line for the wire being loaded is the same as the graph line for the wire being unloaded.

Which statement describes the deformation of the wire?

- A** It is both elastic and plastic.
- B** It is elastic only.
- C** It is neither elastic nor plastic.
- D** It is plastic only.

202. 9702/11/M/J/23/Q18

A lamp is suspended in equilibrium from a fixed support by three long identical wires.



The weight of the lamp causes each wire to have an extension of 0.40 cm. The height h of the lamp above the floor is measured.

The middle wire suddenly breaks and the lamp falls a small distance as the extensions of the remaining two wires increase. The wires obey Hooke's law.

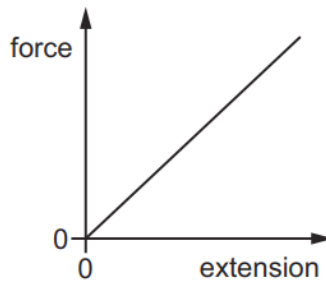
When the lamp is in equilibrium, the height h of the lamp above the floor is measured again.

What is the difference between the two values of h ?

- A** 0.20 cm
- B** 0.27 cm
- C** 0.40 cm
- D** 0.60 cm

203. 9702/11/M/J/23/Q19

The force–extension graph for a spring is shown.



What represents the work done to extend the spring?

- A** the area under the graph
- B** the gradient of the graph
- C** the reciprocal of the gradient of the graph
- D** twice the area under the graph

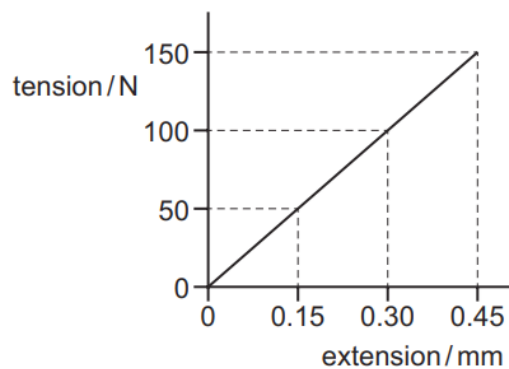
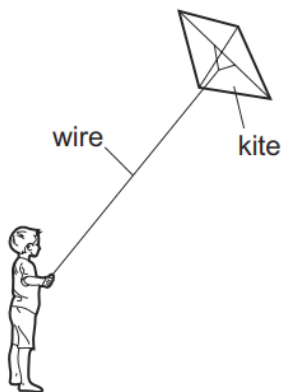
204. 9702/12/M/J/23/Q18

What is meant by the spring constant of a spring?

- A** extension per unit force
- B** $\frac{1}{2} \times \text{force} \times \text{extension}$
- C** force $\times$ extension
- D** force per unit extension

205. 9702/12/M/J/23/Q19

A child holds a thin metal wire that is attached to a kite. The graph shows how the extension of the wire varies with its tension.



A gust of wind increases the tension from 100 N to 150 N.

What is the change in the elastic potential energy of the wire caused by the gust of wind?

- A** 3.8 mJ
- B** 19 mJ
- C** 34 mJ
- D** 38 mJ

206. 9702/13/M/J/23/Q18

A metal wire has length 5.2 m and diameter 1.0 mm. The metal has Young modulus 360 GPa.

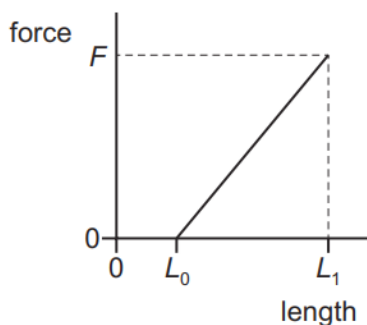
The wire is fixed at one end and a force is applied to the other end. The force extends the wire by 7.2 mm. The wire obeys Hooke's law.

What is the force applied to the wire?

- A** $1.2 \times 10^2 \text{ N}$
- B** $3.9 \times 10^2 \text{ N}$
- C** $5.0 \times 10^2 \text{ N}$
- D** $1.6 \times 10^3 \text{ N}$

207. 9702/13/M/J/23/Q19

The graph shows how the length of a spring varies with the force applied to it.



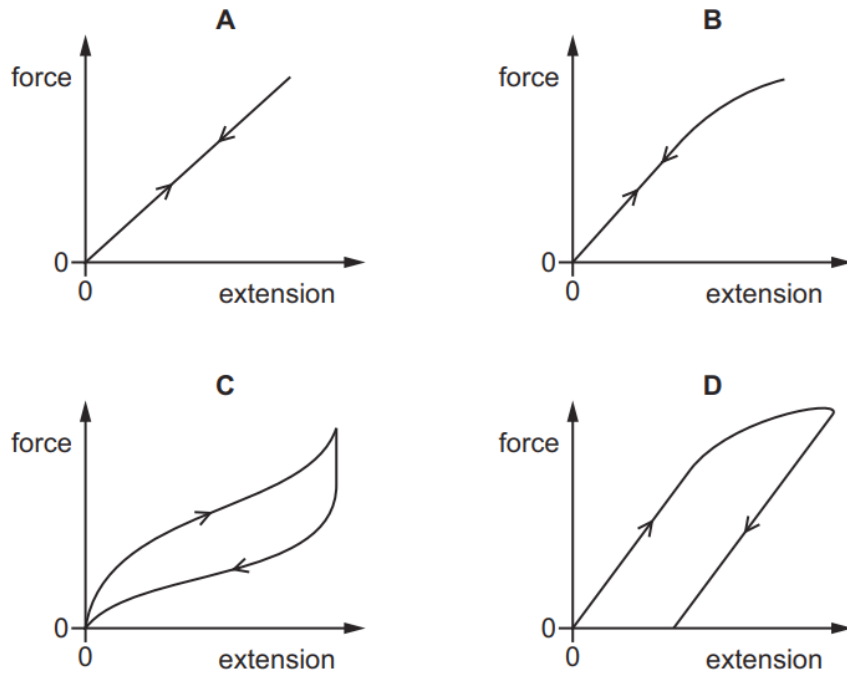
The spring has unstretched length L_0 . When a force F is applied, the spring has length L_1 .

What is the work done in stretching the spring to length L_1 ?

- A** $\frac{1}{2} FL_1$
- B** $\frac{1}{2} F(L_1 - L_0)$
- C** FL_1
- D** $F(L_1 - L_0)$

208. 9702/11/O/N/23/Q19

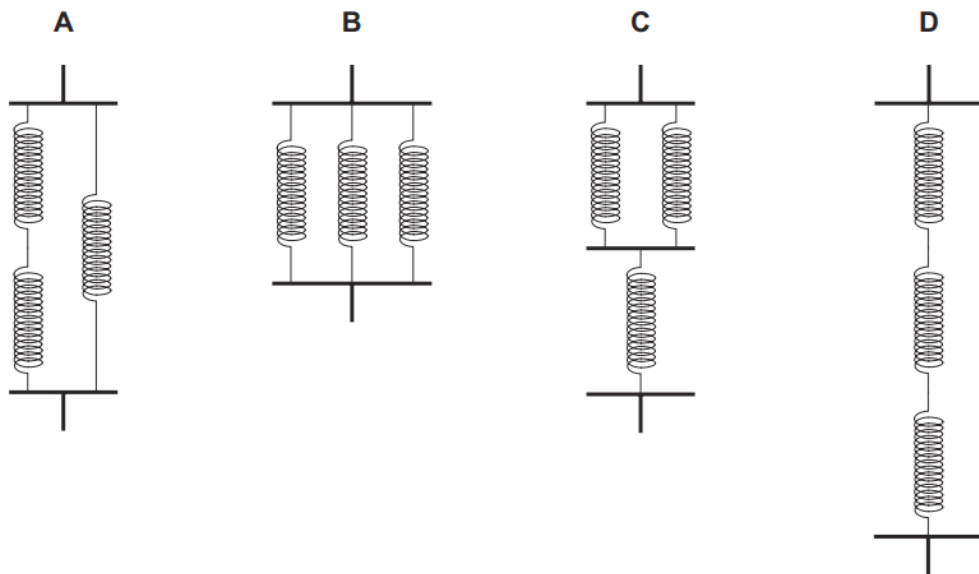
Which force–extension graph shows plastic deformation of a sample of material?



209. 9702/12/O/N/23/Q18

Three identical springs, each with the same spring constant, are connected together in four different arrangements, as shown.

Which arrangement has the largest combined spring constant?



210. 9702/13/O/N/23/Q18

A copper wire of diameter 1.6 mm is stretched within its limit of proportionality by a tensile force of 430 N.

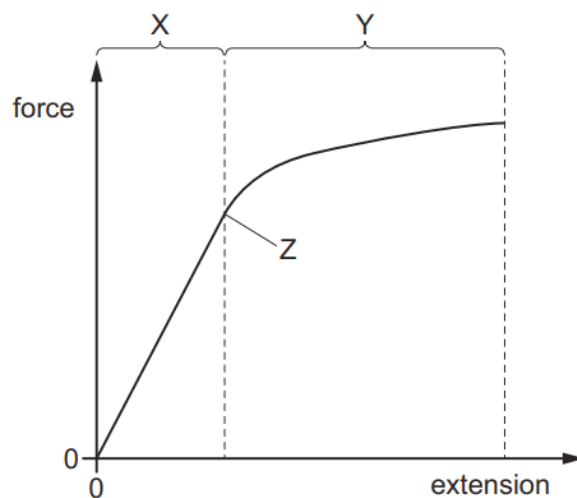
The Young modulus of copper is 130 GPa.

What is the strain in the wire?

- A** 4.1×10^{-4} **B** 1.3×10^{-3} **C** 1.6×10^{-3} **D** 5.2×10^{-3}

211. 9702/12/O/N/23/Q19

The force–extension graph for a wire is shown.

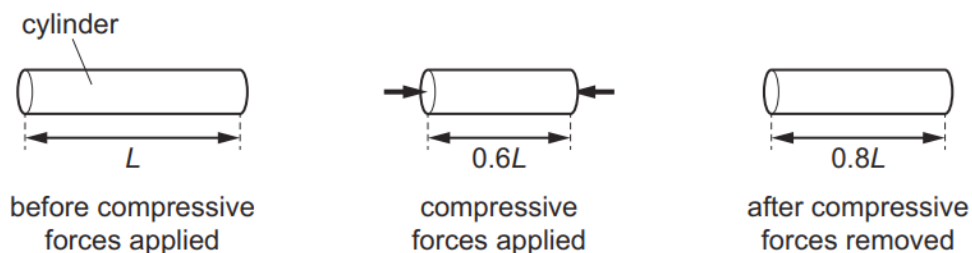


Which row could identify the labels X, Y and Z?

	limit of proportionality	region of elastic deformation	region of plastic deformation
A	X	Y	Z
B	Z	Y	X
C	Y	Z	X
D	Z	X	Y

212. 9702/13/O/N/23/Q19

Compressive forces are applied normally to the end faces of a cylinder of initial length L . The cylinder is compressed by the forces so that its length decreases to $0.6L$. After the compressive forces are removed, the cylinder's length increases to $0.8L$.



What describes the deformation of the cylinder when its length was $0.6L$?

- A** both elastic and plastic
- B** elastic only
- C** plastic only
- D** neither elastic nor plastic

ANSWERS

1. C	49. A	97. C	145. A	191. A
2. B	50. B	98. B	146. A	192. C
3. B	51. C	99. D	147. B	193. B
4. A	52. A	100. A	148. B	194. C
5. D	53. B	101. A	149. C	195. D
6. C	54. A	102. B	150. D	196. B
7. B	55. A	103. C	151. D	197. A
8. B	56. B	104. B	152. A	198. C
9. B	57. C	105. C	153. B	199. A
10. D	58. B	106. A	154. D	200. C
11. C	59. B	107. C	155. C	201. B
12. A	60. B	108. B	156. B	202. A
13. C	61. C	109. A	157. C	203. A
14. A	62. B	110. B	158. B	204. D
15. A	63. C	111. C	159. D	205. B
16. A	64. A	112. B	160. C	206. B
17. C	65. B	113. A	161. B	207. B
18. A	66. B	114. C	162. B	208. D
19. D	67. C	115. D	163. A	209. B
20. D	68. B	116. A	164. A	210. C
21. B	69. B	117. D	165. C	211. D
22. C	70. C	118. D	166. C	212. A
23. B	71. B	119. A	167. C	
24. C	72. D	120. B	168. C	
25. D	73. D	121. A	169. C	
26. A	74. D	122. D	170. C	
27. B	75. B	123. D	171. B	
28. B	76. B	124. A	172. C	
29. C	77. C	125. D	173. B	
30. B	78. A	126. A	174. C	
31. D	79. A	127. B	175. A	
32. A	80. C	128. B	176. D	
33. B	81. A	129. D	177. D	
34. B	82. B	130. D	178. D	
35. A	83. B	131. D	179. C	
36. B	84. C	132. C	180. C	
37. A	85. D	133. A	181. A	
38. C	86. A	134. C	182. A	
39. C	87. D	135. B	183. D	
40. A	88. B	136. B	184. D	
41. C	89. B	137. B	185. C	
42. C	90. C	138. C	186. D	
43. A	91. B	139. D	187. C	
44. D	92. C	140. D	188. A	
45. D	93. A	141. D	189. D	
46. C	94. C	142. B	190. B	
47. B	95. C	143. A		
48. D	96. A	144. B		

- 1 A spring having spring constant k hangs vertically from a fixed point. A load of weight L , when hung from the spring, causes an extension e . The elastic limit of the spring is not exceeded.

(a) State

- (i) what is meant by an *elastic deformation*,

.....
 [2]

- (ii) the relation between k , L and e .

..... [1]

- (b) Some identical springs, each with spring constant k , are arranged as shown in Fig. 4.1.

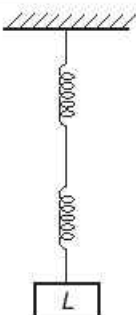
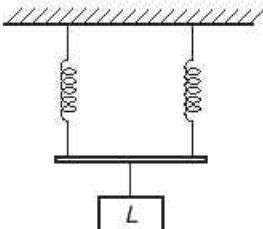
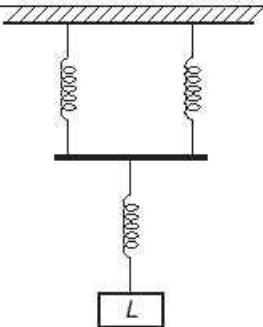
arrangement	total extension	spring constant of arrangement
		
		
		

Fig. 4.1

The load on each of the arrangements is L .

For each arrangement in Fig. 4.1, complete the table by determining

- (i) the total extension in terms of e ,

- (ii) the spring constant in terms of k .

[5]

- 2 A uniform wire has length L and area of cross-section A .

9702/21/O/N/09/Q4

The wire is fixed at one end so that it hangs vertically with a load attached to its free end, as shown in Fig. 4.1.

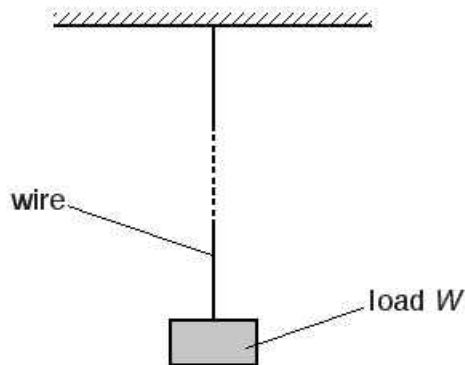


Fig. 4.1

When the load of magnitude W is attached to the wire, it extends by an amount e . The elastic limit of the wire is not exceeded.

The material of the wire has resistivity ρ .

- (a) (i)** Explain what is meant by extends *elastically*.

.....

 [2]

- (ii)** Write down expressions, in terms of L , A , W , ρ and e for

1. the resistance R of the unstretched wire,

$$R = \dots\dots\dots [1]$$

2. the Young modulus E of the wire.

$$E = \dots\dots\dots [1]$$

- (b)** A steel wire has resistance $0.44 \, \Omega$. Steel has resistivity $9.2 \times 10^{-8} \, \Omega \text{ m}$.

A load of 34 N hung from the end of the wire causes an extension of $7.7 \times 10^{-4} \text{ m}$.

Using your answers in **(a)(ii)**, calculate the Young modulus E of steel.

$$E = \dots\dots\dots \text{ Pa } [3]$$

- 3 (a) Explain what is meant by *strain energy* (*elastic potential energy*).

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.....
.....
..... [2]

- (b) A spring that obeys Hooke's law has a spring constant k .

Show that the energy E stored in the spring when it has been extended elastically by an amount x is given by

$$E = \frac{1}{2}kx^2.$$

[3]

- (c) A light spring of unextended length 14.2 cm is suspended vertically from a fixed point, as illustrated in Fig. 4.1.

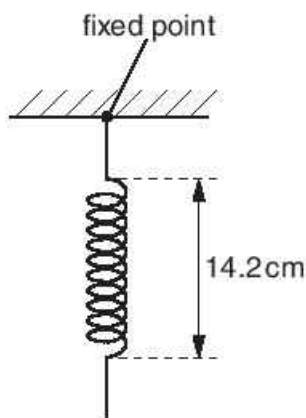


Fig. 4.1

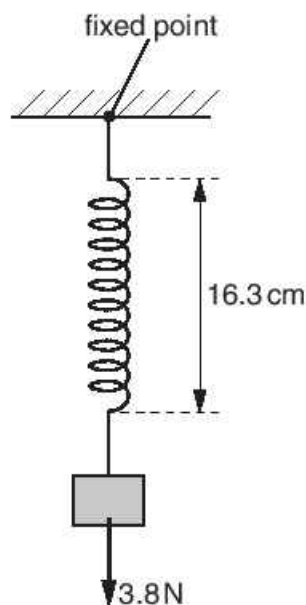


Fig. 4.2

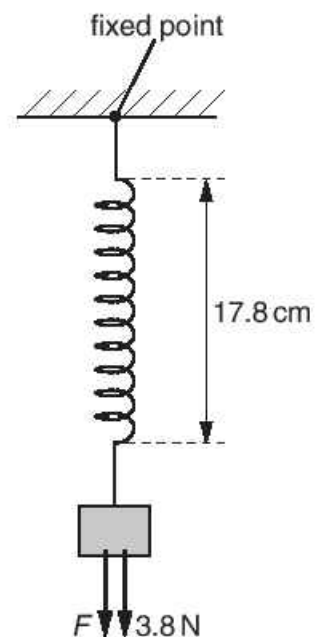


Fig. 4.3

A mass of weight 3.8 N is hung from the end of the spring, as shown in Fig. 4.2. The length of the spring is now 16.3 cm.

An additional force F then extends the spring so that its length becomes 17.8 cm, as shown in Fig. 4.3.

The spring obeys Hooke's law and the elastic limit of the spring is not exceeded.

- (i) Show that the spring constant of the spring is 1.8 N cm^{-1} .

[1]

- (ii) For the extension of the spring from a length of 16.3cm to a length of 17.8cm,
1. calculate the change in the gravitational potential energy of the mass on the spring,

change in energy = J [2]

2. show that the change in elastic potential energy of the spring is 0.077 J,

[1]

3. determine the work done by the force F .

work done = J [1]

- 4 (a) Tensile forces are applied to opposite ends of a copper rod so that the rod is stretched. The variation with stress of the strain of the rod is shown in Fig. 5.1.

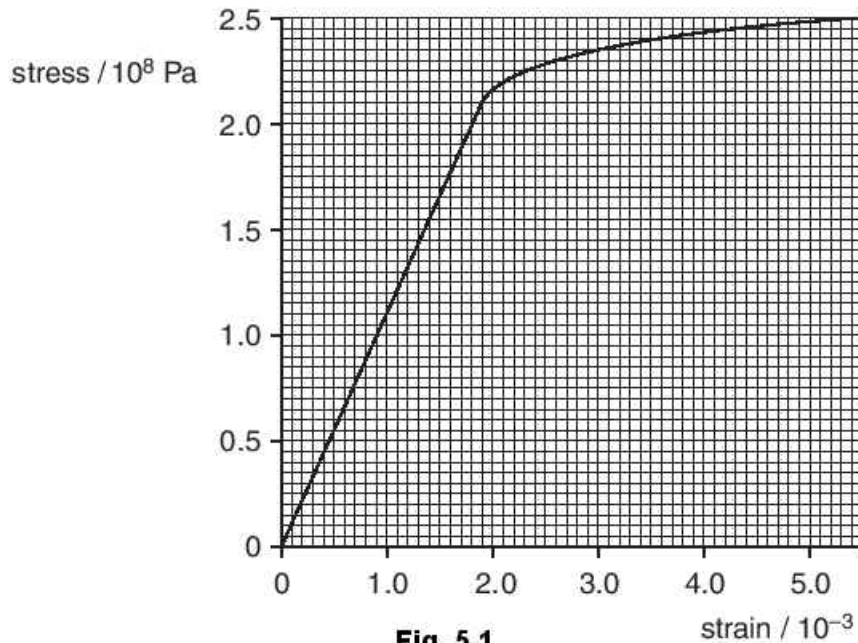


Fig. 5.1

- (i) Use Fig. 5.1 to determine the Young modulus of copper.

Young modulus = Pa [3]

- (ii) On Fig. 5.1, sketch a line to show the variation with stress of the strain of the rod as the stress is reduced from 2.5×10^6 Pa to zero. No further calculations are expected. [1]
- (b) The walls of the tyres on a car are made of a rubber compound. The variation with stress of the strain of a specimen of this rubber compound is shown in Fig. 5.2.

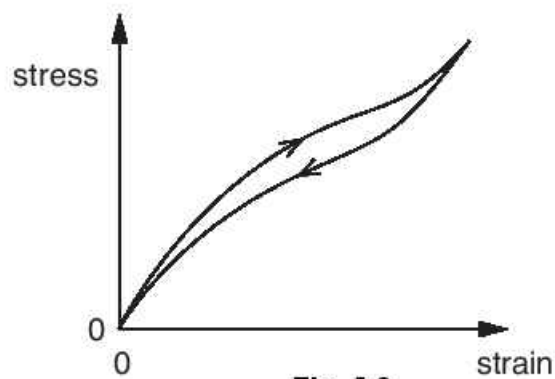


Fig. 5.2

As the car moves, the walls of the tyres bend and straighten continuously. Use Fig. 5.2 to explain why the walls of the tyres become warm.

.....

.....

..... [3]

- 5 (a) A uniform wire has length L and constant area of cross-section A . The material of the wire has Young modulus E and resistivity ρ . A tension F in the wire causes its length to increase by ΔL .

For this wire, state expressions, in terms of L , A , F , ΔL and ρ for

- (i) the stress σ ,

..... [1]

- (ii) the strain ε ,

..... [1]

- (iii) the Young modulus E ,

..... [1]

- (iv) the resistance R .

..... [1]

- (b) One end of a metal wire of length 2.6 m and constant area of cross-section $3.8 \times 10^{-7} \text{ m}^2$ is attached to a fixed point, as shown in Fig. 4.1.

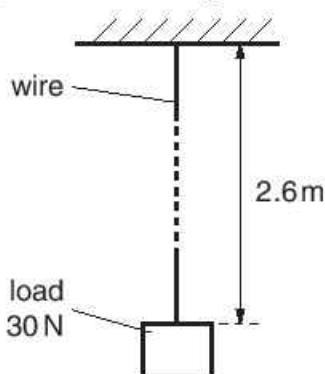


Fig. 4.1

The Young modulus of the material of the wire is $7.0 \times 10^{10} \text{ Pa}$ and its resistivity is $2.6 \times 10^{-8} \Omega \text{ m}$.

A load of 30 N is attached to the lower end of the wire. Assume that the area of cross-section of the wire does not change.

For this load of 30 N,

- (i) show that the extension of the wire is 2.9 mm,

[1]

- (ii) calculate the change in resistance of the wire.

change = Ω [2]

- (c) The resistance of the wire changes with the applied load.

Comment on the suggestion that this change of resistance could be used to measure the magnitude of the load on the wire.

.....

..... [2]

- 6 (a) A metal wire has spring constant k . Forces are applied to the ends of the wire to extend it within the limit of Hooke's law.
Show that, for an extension x , the strain energy E stored in the wire is given by

$$E = \frac{1}{2} kx^2.$$

[4]

- (b) The wire in (a) is now extended beyond its elastic limit. The forces causing the extension are then removed.
The variation with extension x of the tension F in the wire is shown in Fig. 4.1.

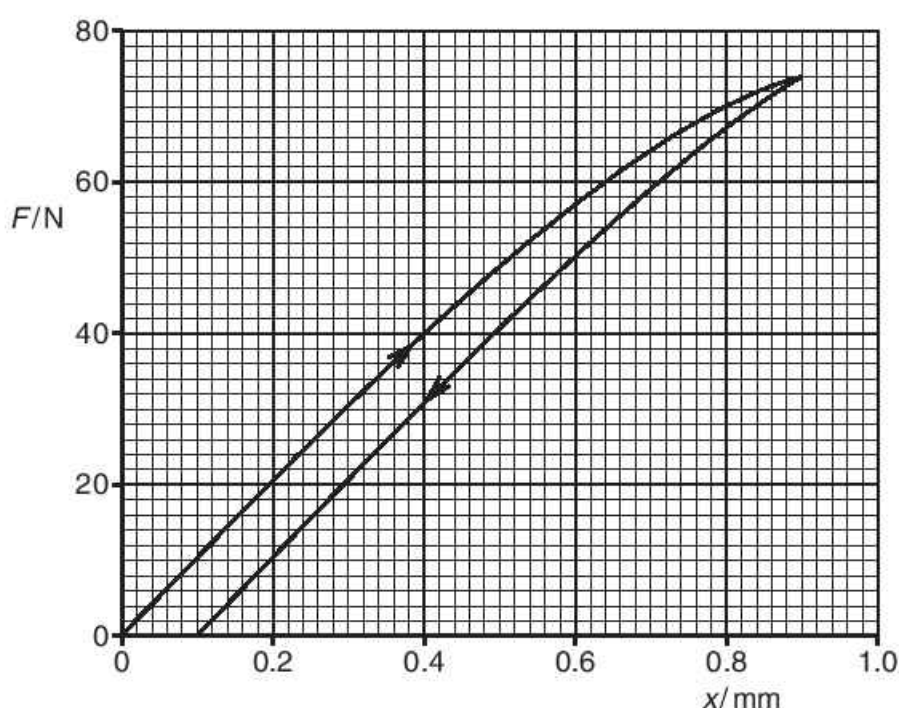


Fig. 4.1

Energy E_S is expended to cause a permanent extension of the wire.

- (i) On Fig. 4.1, shade the area that represents the energy E_S . [1]
- (ii) Use Fig. 4.1 to calculate the energy E_S .

$$E_S = \dots\dots\dots \text{mJ} [3]$$

- (iii) Suggest the change in the structure of the wire that is caused by the energy E_S .

.....

..... [1]

7 (a) Define, for a wire,

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(i) *stress*,

.....
..... [1]

(ii) *strain*.

.....
..... [1]

(b) A wire of length 1.70 m hangs vertically from a fixed point, as shown in Fig. 4.1.

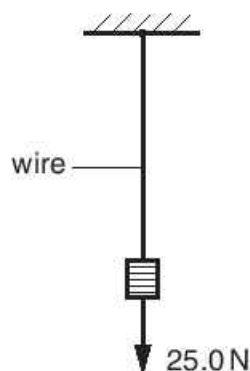


Fig. 4.1

The wire has cross-sectional area $5.74 \times 10^{-8} \text{ m}^2$ and is made of a material that has a Young modulus of $1.60 \times 10^{11} \text{ Pa}$. A load of 25.0 N is hung from the wire.

(i) Calculate the extension of the wire.

extension = m [3]

(ii) The same load is hung from a second wire of the same material. This wire is twice the length but the **same volume** as the first wire. State and explain how the extension of the second wire compares with that of the first wire.

.....
.....
..... [3]

[2]

[4]

- 9 (a) State Hooke's law.

.....
 [1]

- (b) The variation with extension x of the force F for a spring A is shown in Fig. 6.1.

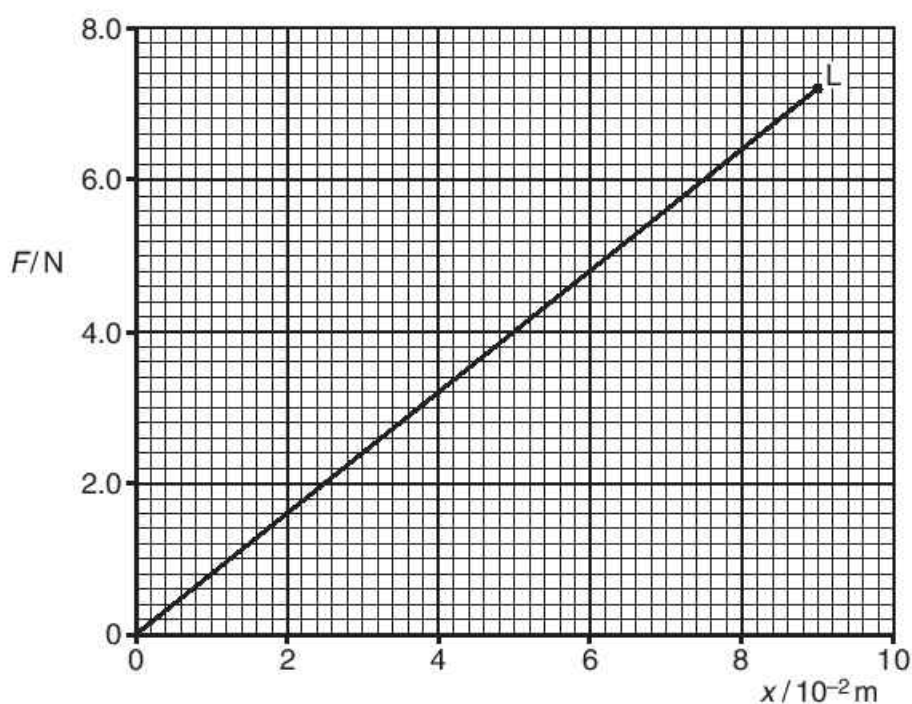


Fig. 6.1

The point L on the graph is the elastic limit of the spring.

- (i) Describe the meaning of *elastic limit*.

.....

 [1]

- (ii) Calculate the spring constant k_A for spring A.

$$k_A = \dots\dots\dots \text{Nm}^{-1} \text{ [1]}$$

- (iii) Calculate the work done in extending the spring with a force of 6.4 N.

work done = J [2]

- (c) A second spring B of spring constant $2k_A$ is now joined to spring A, as shown in Fig. 6.2.

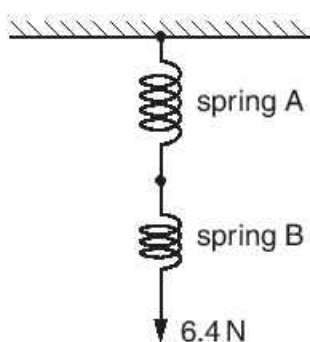


Fig. 6.2

A force of 6.4 N extends the combination of springs.

For the combination of springs, calculate

- (i) the total extension,

extension = m [1]

- (ii) the spring constant.

spring constant = Nm^{-1} [1]

10 (a) Define

9702/22/O/N/11/Q3

(i) *stress*,

.....
.....[1]

(ii) *strain*.

.....
.....[1]

(b) Explain the term *elastic limit*.

.....
.....[1]

(c) Explain the term *ultimate tensile stress*.

.....
.....
.....[2]

(d) (i) A **ductile** material in the form of a wire is stretched up to its breaking point. On Fig. 3.1, sketch the variation with extension x of the stretching force F .



Fig. 3.1

[2]

- (ii) On Fig. 3.2, sketch the variation with x of F for a **brittle** material up to its breaking point.



Fig. 3.2

[1]

- (e) (i) Explain the features of the graphs in (d) that show the characteristics of ductile and brittle materials.

.....

.....

.....

.....

..... [2]

- (ii) The force F is removed from the materials in (d) just before the breaking point is reached. Describe the subsequent change in the extension for

1. the ductile material,

.....

..... [1]

2. the brittle material.

.....

..... [1]

- 11 One end of a spring is fixed to a support. A mass is attached to the other end of the spring. The arrangement is shown in Fig. 3.1.

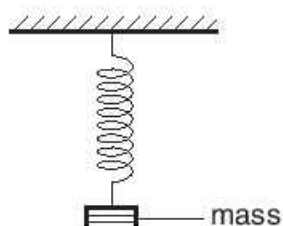


Fig. 3.1

- (a) The mass is in equilibrium. Explain, by reference to the forces acting on the mass, what is meant by equilibrium.

.....

 [2]

- (b) The mass is pulled down and then released at time $t = 0$. The mass oscillates up and down. The variation with t of the displacement of the mass d is shown in Fig. 3.2.

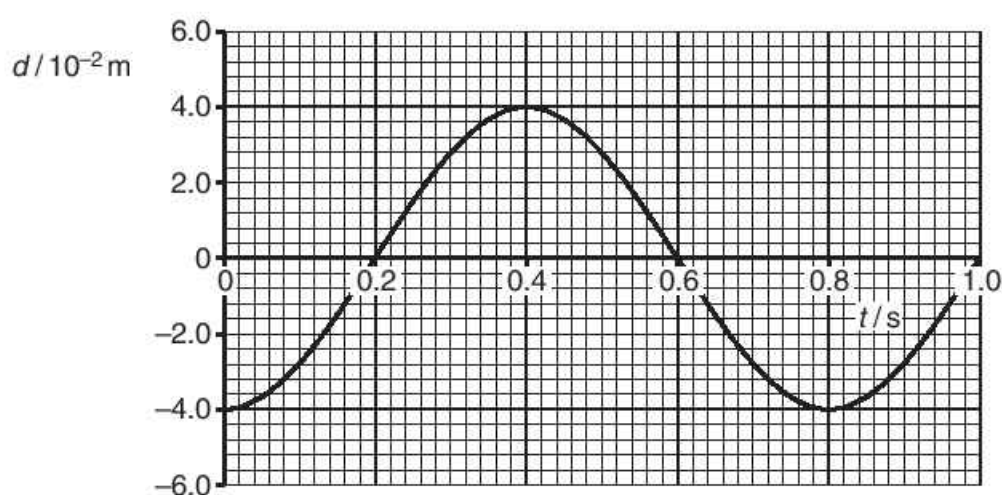


Fig. 3.2

Use Fig. 3.2 to state a time, one in each case, when

- (i) the mass is at maximum speed,

time = s [1]

- (ii) the elastic potential energy stored in the spring is a maximum,

time = s [1]

- (iii) the mass is in equilibrium.

time = s [1]

- (c) The arrangement shown in Fig. 3.3 is used to determine the length l of a spring when different masses M are attached to the spring.

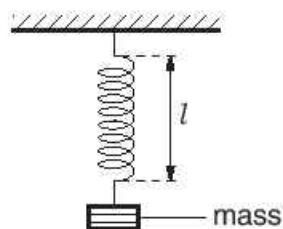


Fig. 3.3

The variation with mass M of l is shown in Fig. 3.4.

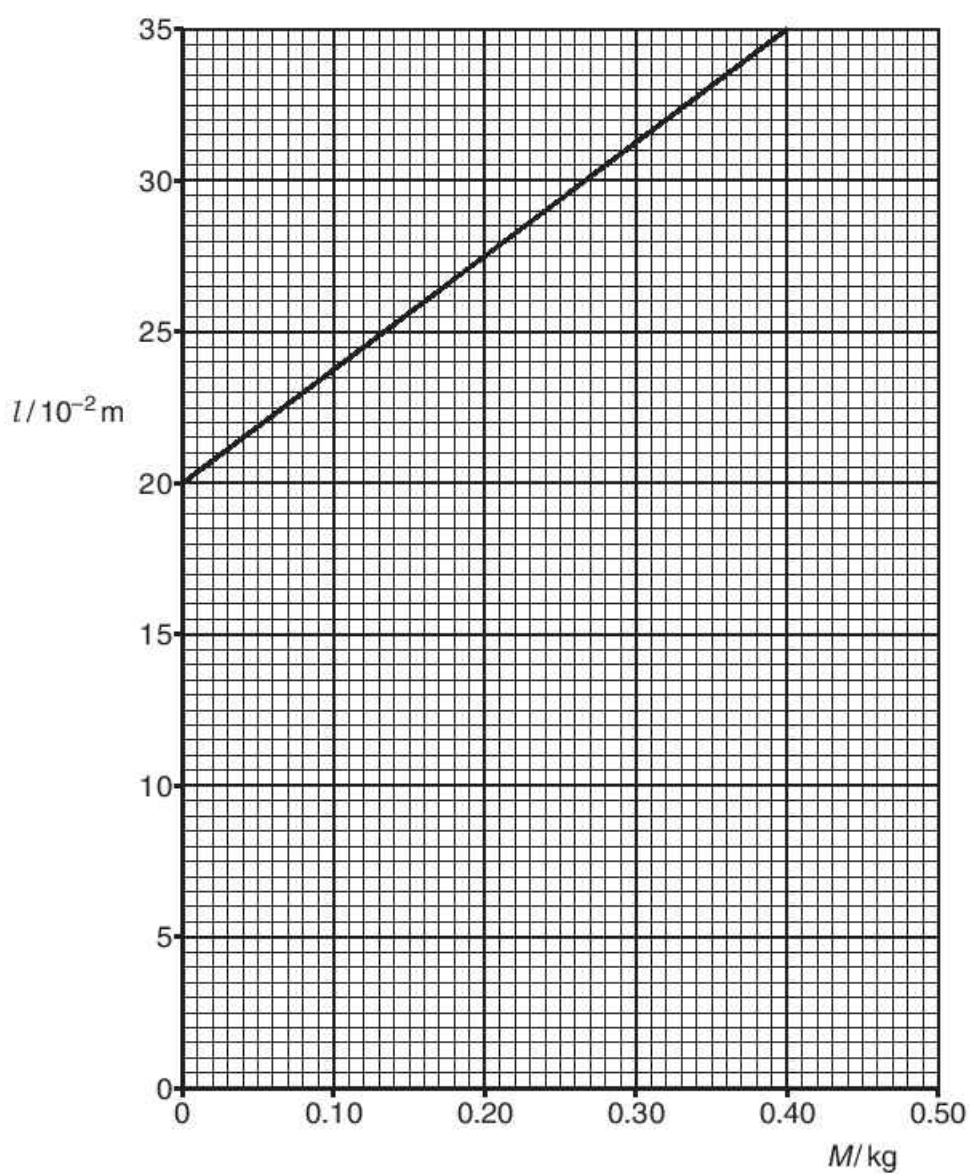


Fig. 3.4

- (i) State and explain whether the spring obeys Hooke's law.

.....
.....
..... [2]

- (ii) Show that the force constant of the spring is 26 N m^{-1} .

[2]

- (iii) A mass of 0.40 kg is attached to the spring. Calculate the energy stored in the spring.

energy = J [3]

12 (a) Define the *Young modulus*.

9702/22/M/J/12/Q5

.....
 [1]

- (b) A load F is suspended from a fixed point by a steel wire. The variation with extension x of F for the wire is shown in Fig. 5.1.

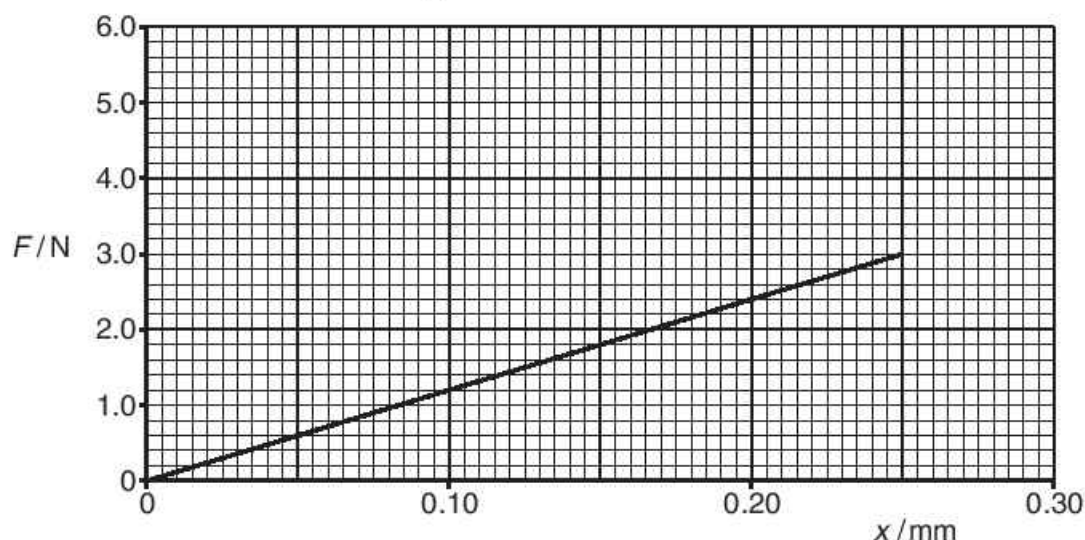


Fig. 5.1

- (i) State two quantities, other than the gradient of the graph in Fig. 5.1, that are required in order to determine the Young modulus of steel.

1.
 2. [1]

- (ii) Describe how the quantities you listed in (i) may be measured.

.....
 [2]

- (iii) A load of 3.0 N is applied to the wire. Use Fig. 5.1 to calculate the energy stored in the wire.

energy = J [2]

- (c) A copper wire has the same original dimensions as the steel wire. The Young modulus for steel is $2.2 \times 10^{11} \text{ N m}^{-2}$ and for copper is $1.1 \times 10^{11} \text{ N m}^{-2}$.

On Fig. 5.1, sketch the variation with x of F for the copper wire for extensions up to 0.25 mm. The copper wire is not extended beyond its limit of proportionality.

[2]

13 (a) Explain what is meant by *plastic deformation*.

9702/21/O/N/12/Q5

.....
..... [1]

(b) A copper wire of uniform cross-sectional area $1.54 \times 10^{-6} \text{ m}^2$ and length 1.75 m has a breaking stress of $2.20 \times 10^8 \text{ Pa}$. The Young modulus of copper is $1.20 \times 10^{11} \text{ Pa}$.

(i) Calculate the breaking force of the wire.

breaking force = N [2]

(ii) A stress of $9.0 \times 10^7 \text{ Pa}$ is applied to the wire. Calculate the extension.

extension = m [2]

(c) Explain why it is not appropriate to use the Young modulus to determine the extension when the breaking force is applied.

.....
..... [1]

.....
 [1]

- (b) A spring is attached to a support and hangs vertically, as shown in Fig. 6.1. An object M of mass 0.41 kg is attached to the lower end of the spring. The spring extends until M is at rest at R.

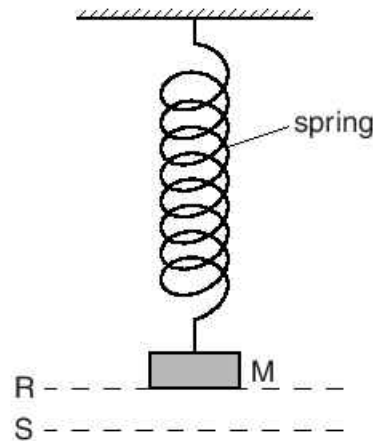


Fig. 6.1

The spring constant of the spring is 25 N m^{-1} . Show that the extension of the spring is about 0.16 m.

[2]

- (c) The object M in Fig. 6.1 is pulled down a further 0.060 m to S and is then released.

For M, just as it is released,

- (i) state the forces acting on M,

..... [1]

- (ii) calculate the acceleration of M.

acceleration = ms^{-2} [3]

- (d) Describe and explain the energy changes from the time the object M in Fig. 6.1 is released to the time it first returns to R.

.....

 [2]

15 Energy is stored in a metal wire that is extended elastically.

9702/21/M/J/13/Q1

(a) Explain what is meant by *extended elastically*.

.....
..... [2]

(b) Show that the SI units of energy per unit volume are $\text{kg m}^{-1} \text{s}^{-2}$.

[2]

(c) For a wire extended elastically, the elastic energy per unit volume X is given by

$$X = C\varepsilon^2 E$$

where C is a constant,

ε is the strain of the wire,

and E is the Young modulus of the wire.

Show that C has no units.

[3]

16 (a) Define

9702/23/M/J/13/Q4

(i) stress,

..... [1]

(ii) strain.

..... [1]

(b) The Young modulus of the metal of a wire is 0.17 TPa. The cross-sectional area of the wire is 0.18 mm^2 .

The wire is extended by a force F . This causes the length of the wire to be increased by 0.095%.

Calculate

(i) the stress,

stress = Pa [4]

(ii) the force F .

$F =$ N [2]

17 (a) Define the *Young modulus*.

9702/22/M/J/14/Q5

.....
..... [1]

- (b) Two wires P and Q of the same material and same original length l_0 are fixed so that they hang vertically, as shown in Fig. 5.1.

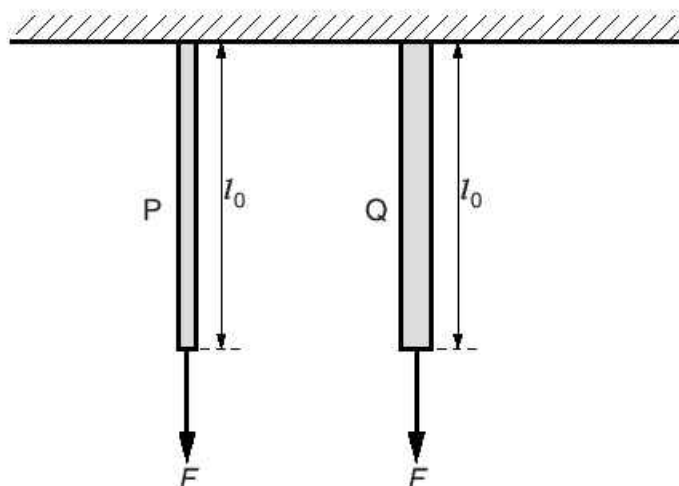


Fig. 5.1 (not to scale)

The diameter of P is d and the diameter of Q is $2d$. The same force F is applied to the lower end of each wire.

Show your working and determine the ratio

(i) $\frac{\text{stress in P}}{\text{stress in Q}}$,

ratio = [2]

(ii) $\frac{\text{strain in P}}{\text{strain in Q}}$.

ratio = [2]

18 A spring hangs vertically from a point P, as shown in Fig. 4.1.

9702/23/M/J/14/Q4

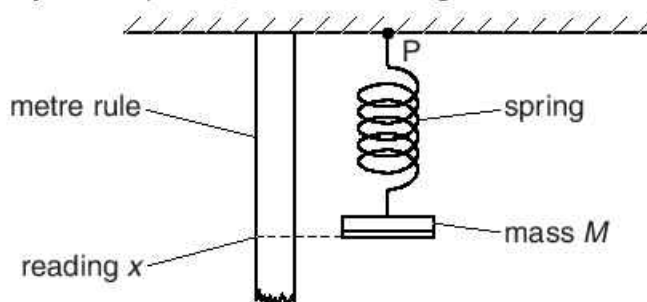


Fig. 4.1

A mass M is attached to the lower end of the spring. The reading x from the metre rule is taken, as shown in Fig. 4.1. Fig. 4.2 shows the relationship between x and M .

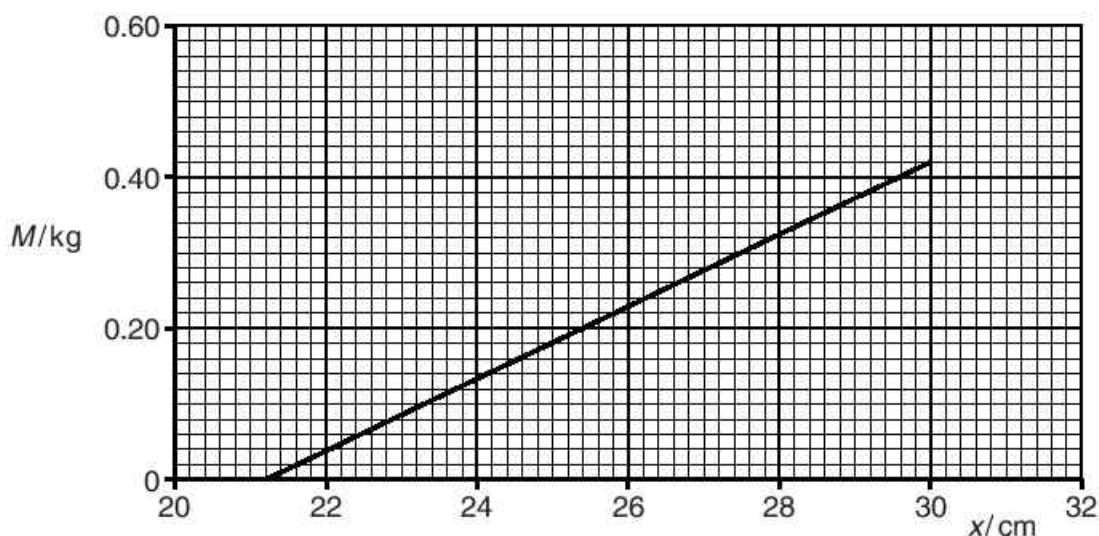


Fig. 4.2

- (a) Explain how the apparatus in Fig. 4.1 may be used to determine the load on the spring at the elastic limit.

.....

 [2]

- (b) State and explain whether Fig. 4.2 suggests that the spring obeys Hooke's law.

.....

 [2]

- (c) Use Fig. 4.2 to determine the spring constant, in Nm^{-1} , of the spring.

spring constant = Nm^{-1} [3]

19 (a) Compare the molecular motion of a liquid with

9702/22/O/N/14/Q4

(i) a solid,

.....
.....[2]

(ii) a gas.

.....
.....[1]

(b) (i) A ductile material in the form of a wire is stretched up to its breaking point. On Fig. 4.1, sketch the variation with extension x of the stretching force F .

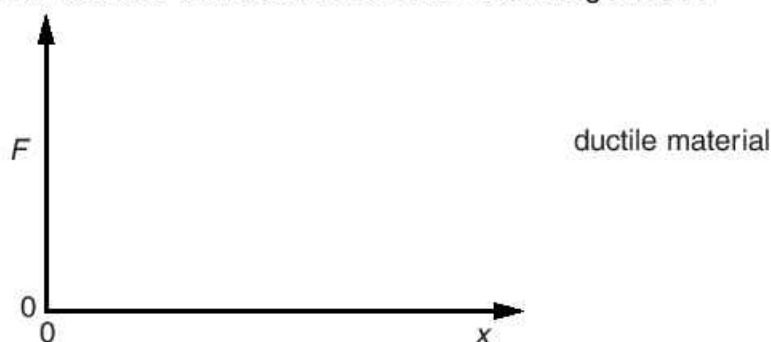


Fig. 4.1

[1]

(ii) On Fig. 4.2, sketch the variation with extension x of the stretching force F for a brittle material up to its breaking point.

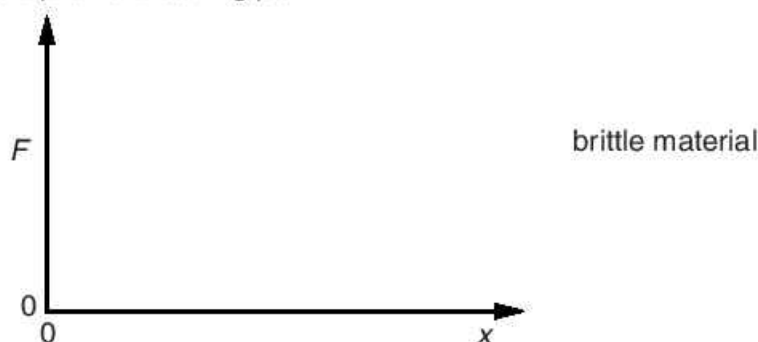


Fig. 4.2

[1]

(c) Describe a similarity and a difference between ductile and brittle materials.

similarity:
.....
difference:
.....

[2]

- 20 A spring is kept horizontal by attaching it to points A and B, as shown in Fig. 4.1.

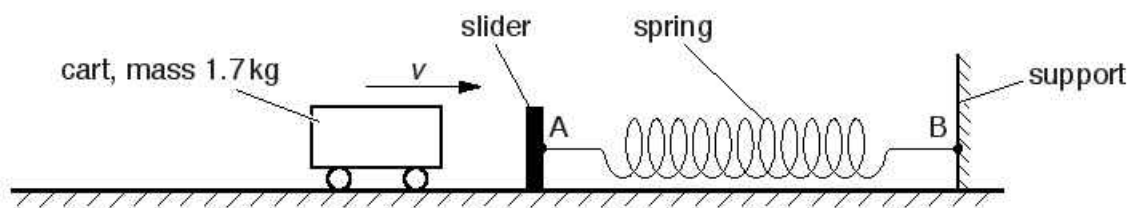


Fig. 4.1

Point A is on a movable slider and point B is on a fixed support. A cart of mass 1.7 kg has horizontal velocity v towards the slider. The cart collides with the slider. The spring is compressed as the cart comes to rest. The variation of compression x of the spring with force F exerted on the spring is shown in Fig. 4.2.

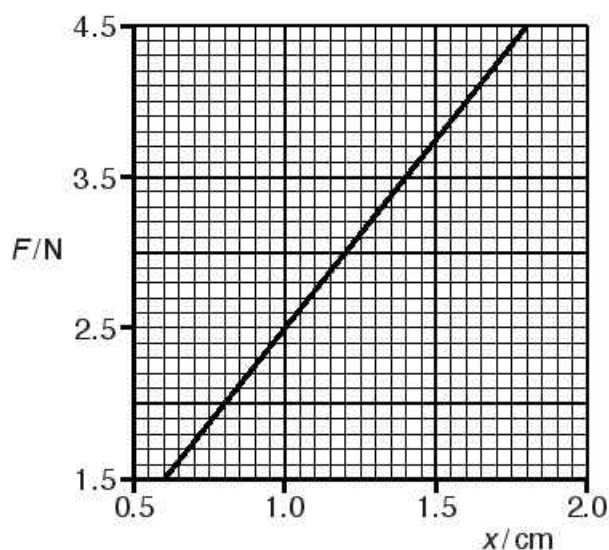


Fig. 4.2

Fig. 4.2 shows the compression of the spring for $F = 1.5\text{ N}$ to $F = 4.5\text{ N}$. The cart comes to rest when F is 4.5 N.

(a) Use Fig. 4.2 to

- (i) show that the compression of the spring obeys Hooke's law,

.....
 [2]

- (ii) determine the spring constant of the spring,

spring constant = Nm^{-1} [2]

- (iii) determine the elastic potential energy E_p stored in the spring due to the cart being brought to rest.

$$E_p = \dots\dots\dots \text{ J [3]}$$

- (b) Calculate the speed v of the cart as it makes contact with the slider. Assume that all the kinetic energy of the cart is converted to the elastic potential energy of the spring.

$$\text{speed} = \dots\dots\dots \text{ m s}^{-1} \text{ [2]}$$

21 (a) Define the *Young modulus*.

.....
[1]

(b) The Young modulus of steel is 1.9×10^{11} Pa. The Young modulus of copper is 1.2×10^{11} Pa.

A steel wire and a copper wire each have the same cross-sectional area and length. The two wires are each extended by equal forces.

(i) Use the definition of the Young modulus to determine the ratio

$$\frac{\text{extension of the copper wire}}{\text{extension of the steel wire}}.$$

ratio =[3]

(ii) The two wires are each extended by a force. Both wires obey Hooke's law.

On Fig. 3.1, sketch a graph for each wire to show the variation with extension of the force.

Label the line for steel with the letter **S** and the line for copper with the letter **C**.



Fig. 3.1

[1]

22 (a) State Hooke's law.

.....
 [1]

(b) The variation with compression x of the force F acting on a spring is shown in Fig. 3.1.

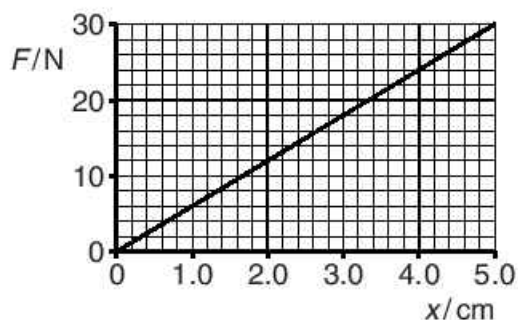


Fig. 3.1

The spring is fixed to the closed end of a horizontal tube. A block is pushed into the tube so that the spring is compressed, as shown in Fig. 3.2.

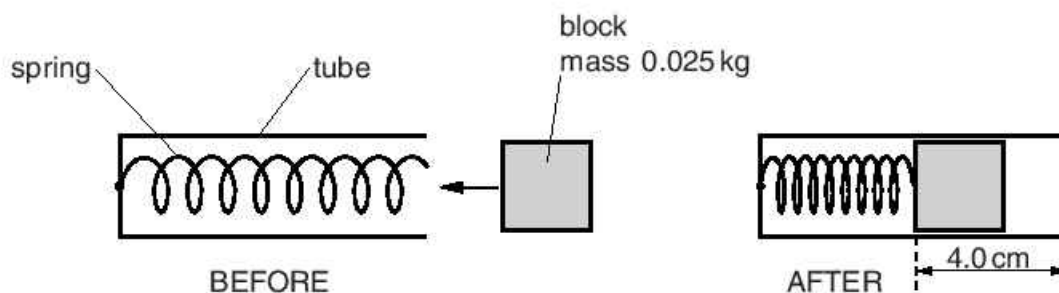


Fig. 3.2 (not to scale)

The compression of the spring is 4.0 cm. The mass of the block is 0.025 kg.

(i) Calculate the spring constant of the spring.

spring constant = Nm^{-1} [2]

(ii) Show that the work done to compress the spring by 4.0 cm is 0.48 J.

(iii) The block is now released and accelerates along the tube as the spring returns to its original length. The block leaves the end of the tube with a speed of 6.0 m s^{-1} . [2]

1. Calculate the kinetic energy of the block as it leaves the end of the tube.

kinetic energy = J [2]

2. Assume that the spring has negligible kinetic energy as the block leaves the tube. Determine the average resistive force acting against the block as it moves along the tube.

resistive force = N [3]

(iv) Determine the efficiency of the transfer of elastic potential energy from the spring to the kinetic energy of the block.

efficiency = [2]

- 23 The Young modulus of the material of a wire can be determined using the apparatus shown in Fig. 3.1.

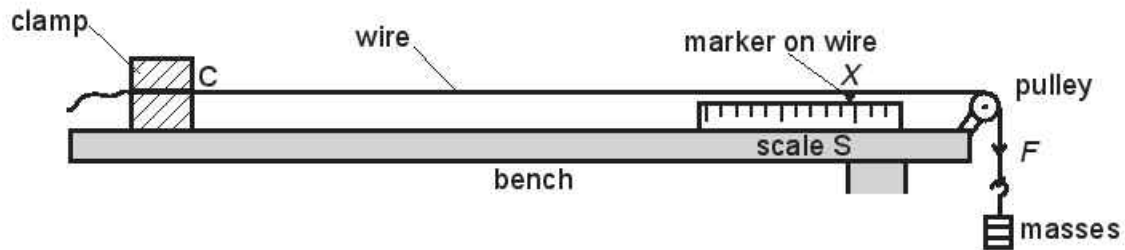


Fig. 3.1

One end of the wire is clamped at C and a marker is attached to the wire above a scale S. A force to extend the wire is applied by attaching masses to the other end of the wire.

The reading X of the marker on the scale S is determined for different forces F applied to the end of the wire. The variation with X of F is shown in Fig. 3.2.

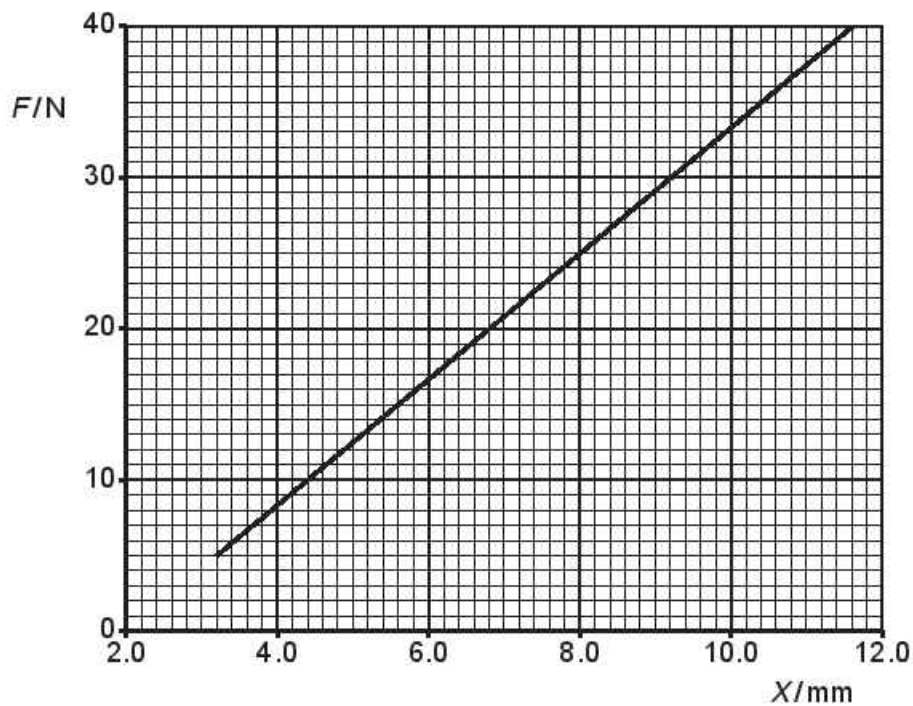


Fig. 3.2

- (a) The length of the wire from C to the marker for $F = 0$ is 3.50m. The diameter of the wire is 0.38mm.

Use the gradient of the line in Fig. 3.2 to determine the Young modulus E of the material of the wire in TPa.

$E = \dots\dots\dots$ TPa [3]

- (b) The experiment is repeated with a thicker wire of the same material and length.

State how the range of the force F must be changed to obtain the same range of scale readings as in Fig. 3.2.

.....
.....
..... [1]

24 (a) Define *strain*.

.....
..... [1]

- (b) A wire is designed to ensure that its strain does not exceed 4.0×10^{-4} when a force of 8.0 kN is applied. The Young modulus of the metal of the wire is 2.1×10^{11} Pa. It may be assumed that the wire obeys Hooke's law.

For a force of 8.0 kN, calculate, for the wire,

- (i) the maximum stress,

maximum stress = Pa [2]

- (ii) the minimum cross-sectional area.

minimum cross-sectional area = m^2 [2]

- 25 A spring is attached at one end to a fixed point and hangs vertically with a cube attached to the other end. The cube is initially held so that the spring has zero extension, as shown in Fig. 3.1.

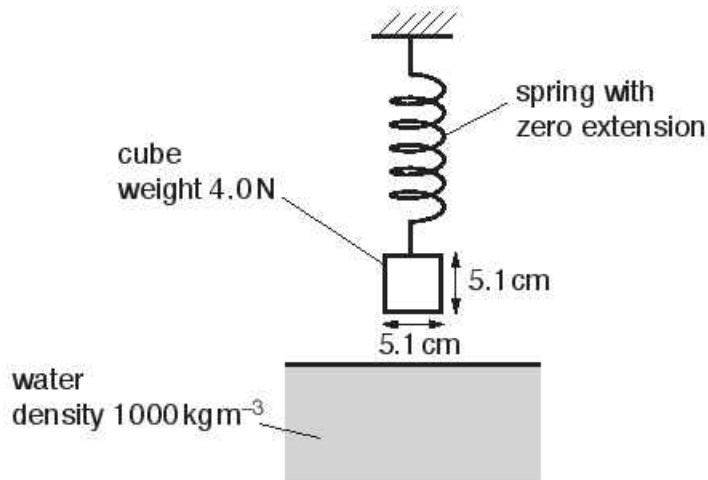


Fig. 3.1

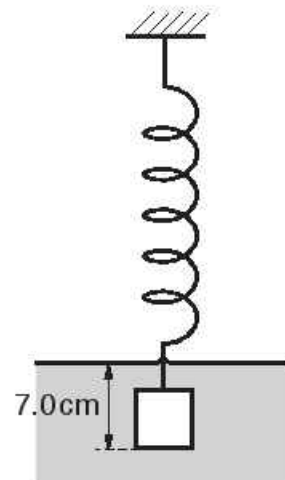


Fig. 3.2

The cube has weight 4.0 N and sides of length 5.1 cm. The cube is released and sinks into water as the spring extends. The cube reaches equilibrium with its base at a depth of 7.0 cm below the water surface, as shown in Fig. 3.2.

The density of the water is 1000 kg m⁻³.

- (a) Calculate the difference in the pressure exerted by the water on the bottom face and on the top face of the cube.

difference in pressure = Pa [2]

- (b) Use your answer in (a) to show that the upthrust on the cube is 1.3 N.

- (c) Calculate the force exerted on the spring by the cube when it is in equilibrium in the water.

force = N [1]

- (d) The spring obeys Hooke's law and has a spring constant of 30 N m^{-1} .

Determine the initial height above the water surface of the base of the cube before it was released.

height above surface = cm [3]

- (e) The cube in the water is released from the spring.

- (i) Determine the initial acceleration of the cube.

acceleration = ms^{-2} [2]

- (ii) Describe and explain the variation, if any, of the acceleration of the cube as it sinks in the water.

.....
.....
..... [2]

- 26** A solid cylinder is lifted out of oil by a wire attached to a motor. Fig. 5.1 shows two different positions X and Y of the cylinder during the lifting process.

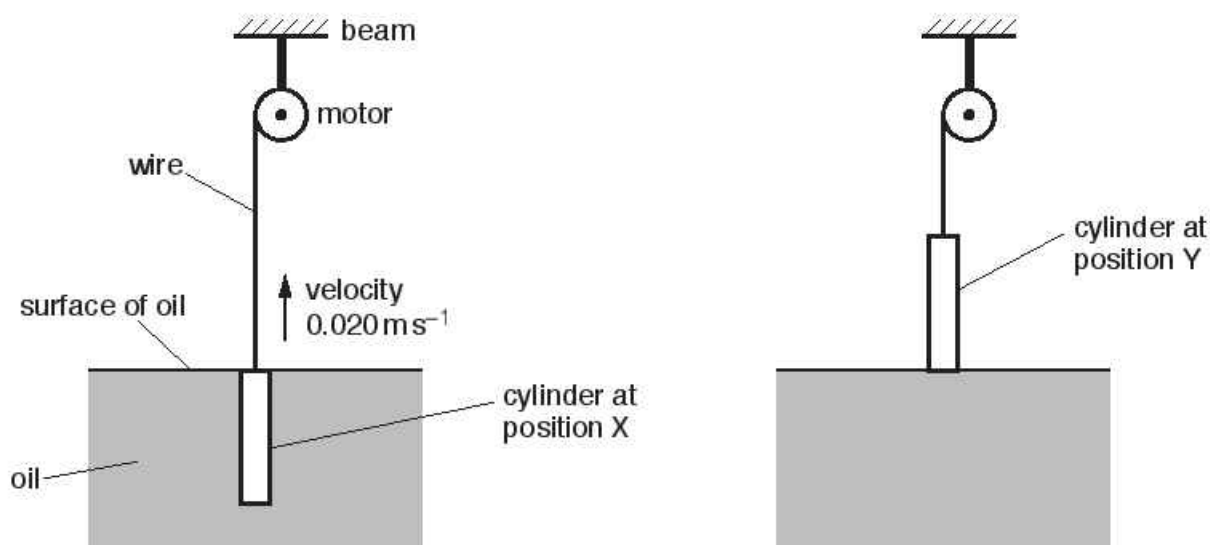


Fig. 5.1

The motor is fixed to an overhead beam.

The cylinder has cross-sectional area 0.018 m^2 , length 1.2 m and weight 560 N .

The density of the oil is 940 kg m^{-3} .

Throughout the lifting process, the cylinder moves vertically upwards with a constant velocity of 0.020 m s^{-1} . The viscous force of the oil acting on the cylinder is negligible.

- (a)** Calculate the density of the cylinder.

density = kg m^{-3} [2]

- (b)** For the cylinder at position X, show that the upthrust due to the oil is 200 N .

(c) Calculate, for the moving cylinder at position X,

(i) the tension in the wire,

tension = N [1]

(ii) the power output of the motor.

power = W [2]

(d) The cylinder is raised with constant velocity from position X to position Y.

(i) State and explain the variation, if any, of the power output of the motor as the cylinder is raised. Numerical values are not required.

.....
.....
.....
.....
.....[3]

(ii) The rate of energy output of the motor is less than the rate of increase of gravitational potential energy of the cylinder. Without calculation, explain this difference.

.....
.....[1]

27 (a) Define the *Young modulus* of a material.

.....
 [1]

(b) A metal rod is compressed, as shown in Fig. 4.1.

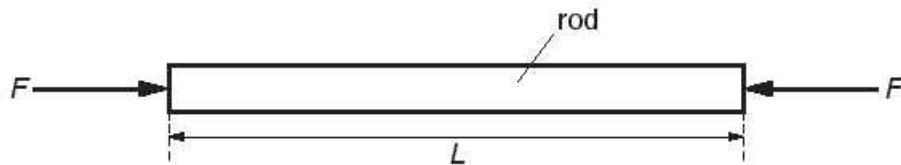


Fig. 4.1

The variation with compressive force F of the length L of the rod is shown in Fig. 4.2.

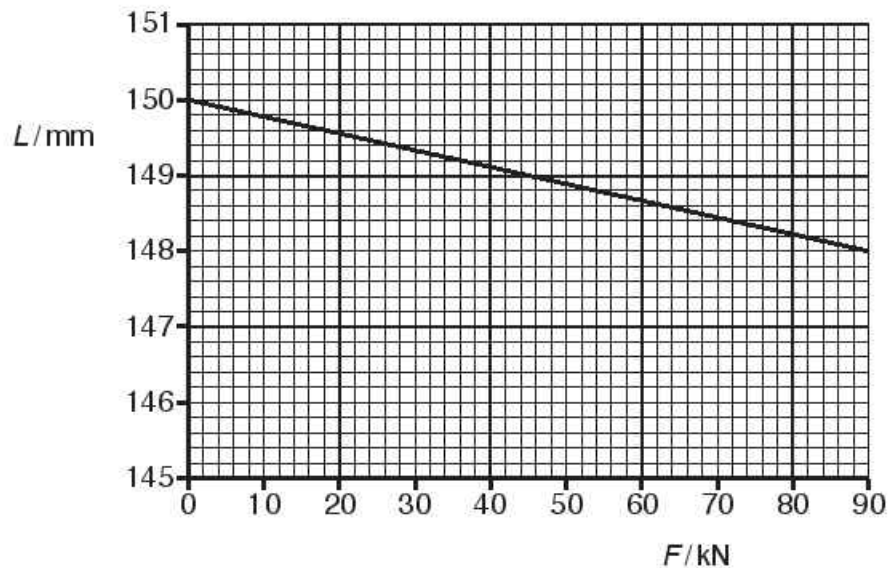


Fig. 4.2

Use Fig. 4.2 to

(i) determine the spring constant k of the rod,

$k = \dots\dots\dots \text{N m}^{-1}$ [2]

- (ii) determine the strain energy stored in the rod for $F = 90\text{ kN}$.

strain energy = J [3]

- (c) The rod in (b) has cross-sectional area A and is made of metal of Young modulus E . It is now replaced by a new rod of the same original length. The new rod has cross-sectional area $A/3$ and is made of metal of Young modulus $2E$. The compression of the new rod obeys Hooke's law.

On Fig. 4.2, sketch the variation with F of the length L for the new rod from $F = 0$ to $F = 90\text{ kN}$. [2]

9702/23/O/N/18/Q1

- 28 (a) Mass, length and time are all SI base quantities.

State two other SI base quantities.

1.

2.

[2]

- (b) A wire hangs between two fixed points, as shown in Fig. 1.1.

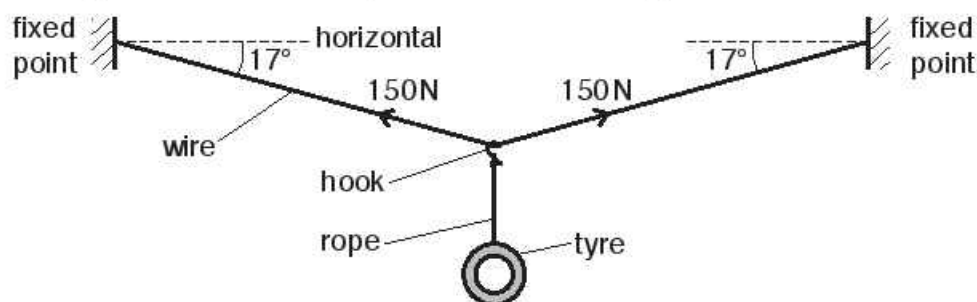


Fig. 1.1 (not to scale)

A child's swing is made by connecting a car tyre to the wire using a rope and a hook. The system is in equilibrium with the wire hanging at an angle of 17° to the horizontal. The tension in the wire is 150 N . Assume that the rope and hook have negligible weight.

- (i) Determine the weight of the tyre.

weight = N [2]

- (ii) The wire has a cross-sectional area of 7.5 mm^2 and is made of metal of Young modulus $2.1 \times 10^{11} \text{ Pa}$. The wire obeys Hooke's law.

Calculate, for the wire,

1. the stress,

stress = Pa [2]

2. the strain.

strain = [2]

9702/23/O/N/19/Q4

- 29 A ball X moves along a horizontal frictionless surface and collides with another ball Y, as illustrated in Fig. 4.1.

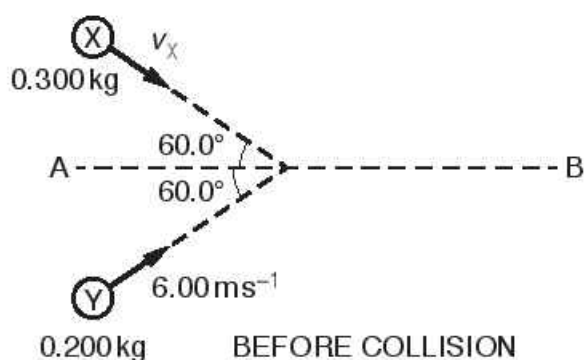


Fig. 4.1 (not to scale)

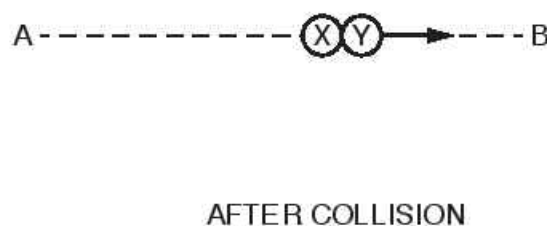


Fig. 4.2 (not to scale)

Ball X has mass 0.300 kg and initial velocity v_X at an angle of 60.0° to line AB.

Ball Y has mass 0.200 kg and initial velocity 6.00 ms^{-1} at an angle of 60.0° to line AB.

The balls stick together during the collision and then travel along line AB, as illustrated in Fig. 4.2.

- (a) (i) Calculate, to three significant figures, the component of the initial momentum of ball Y that is perpendicular to line AB.

component of momentum = kgms^{-1} [2]

- (ii) By considering the component of the initial momentum of each ball perpendicular to line AB, calculate, to three significant figures, v_X .

$v_X = \dots\dots\dots \text{ms}^{-1}$ [1]

- (iii) Show that the speed of the two balls after the collision is 2.4 ms^{-1} .

- (b) The two balls continue moving together along the horizontal frictionless surface towards a spring, as illustrated in Fig. 4.3.

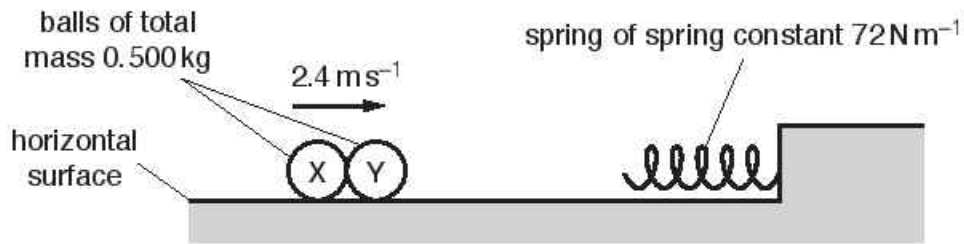


Fig. 4.3

The balls hit the spring and remain stuck together as they decelerate to rest. All the kinetic energy of the balls is converted into elastic potential energy of the spring. The energy E stored in the spring is given by

$$E = \frac{1}{2}kx^2$$

where k is the spring constant of the spring and x is its compression.
The spring obeys Hooke's law and has a spring constant of 72 N m^{-1} .

- (i) Determine the maximum compression of the spring caused by the two balls.

maximum compression = m [3]

(ii) On Fig. 4.4, sketch graphs to show the variation with compression x of the spring, from zero to maximum compression, of:

1. the magnitude of the deceleration a of the balls
2. the kinetic energy E_k of the balls.

Numerical values are not required.

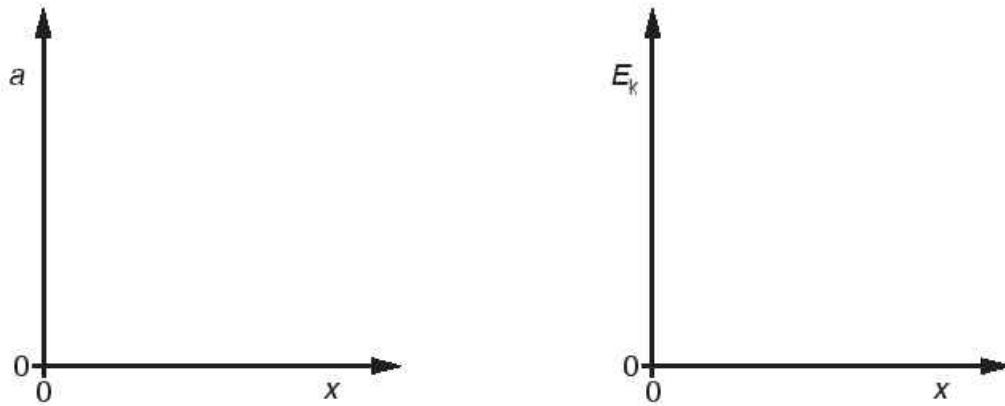


Fig. 4.4

[3]

- 30 One end of a wire is attached to a fixed point. A force F is applied to the wire to cause extension x . The variation with F of x is shown in Fig. 5.1.

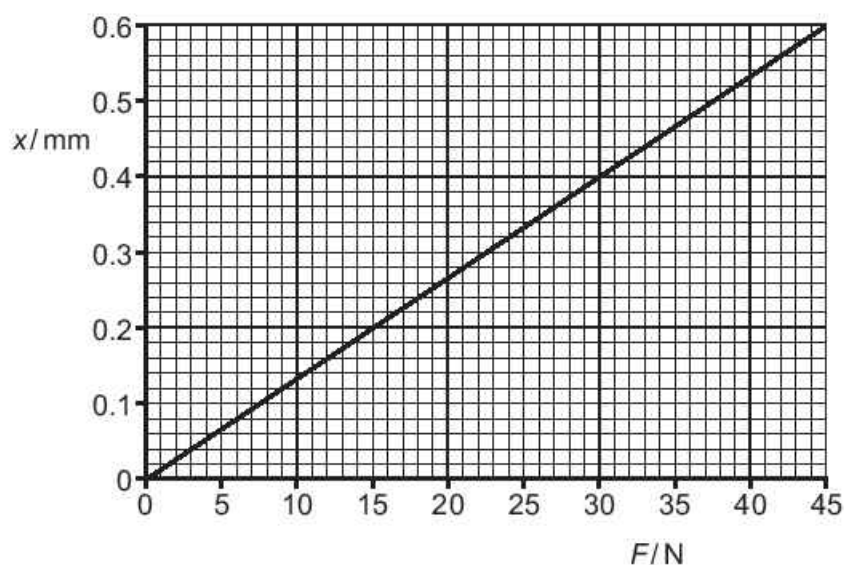


Fig. 5.1

The wire has a cross-sectional area of $4.1 \times 10^{-7} \text{ m}^2$ and is made of metal of Young modulus $1.7 \times 10^{11} \text{ Pa}$. Assume that the cross-sectional area of the wire remains constant as the wire extends.

- (a) State the name of the law that describes the relationship between F and x shown in Fig. 5.1.

..... [1]

- (b) The wire has an extension of 0.48 mm.

Determine:

- (i) the stress

stress = Pa [2]

- (ii) the strain.

strain = [2]

- (c) The resistivity of the metal of the wire is $3.7 \times 10^{-7} \Omega \text{ m}$.

Determine the change in resistance of the wire when the extension x of the wire changes from $x = 0.48 \text{ mm}$ to $x = 0.60 \text{ mm}$.

change in resistance = Ω [3]

- (d) A force of greater than 45 N is now applied to the wire.

Describe how it may be checked that the elastic limit of the wire has not been exceeded.

.....
..... [1]

- 31 (a) State the principle of moments.

.....

.....

..... [2]

- (b) In a bicycle shop, two wheels hang from a horizontal uniform rod AC, as shown in Fig. 3.1.

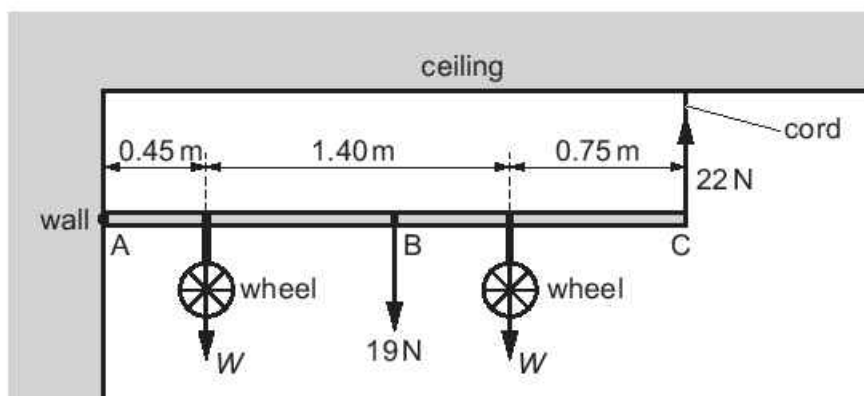


Fig. 3.1 (not to scale)

The rod has weight 19 N and is freely hinged to a wall at end A. The other end C of the rod is attached by a vertical elastic cord to the ceiling. The centre of gravity of the rod is at point B. The weight of each wheel is W and the tension in the cord is 22 N.

- (i) By taking moments about end A, show that the weight W of each wheel is 14 N.

[2]

- (ii) Determine the magnitude and the direction of the force acting on the rod at end A.

magnitude = N

direction

[2]

- (c) The unstretched length of the cord in (b) is 0.25 m. The variation with length L of the tension F in the cord is shown in Fig. 3.2.

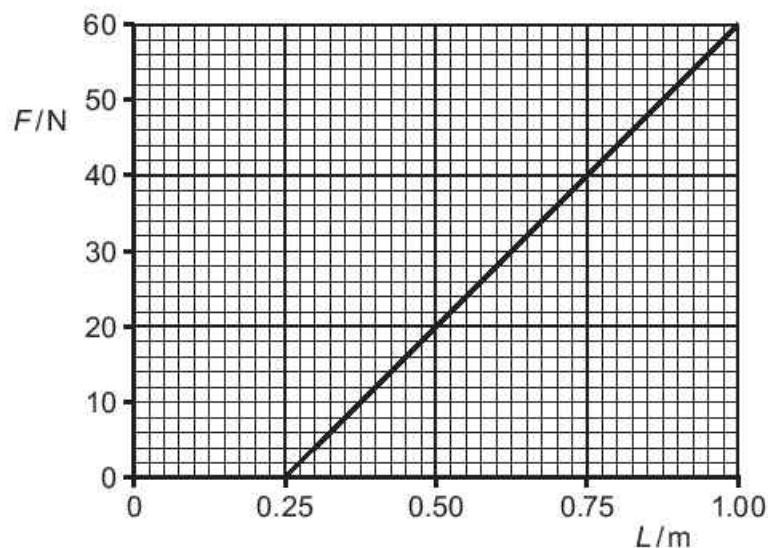


Fig. 3.2

- (i) State and explain whether Fig. 3.2 suggests that the cord obeys Hooke's law.

.....

 [2]

- (ii) Calculate the spring constant k of the cord.

$k = \dots\dots\dots \text{Nm}^{-1}$ [2]

- (iii) On Fig. 3.2, shade the area that represents the work done to extend the cord when the tension is increased from $F = 0$ to $F = 40$ N. [1]

32. 9702/21/O/N/20/Q4

(a) Define, for a wire:

(i) *stress*

.....
 [1]

(ii) *strain*.

.....
 [1]

(b) (i) A school experiment is performed on a metal wire to determine the Young modulus of the metal. A force is applied to one end of the wire which is fixed at the other end. The variation of the force F with extension x of the wire is shown in Fig. 4.1.

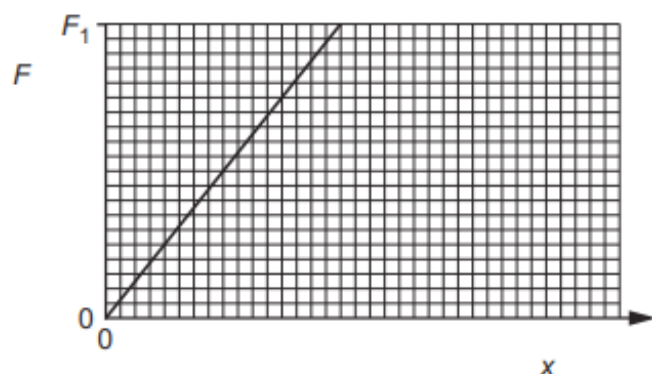


Fig. 4.1

The maximum force applied to the wire is F_1 .

The gradient of the graph line in Fig. 4.1 is G . The wire has initial length L and cross-sectional area A .

Determine an expression, in terms of A , G and L , for the Young modulus E of the metal.

$E =$ [2]

- (ii) A student repeats the experiment in **(b)(i)** using a new wire that has twice the diameter of the first wire. The initial length of the wire and the metal of the wire are unchanged.

On Fig. 4.1, draw the graph line representing the new wire for the force increasing from $F = 0$ to $F = F_1$. [2]

- (iii) Another student repeats the original experiment in **(b)(i)**, increasing the force beyond F_1 to a new maximum force F_2 . The new graph obtained is shown in Fig. 4.2.

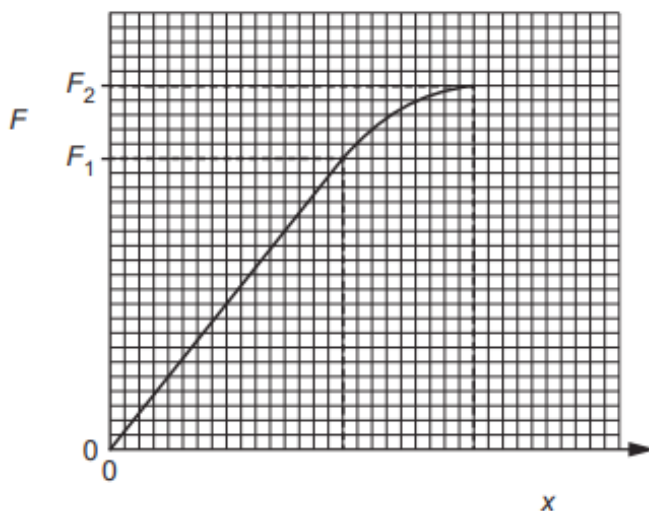


Fig. 4.2

- On Fig. 4.2, shade an area that represents the work done to extend the wire when the force is increased from F_1 to F_2 . [1]
- Explain how the student can check that the elastic limit of the wire was not exceeded when force F_2 was applied.

.....

 [1]

- (iv) Each student in the class performs the experiment in **(b)(i)**. The teacher describes the values of the Young modulus calculated by the students as having high accuracy and low precision.

Explain what is meant by *low precision*.

.....
 [1]

[Total: 9]

33. 9702/22/O/N/20/Q3

- (a) A spring is fixed at one end and is compressed by applying a force to the other end. The variation of the force F acting on the spring with its compression x is shown in Fig. 3.1.

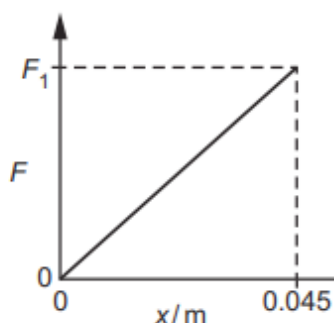


Fig. 3.1

A compression of 0.045 m is produced when a force F_1 acts on the spring. The spring has a spring constant of 800 N m^{-1} .

- (i) Determine F_1 .

$$F_1 = \dots\dots\dots \text{ N [2]}$$

- (ii) Use Fig. 3.1 to show that, for a compression of 0.045 m, the elastic potential energy of the spring is 0.81 J.

[2]

- (b) A child's toy uses the spring in (a) to launch a ball of mass 0.020 kg vertically into the air. The ball is initially held against one end of the spring which has a compression of 0.045 m. The spring is then released to launch the ball. The kinetic energy of the ball as it leaves the toy is 0.72 J.

- (i) The toy converts the elastic potential energy of the spring into the kinetic energy of the ball. Use the information in (a)(ii) to calculate the percentage efficiency of this conversion.

$$\text{efficiency} = \dots\dots\dots \% [1]$$

- (ii) Determine the initial momentum of the ball as it leaves the toy.

momentum = N s [3]

- (c) The ball in (b) leaves the toy at point A and moves vertically upwards through the air. Point B is the position of the ball when it is at maximum height h above point A, as illustrated in Fig. 3.2.

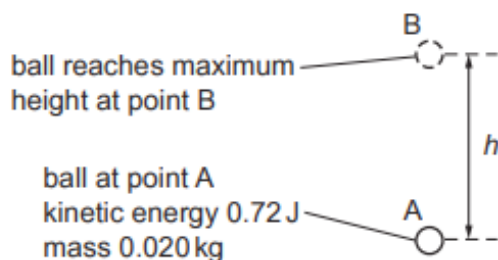


Fig. 3.2 (not to scale)

The gravitational potential energy of the ball increases by 0.60 J as it moves from A to B.

- (i) Calculate h .

h = m [2]

- (ii) Determine the average force due to air resistance acting on the ball for its movement from A to B.

average force = N [2]

- (iii) When there is air resistance, the ball takes time T to move from A to B.

State and explain whether the time taken for the ball to move from A to its maximum height will be more than, less than or equal to time T if there is **no** air resistance.

.....

..... [1]

[Total: 13]

34. 9702/23/O/N/20/Q4

(a) State Hooke's law.

[1]

(b) A spring is fixed at one end. A compressive force F is applied to the other end. The variation of the force F with the compression x of the spring is shown in Fig. 4.1.

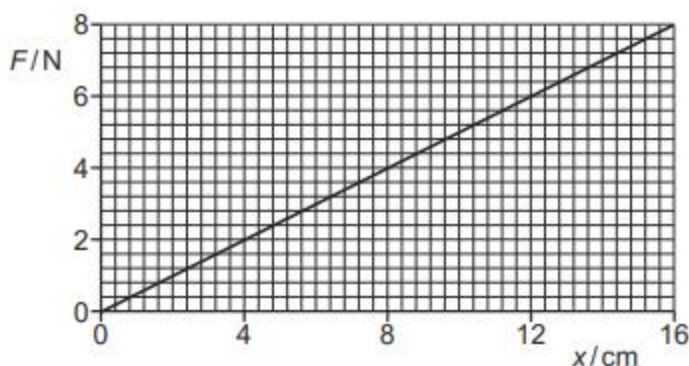


Fig. 4.1

Show that the elastic potential energy of the spring is 0.64 J when its compression is 16.0 cm.

[2]

(c) The spring in (b) is used to project a toy car along a track from point X to point Y, as illustrated in Fig. 4.2.

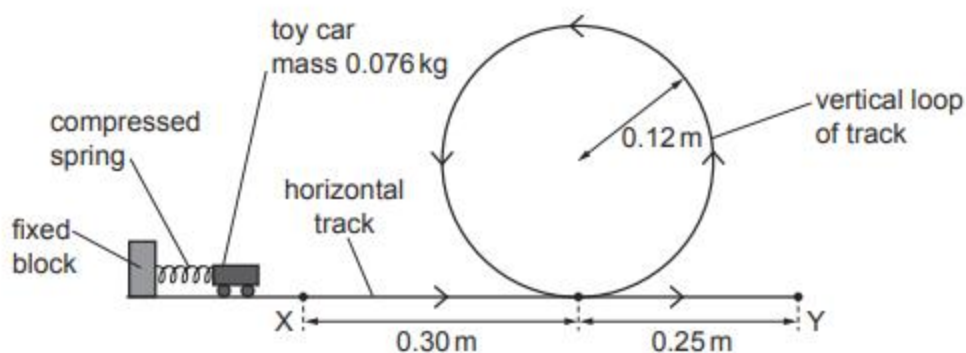


Fig. 4.2 (not to scale)

The spring is initially given a compression of 16.0 cm. The car of mass 0.076 kg is held against one end of the compressed spring. When the spring is released it projects the car forward. The car leaves the spring at point X with kinetic energy that is equal to the initial elastic potential energy of the compressed spring.

The car follows the track around a vertical loop of radius 0.12 m and then passes point Y. Assume that friction and air resistance are negligible.

Calculate:

- (i) the speed of the car at X

speed = ms^{-1} [2]

- (ii) the kinetic energy of the car when it is at the top of the loop

kinetic energy = J [3]

- (iii) the speed of the car at Y.

speed = ms^{-1} [1]

- (d) In practice, a resistive force due to friction and air resistance acts on the car so that its kinetic energy at Y is 0.23 J less than its kinetic energy at X.

Determine the average resistive force acting on the car for its movement from X to Y.

average resistive force = N [3]

[Total: 12]

35. 9702/22/F/M/21/Q3

A spring is extended by a force. The variation with extension x of the force F is shown in Fig. 3.1.

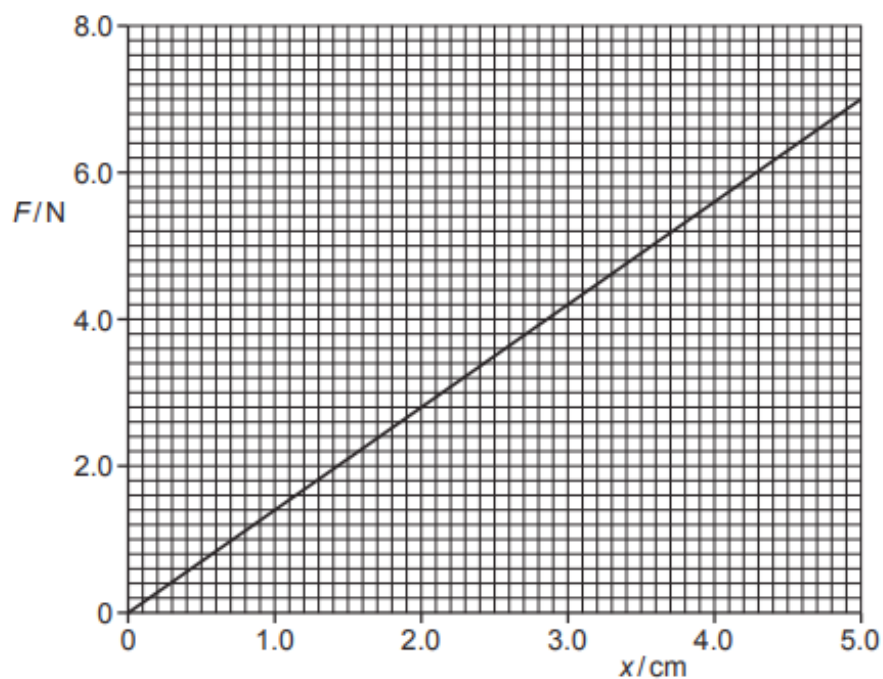


Fig. 3.1

(a) State the name of the law that relates the force and extension of the spring shown in Fig. 3.1.

..... [1]

(b) Determine:

(i) the spring constant, in Nm^{-1} , of the spring

spring constant = Nm^{-1} [2]

(ii) the strain energy (elastic potential energy) in the spring when the extension is 4.0 cm.

strain energy = J [2]

- (c) One end of the spring is attached to a fixed point. A cylinder that is submerged in a liquid is now suspended from the other end of the spring, as shown in Fig. 3.2.

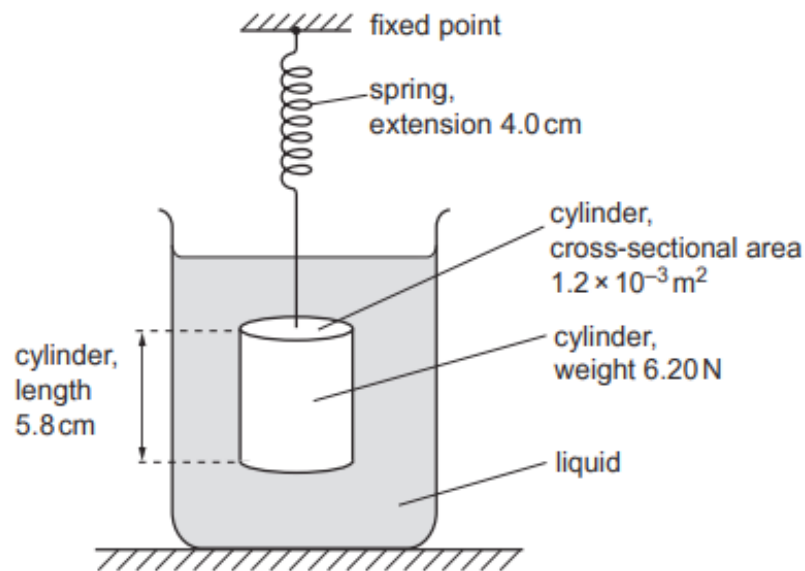


Fig. 3.2

The cylinder has length 5.8 cm, cross-sectional area $1.2 \times 10^{-3} \text{ m}^2$ and weight 6.20 N. The cylinder is in equilibrium when the extension of the spring is 4.0 cm.

- (i) Show that the upthrust acting on the cylinder is 0.60 N.

[1]

- (ii) Calculate the difference in pressure between the bottom face and the top face of the cylinder.

difference in pressure = Pa [2]

(iii) Calculate the density of the liquid.

density = kg m^{-3} [2]

(d) The liquid in (c) is replaced by another liquid of greater density.

State the effect, if any, of this change on:

(i) the upthrust acting on the cylinder

..... [1]

(ii) the extension of the spring.

..... [1]

[Total: 12]

36. 9702/23/M/J/21/Q3

(a) Define the *moment* of a force about a point.

.....

.....

..... [2]

(b) Fig. 3.1 shows a type of balance that is used for measuring mass.

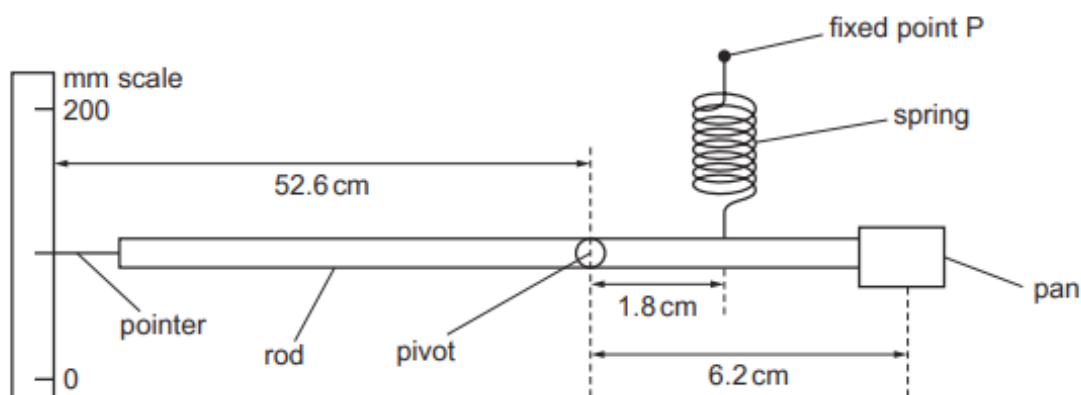


Fig. 3.1 (not to scale)

A rigid rod is pivoted about a point 6.2 cm from the centre of a pan which is attached to one end. The object being measured is placed on the centre of this pan.

A spring, attached to the rod 1.8 cm from the pivot, is attached at its other end to a fixed point P. The spring obeys Hooke's law over the full range of operation of the balance.

A pointer, on the other side of the pivot, is set against a millimetre scale which is a distance 52.6 cm from the pivot.

When the system is in equilibrium with no mass on the pan, the rod is horizontal and the pointer indicates a reading on the scale of 86 mm.

An object of mass 0.472 kg is now placed on the pan. As a result, the pointer moves to indicate a reading of 123 mm on the scale when the system is again in equilibrium.

(i) Show that the increase in the length of the spring is approximately 1.3 mm.

- (ii) Calculate the magnitude of the moment about the pivot of the weight of the object.

moment = N m [2]

- (iii) Use your answer in (b)(ii) to determine the increase in the tension in the spring due to the 0.472 kg mass.

increase in tension = N [2]

- (iv) Use the information in (b)(i) and your answer in (b)(iii) to determine the spring constant k of the spring. Give a unit with your answer.

k = unit [2]

[Total: 10]

37. 9702/21/O/N/21/Q3

- (a) A uniform metal bar, initially unstretched, has sides of length w , x and y , as shown in Fig. 3.1.

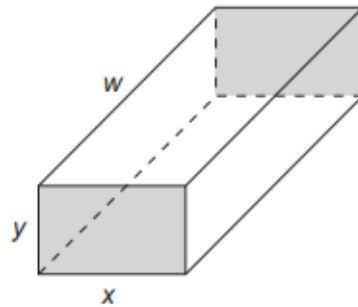


Fig. 3.1

The bar is now stretched by a tensile force F applied to the shaded ends. The changes in the lengths x and y are negligible. The bar now has sides of length x , y and z , as shown in Fig. 3.2.

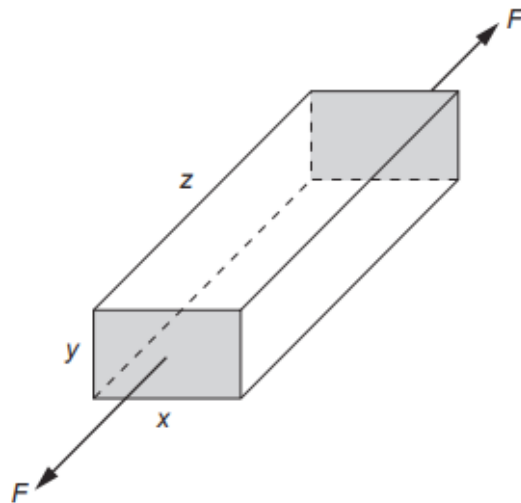


Fig. 3.2

Determine expressions, in terms of some or all of F , w , x , y and z , for:

- (i) the stress σ applied to the bar by the tensile force

$$\sigma = \dots\dots\dots [1]$$

- (ii) the strain ε in the bar due to the tensile force

$$\varepsilon = \dots\dots\dots [1]$$

- (iii) the Young modulus E of the metal from which the bar is made.

$$E = \dots\dots\dots [2]$$

- (b) A copper wire is stretched by a tensile force that gradually increases from 0 to 280 N. The variation with extension of the tensile force is shown in Fig. 3.3.

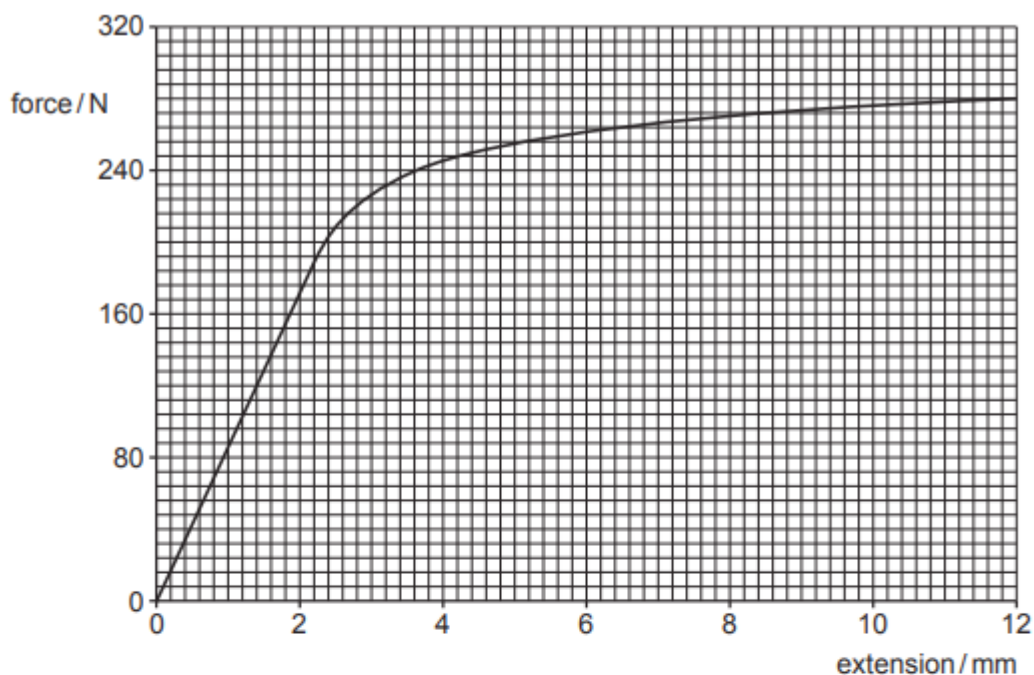


Fig. 3.3

- (i) State the maximum extension of the wire for which it obeys Hooke's law.

extension = mm [1]

- (ii) Use Fig. 3.3 to determine the strain energy in the wire when the tensile force is 120 N.

strain energy = J [3]

- (iii) Explain why the work done in stretching the wire to an extension of 12 mm is not equal to the energy recovered when the tensile force is removed.

.....

 [2]

[Total: 10]

38. 9702/21/M/J/22/Q4

A spring is suspended from a fixed point at one end and a vertical force is applied to the other end, as shown in Fig. 4.1.

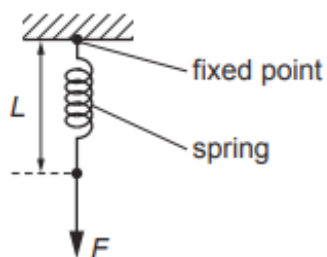


Fig. 4.1

The variation of the applied force F with the length L of the spring is shown in Fig. 4.2.

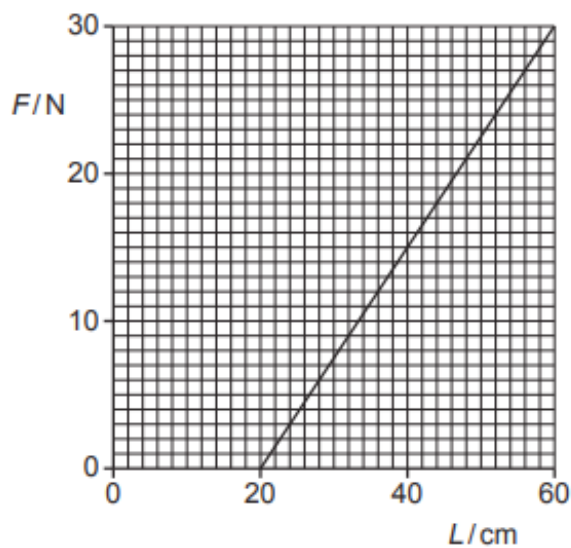


Fig. 4.2

(a) Determine the spring constant k of the spring.

$k = \dots\dots\dots \text{Nm}^{-1}$ [2]

(b) Determine the elastic potential energy in the spring when the applied force F is 15 N.

elastic potential energy = J [3]

[Total: 5]

39. 9702/21/O/N/22/Q2

A steel ball is projected horizontally from the top of a table, as shown in Fig. 2.1.

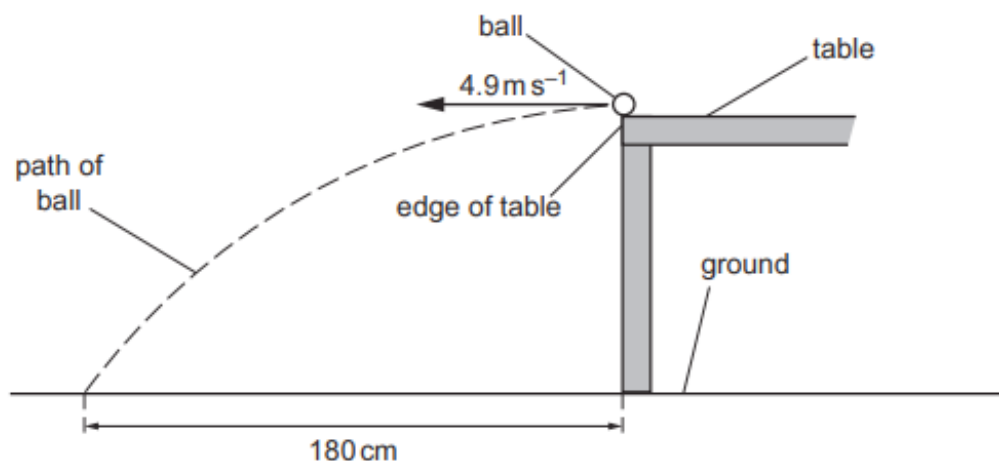


Fig. 2.1 (not to scale)

The ball is projected horizontally at a speed of 4.9 m s^{-1} . The ball lands on the ground a horizontal distance of 180 cm from the edge of the table.

Assume that air resistance is negligible.

- (a) (i)** Calculate the time taken for the ball to reach the ground.

time = s [1]

- (ii)** Calculate the vertical component of the velocity of the ball as it hits the ground.

velocity = m s^{-1} [2]

- (iii) Determine the magnitude and the angle to the horizontal of the velocity of the ball as it hits the ground.

magnitude of velocity = ms^{-1}

angle to the horizontal = $^{\circ}$
[3]

- (b) The ball is projected by means of a compressed spring which is attached to a fixed block as shown in Fig. 2.2.

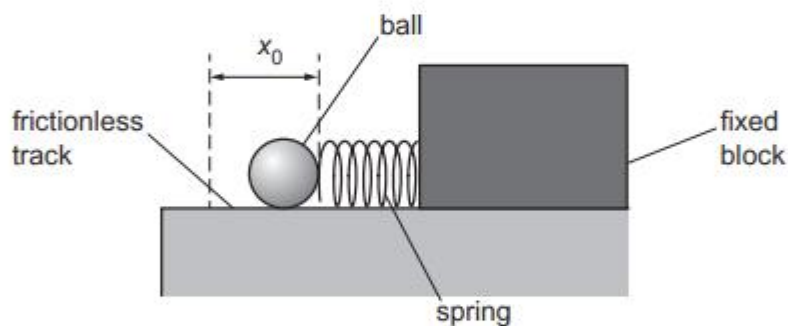


Fig. 2.2

The ball is placed on a frictionless track in front of the spring. The ball is then pulled back so that the spring has compression x_0 .

When the spring is released, the ball is projected horizontally as shown in Fig. 2.3.

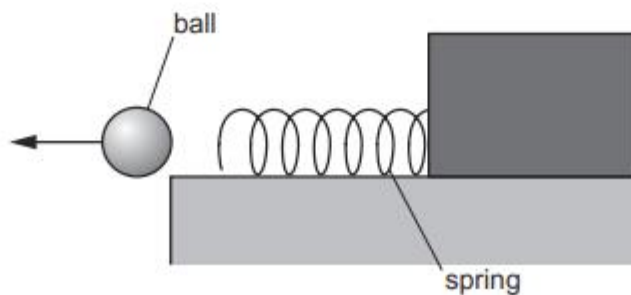


Fig. 2.3

The variation with compression x of the applied force F for the spring is shown in Fig. 2.4.

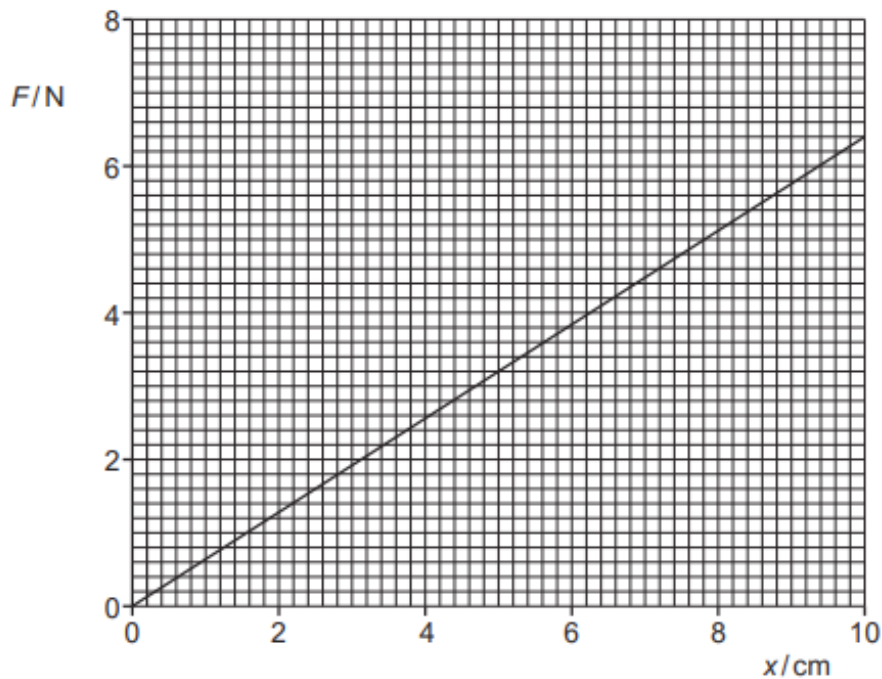


Fig. 2.4

The ball is a uniform sphere of steel of diameter 0.016 m and mass 0.017 kg.

- (i) Calculate the density of the steel.

density = kg m^{-3} [3]

- (ii) All of the elastic potential energy in the spring is converted into kinetic energy of the ball. The speed of the ball as it leaves the spring is 4.9 m s^{-1} .

Show that the maximum elastic potential energy of the spring is 0.20 J.

(iii) Use Fig. 2.4 to determine the spring constant k of the spring.

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

(iv) Use your answer in (b)(iii) and the value of energy given in (b)(ii) to determine the compression x_0 of the spring.

$$x_0 = \dots\dots\dots \text{m} \quad [2]$$

(c) The steel ball is replaced by a polystyrene ball of the same diameter but of much lower mass. The spring is given compression x_0 and is then released.

Air resistance on this ball is **not** negligible after it leaves the spring.

Explain:

(i) why this ball leaves the spring with a greater speed than that of the steel ball

.....
.....
..... [1]

(ii) why this ball takes a longer time to reach the ground than the steel ball.

.....
.....
..... [1]

[Total: 17]

40. 9702/22/O/N/22/Q4

A horizontal spring is fixed at one end. A block is pushed against the other end of the spring so that the spring is compressed, as shown in Fig. 4.1.

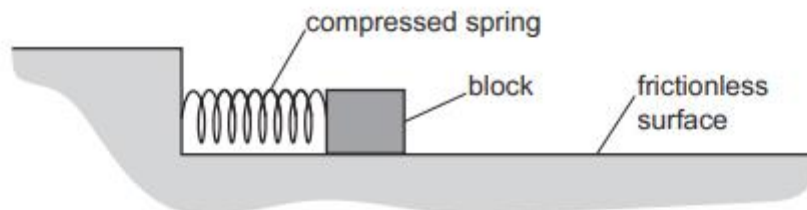


Fig. 4.1

The block is released and accelerates along a horizontal frictionless surface as the spring returns to its original length. The block leaves the end of the spring with a speed of 2.3 m s^{-1} , as shown in Fig. 4.2.

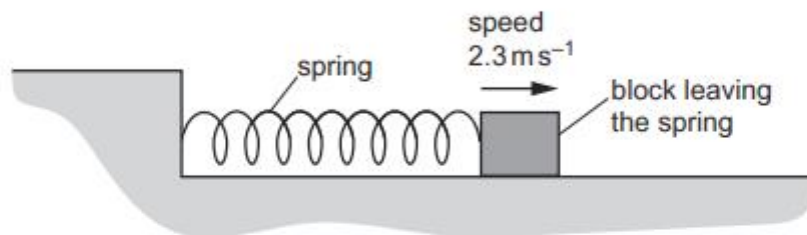


Fig. 4.2

The block has a mass of 250 g and the spring has a spring constant of 420 N m^{-1} .

Assume that the spring always obeys Hooke's law and that all the elastic potential energy of the spring is transferred to the kinetic energy of the block.

(a) Calculate the kinetic energy of the block as it leaves the spring.

kinetic energy = J [2]

(b) Calculate the compression of the spring immediately before the block is released.

compression = m [2]

- (c) After leaving the spring, the block moves along the surface until it hits a barrier at a speed of 2.3 m s^{-1} . The block then rebounds at a speed of 1.5 m s^{-1} and moves back along its original path. The block is in contact with the barrier for a time of 0.086 s .

Calculate:

- (i) the change in momentum of the block during the collision

change in momentum = N s [2]

- (ii) the average resultant force exerted on the block during the collision.

average resultant force = N [1]

- (d) The maximum compression x of the spring is now varied in order to vary the kinetic energy E_K of the block as it leaves the spring. Assume that all the elastic potential energy in the spring is always transferred to the kinetic energy of the block.

On Fig. 4.3, sketch a graph to show the variation with x of E_K .

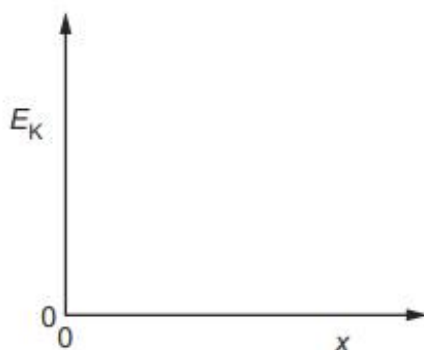


Fig. 4.3

[1]

[Total: 8]

41. 9702/23/O/N/22/Q3

(a) State the principle of moments.

.....

.....

..... [2]

(b) A hollow plastic sphere is attached at one end of a bar. The sphere is partially submerged in water and the bar is attached to a fixed vertical support by a pivot P, as shown in Fig. 3.1.

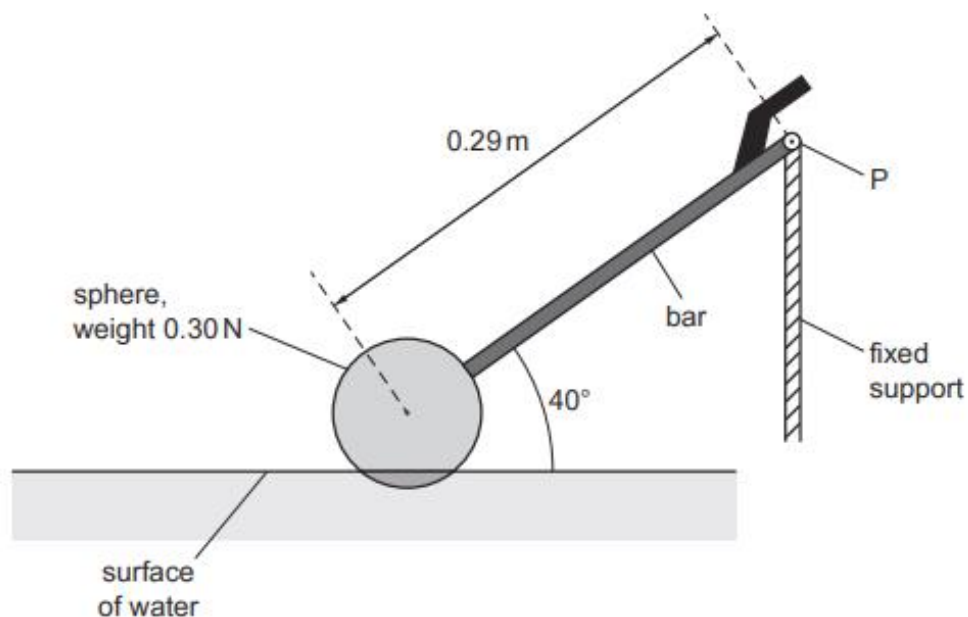


Fig. 3.1 (not to scale)

The sphere has weight 0.30 N. The distance from P to the centre of gravity of the sphere is 0.29 m. Assume that the weight of the bar is negligible.

Calculate the moment of the weight of the sphere about P.

moment = Nm [2]

- (c) The system shown in Fig. 3.1 is part of a mechanism that controls the amount of water in a tank.

Water enters the tank and causes the sphere to rise. This results in the bar becoming horizontal. Fig. 3.2 shows the system in its new position.

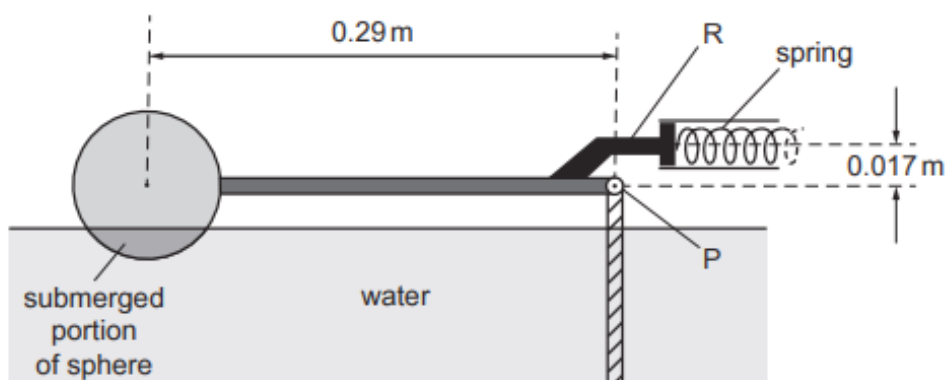
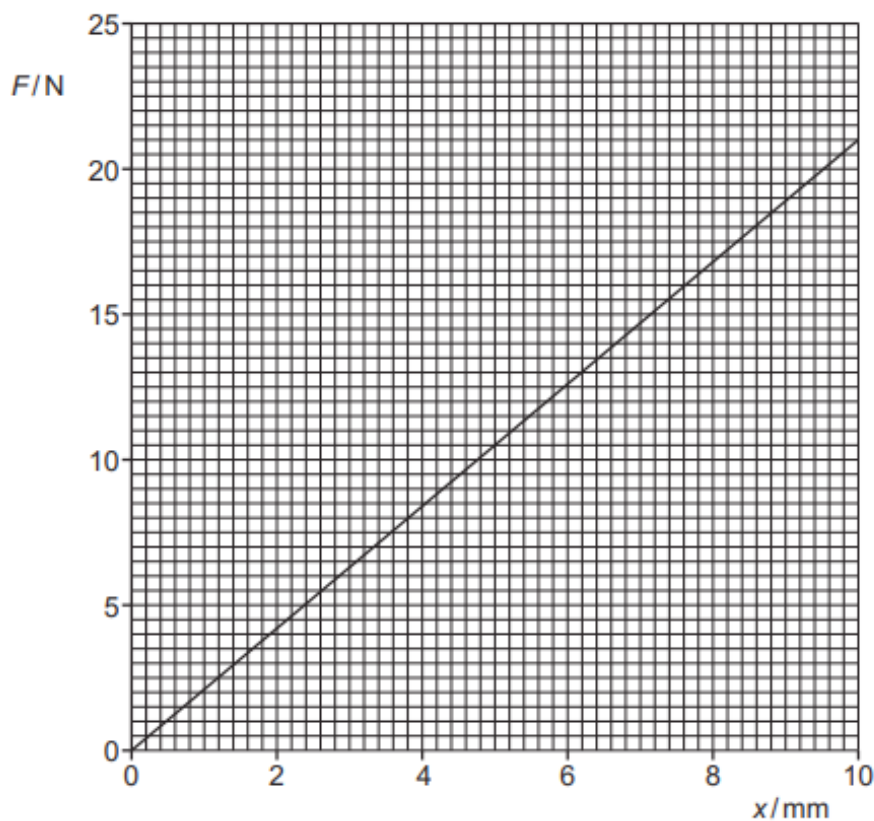


Fig. 3.2 (not to scale)

In this position the rod R exerts a force to compress a horizontal spring that controls the water supply to the tank. R is positioned at a perpendicular distance of 0.017 m above P.

The variation of the force F applied to the spring with compression x of the spring is shown in Fig. 3.3.



- (i) Use Fig. 3.3 to calculate the spring constant k of the spring.

$$k = \dots\dots\dots \text{Nm}^{-1} \quad [2]$$

- (ii) At the position shown in Fig. 3.2, the system is stationary and in equilibrium.

The radius of the sphere is 0.0480m and 26.0% of the volume of the sphere is submerged.

The density of water is $1.00 \times 10^3 \text{kgm}^{-3}$.

Show that the upthrust on the sphere is 1.18 N.

[2]

- (iii) By taking moments about P, determine the force exerted on the spring by the rod R.

$$\text{force} = \dots\dots\dots \text{N} \quad [2]$$

- (iv) Calculate the elastic potential energy E_p of the compressed spring.

$$E_p = \dots\dots\dots \text{ J [2]}$$

- (d) When the sphere moves from the position shown in Fig. 3.1 to the position shown in Fig. 3.2, the upthrust on the sphere does work.
Assume that resistive forces are negligible.

Explain why the work done by the upthrust is not equal to the gain in elastic potential energy of the spring.

.....
..... [1]

[Total: 13]

42. 9702/22/M/J/23/Q4

A spring is suspended from a fixed point at one end. The spring is extended by a vertical force applied to the other end. The variation of the applied force F with the length L of the spring is shown in Fig. 4.1.

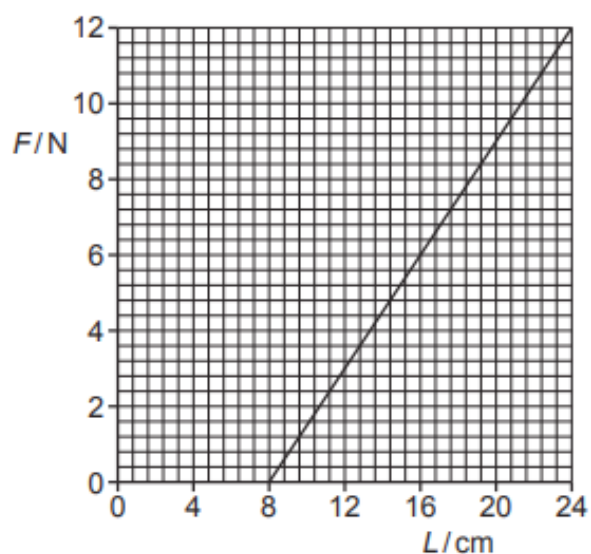


Fig. 4.1

For the spring:

- (a) state the name of the law that gives the relationship between the force and the extension

..... [1]

- (b) determine the spring constant, in Nm^{-1}

spring constant = Nm^{-1} [2]

(c) determine the elastic potential energy when $F = 6.0\text{ N}$.

elastic potential energy = J [2]

[Total: 5]

43. 9702/22/O/N/23/Q3

A vertical rod is fixed to the horizontal surface of a table, as shown in Fig. 3.1.

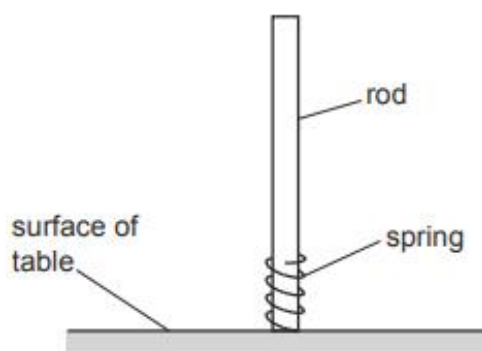


Fig. 3.1 (not to scale)

A spring of mass 7.5g is able to slide along the full length of the rod.

The spring is first pushed against the surface of the table so that it has an initial compression of 2.1 cm. The spring is then suddenly released so that it leaves the surface of the table with a kinetic energy of 0.048 J and then moves up the rod.

Assume that the spring obeys Hooke's law and that the initial elastic potential energy of the compressed spring is equal to the kinetic energy of the spring as it leaves the surface of the table. Air resistance is negligible.

- (a) By using the initial elastic potential energy of the compressed spring, calculate its spring constant.

spring constant = Nm^{-1} [2]

- (b) Calculate the speed of the spring as it leaves the surface of the table.

speed = ms^{-1} [2]

- (c) The spring rises to its maximum height up the rod from the surface of the table. This causes the gravitational potential energy of the spring to increase by 0.039 J.
- (i) Calculate, for this movement of the spring, the increase in height of the spring after leaving the surface of the table.

increase in height = m [2]

- (ii) Calculate the average frictional force exerted by the rod on the spring as it rises.

average frictional force = N [2]

- (d) The rod is replaced by another rod that exerts negligible frictional force on the moving spring. The initial compression x of the spring is now varied in order to vary the maximum increase in height Δh of the spring after leaving the surface of the table. Assume that the spring obeys Hooke's law for all compressions.

On Fig. 3.2, sketch a graph to show the variation with x of Δh . Numerical values are not required.

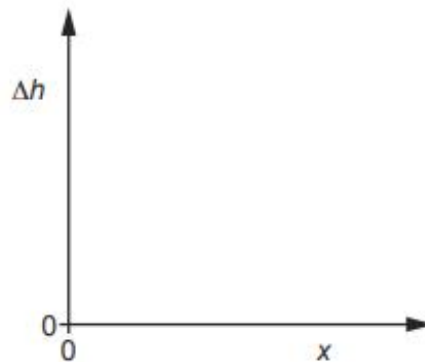


Fig. 3.2

[2]

[Total: 10]

44. 9702/23/O/N/23/Q4

Fig. 4.1 shows the variation with extension x of the tensile force F for two wires, G and H, made from the same material.

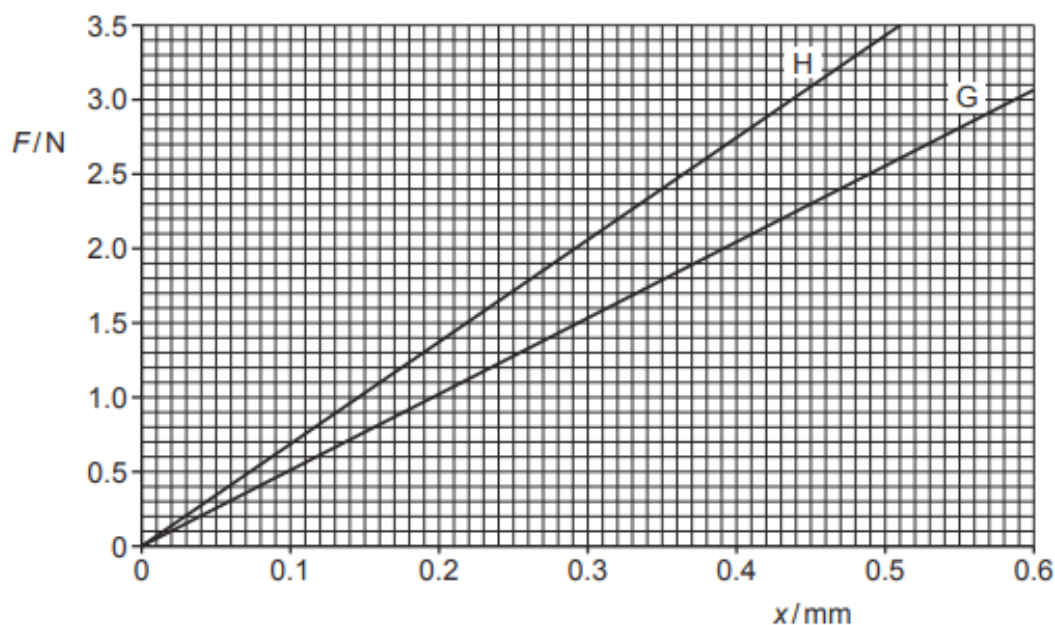


Fig. 4.1

The elastic limit has not been exceeded for G or H.

(a) For the lines in Fig. 4.1:

(i) state what is represented by the gradient

..... [1]

(ii) explain why the area under the line represents the elastic potential energy of the wire.

.....

 [2]

(b) Wires G and H are joined together end-to-end to form a composite wire of negligible weight. The composite wire hangs vertically from a fixed support.

A block of weight of 2.0 N is attached to the end of the wire, as shown in Fig. 4.2.

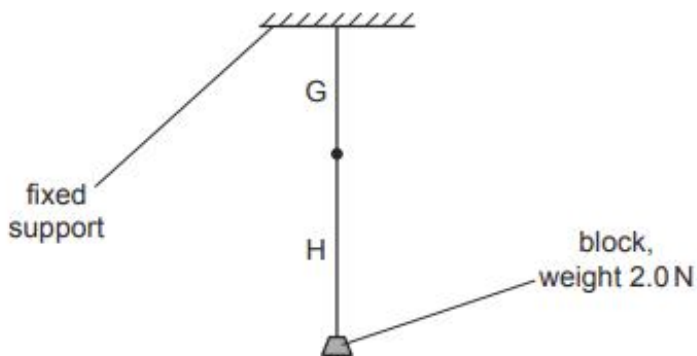


Fig. 4.2

(i) Use Fig. 4.1 to determine:

- the extension x_G of wire G

$$x_G = \dots\dots\dots \text{ mm}$$

- the extension x_H of wire H.

$$x_H = \dots\dots\dots \text{ mm}$$

[1]

(ii) Calculate the total elastic potential energy E_p of the composite wire due to the weight of the block.

$$E_p = \dots\dots\dots \text{ J} \quad [2]$$

(iii) The original length of wire G is L and the original length of wire H is $1.5L$.

Calculate the ratio

$$\frac{\text{cross-sectional area of wire G}}{\text{cross-sectional area of wire H}}.$$

$$\text{ratio} = \dots\dots\dots [3]$$

ANSWERS

- 1 (a) (i) change of shape / size / length / dimensionC1
when (deforming) force is removed, returns to original shape / sizeA1 [2]
(ii) $L = ke$ B1 [1]
- (b) $2e$ B1
 $\frac{1}{2}k$ (allow e.c.f. from extension)B1
 $\frac{1}{2}e$ and $2k$ B1
 $\frac{3}{2}e$ (allow e.c.f. from extension in part 2)B1
 $\frac{2}{3}k$ (allow e.c.f. from extension)B1 [5]
-
- 2 (a) (i) returns to original shape / size / length etc.B1
when load / distorting forces / weight / strain is removedB1 [2]
(ii) 1 $R = \rho L / A$ B1 [1]
2 $E = WL / Ae$ B1 [1]
- (b) $E = WR / e\rho$ C1
 $= (34 \times 0.44) / (7.7 \times 10^{-4} \times 9.2 \times 10^{-8})$ C1
 $= 2.1 \times 10^{11} \text{ Pa}$ A1 [3]
-
- 3 (a) ability to do workB1
as a result of a change of shape of an object/stretched etcB1 [2]
(b) work = average force $\times$ distance moved (in direction of the force)B1
either work = $\frac{1}{2} \times F \times x$
or work is area under F/x graph which is $\frac{1}{2}Fx$ B1
 $F = kx$ B1
so work / energy = $\frac{1}{2}kx^2$ A0 [3]
- (c) (i) spring constant = $\frac{3.8}{2.1}$ M1
 $= 1.8 \text{ N cm}^{-1}$ A0 [1]
(ii) 1 $\Delta E_P = mg\Delta h$ or $W\Delta h$ C1
 $= 3.8 \times 1.5 \times 10^{-2}$
 $= 0.057 \text{ J}$ A1 [2]
2 $\Delta E_S = \frac{1}{2} \times 1.8 \times 10^{-2} (0.036^2 - 0.021^2)$ M1
 $= 0.077 \text{ J}$ A0 [1]
3 work done = $0.077 - 0.057$
 $= 0.020 \text{ J}$ A1 [1]
(allow e.c.f. if $\Delta E_S > \Delta E_P$)
-

4	(a)	(i)	Young modulus = stress/strain	C1	
			data chosen using point in linear region of graph	M1	
			Young modulus = $(2.1 \times 10^8)/(1.9 \times 10^{-3})$ = 1.1×10^{11} Pa	A1	[3]
	(b)	(ii)	This mark was removed from the assessment, owing to a power-of-ten inconsistency in the printed question paper.		
			area between lines represents energy/area under curve represents energy .. when rubber is stretched and then released/two areas are different	M1 A1	
			this energy seen as thermal energy/heating/difference represents energy released as heat	A1	[3]
5	(a)	(i)	F/A	B1	[1]
		(ii)	$\Delta L/L$	B1	[1]
		(iii)	allow $FL/A\Delta L$	B1	[1]
		(iv)	allow $\rho L/A$ or $\rho(L + \Delta L)/A$	B1	[1]
	(b)	(i)	$\Delta L = FL/EA$ = $(30 \times 2.6)/(7.0 \times 10^{10} \times 3.8) \times 10^{-7}$ = 2.93×10^{-3} m = 2.93 mm	M1 A0	[1]
		(ii)	$\Delta R = \rho\Delta L/A$ = $(2.6 \times 10^{-8} \times 2.93 \times 10^{-3})/(3.8 \times 10^{-7})$ = $2.0 \times 10^{-4} \Omega$	C1 A1	[2]
	(c)		change in resistance is (very) small	M1	
			so method is not appropriate	A1	[2]
6	(a)		energy = average force $\times$ extension	B1	
			= $\frac{1}{2} \times F \times x$	B1	
			(Hooke's law) extension proportional to (applied) force	B1	
			hence $F = kx$	B1	
			so $E = \frac{1}{2} kx^2$	A0	[4]
	(b)	(i)	correct area shaded	B1	[1]
		(ii)	1.0 cm ² represents 1.0 mJ or correct units used in calculation $E_s = 6.4 \pm 0.2$ mJ (for answer $> \pm 0.2$ mJ but $\leq \pm 0.4$ mJ, then allow 2/3 marks)	C1 A2	[3]
		(iii)	arrangement of atoms / molecules is changed	B1	[1]
7	(a)	(i)	stress is force / area	B1	[1]
		(ii)	strain is extension / <u>original</u> length	B1	[1]
	(b)	(i)	$E = [F/A] \div [e/l]$ $e = (25 \times 1.7)/(5.74 \times 10^{-8} \times 1.6 \times 10^{11})$ $e = 4.6 \times 10^{-3}$ m	C1 C1 A1	[3]
		(ii)	A becomes A/2 or stress is doubled	B1	
			$e \propto l/A$ or substitution into full formula	B1	
			total extension increase is 4e	A1	[3]

8	(a)	clamped horizontal wire over pulley or vertical wire attached to ceiling with mass attached	B1	
		details: reference mark on wire with fixed scale alongside	B1	[2]
	(b)	measure original length of wire to reference mark with metre ruler / tape	(B1)	
		measure diameter with micrometer / digital calipers	(B1)	
		measure initial and final reading (for extension) with metre ruler or other suitable scale	(B1)	
		measure / record mass or weight used for the extension	(B1)	
		good physics method:		
		measure diameter in several places / remove load and check wire returns to original length / take several readings with different loads	(B1)	
		MAX of 4 points	B4	[4]
9	(a)	extension is proportional to force (for small extensions)	B1	[1]
	(b) (i)	point beyond which (the spring) does not return to its original length when the load is removed	B1	[1]
	(ii)	gradient of graph = 80 Nm^{-1}	A1	[1]
	(iii)	work done is area under graph / $\frac{1}{2} Fx$ / $\frac{1}{2} kx^2$	C1	
		= $0.5 \times 6.4 \times 0.08 = 0.256$ (allow 0.26) J	A1	[2]
	(c) (i)	extension = $0.08 + 0.04 = 0.12 \text{ m}$	A1	[1]
	(ii)	spring constant = $6.4 / 0.12 = 53.3 \text{ Nm}^{-1}$	A1	[1]
10	(a) (i)	stress = force / (cross-sectional) area	B1	[1]
	(ii)	strain = extension / <u>original</u> length or change in length / <u>original</u> length	B1	[1]
	(b)	<u>point</u> beyond which material does not return to the original length / shape / size when the load / force is removed	B1	[1]
	(c)	UTS is the maximum force / <u>original</u> cross-sectional area	M1	
		wire is able to support / before it breaks	A1	[2]
		allow one: maximum stress the wire is able to support / before it breaks		
	(d) (i)	straight line from (0,0)	M1	
		correct shape in plastic region	A1	[2]
	(ii)	only a straight line from (0,0)	B1	[1]
	(e) (i)	ductile: initially force proportional to extension then a large extension for small change in force	B1	
		brittle: force proportional to extension until it breaks	B1	[2]
	(ii) 1.	does not return to its original length / permanent extension (as entered plastic region)	B1	
	2.	returns to original length / no extension (as no plastic region / still in elastic region)	B1	[2]

11 (a)	Resultant force (and resultant torque) is zero	B1
	Weight (down) = force from/due to spring (up)	B1 [2]
(b) (i)	0.2, 0.6, 1.0s (<i>one of these</i>)	A1 [1]
(ii)	0, 0.8s (<i>one of these</i>)	A1 [1]
(iii)	0.2, 0.6, 1.0s (<i>one of these</i>)	A1 [1]
(c) (i)	Hooke's law: extension is proportional to the force (<i>not mass</i>)	B1
	Linear/straight line graph hence obeys Hooke's law	B1 [2]
(ii)	Use of the gradient (<i>not just $F = kx$</i>)	C1
	$K = (0.4 \times 9.8) / 15 \times 10^{-2}$	M1
	$= 26(.1) \text{ Nm}^{-1}$	A0 [2]
(iii)	<i>either</i> energy = area to left of line <i>or</i> energy = $\frac{1}{2} ke^2$	C1
	$= \frac{1}{2} \times [(0.4 \times 9.8) / 15 \times 10^{-2}] \times (15 \times 10^{-2})^2$	C1
	$= 0.294 \text{ J}$ (<i>allow 2 s.f.</i>)	A1 [3]
12 (a)	$E = \text{stress} / \text{strain}$	B1 [1]
(b) (i)	1. diameter / cross sectional area / radius	
	2. original length	B1 [1]
(ii)	measure original length with a <u>metre</u> ruler / tape	B1
	measure the <u>diameter</u> with micrometer (screw gauge)	B1 [2]
	<i>allow digital vernier calipers</i>	
(iii)	energy = $\frac{1}{2} Fe$ or area under graph or $\frac{1}{2} kx^2$	C1
	$= \frac{1}{2} \times 0.25 \times 10^{-3} \times 3 = 3.8 \times 10^{-4} \text{ J}$	A1 [2]
(c)	straight line through origin below original line	M1
	line through (0.25, 1.5)	A1 [2]
13 (a)	when the load is removed then the wire / body object does not return to its original shape / length	B1 [1]
(b) (i)	stress = force / area	C1
	$F = 220 \times 10^6 \times 1.54 \times 10^{-6} = 340 \text{ (338.8) N}$	A1 [2]
(ii)	$E = (F \times l) / (A \times e)$	C1
	$e = (90 \times 10^6) \times 1.75 / (1.2 \times 10^{11}) = 1.31 \times 10^{-3} \text{ m}$	A1 [2]
(c)	the stress is no longer proportional to the extension	B1 [1]

- 14 (a)** extension is proportional to force / load B1 [1]
- (b)** $F = mg$ C1
 $x = (mg / k) = 0.41 \times 9.81 / 25 = (4.02 / 25)$ M1
 $x = 0.16\text{m}$ A0 [2]
- (c) (i)** weight and (reaction) force from spring (which is equal to tension in spring) B1 [1]
- (ii)** $F - \text{weight}$ or $0.06 \times 25 = ma$ C1
 $F = 0.2209 \times 25 = 5.52 \text{ (N)}$ or $0.22 \times 25 = 5.5$
 $a = (5.52 - 0.41 \times 9.81) / 0.41$ or $1.5 / 0.41$ and $(5.5 - 4.02)$ C1
 $a = 3.7 \text{ (3.66) m s}^{-2}$ gives 3.6 m s^{-2} A1 [3]
- (d)** elastic potential energy / strain energy to kinetic energy and gravitational potential energy B1
stretching / extension reduces and velocity increases / height increases B1 [2]

- 15 (a)** the wire returns to its original length (not 'shape') M1
when the load is removed A1 [2]
- (b)** energy: $\text{N m} / \text{kg m}^2 \text{ s}^{-2}$ and volume m^3 C1
energy / volume: $\text{kg m}^2 \text{ s}^{-2} / \text{m}^3$ M1
energy / volume: $\text{kg m}^{-1} \text{ s}^{-2}$ A0 [2]
- (c)** ε has no units B1
 E : $\text{kg m s}^{-2} \text{ m}^{-2}$ M1
units of RHS: $\text{kg m}^{-1} \text{ s}^{-2} = \text{LHS units}$ / satisfactory conclusion to show C has no units A1 [3]

- 16 (a) (i)** stress = force / cross-sectional area B1 [1]
- (ii)** strain = extension / original length B1 [1]
- (b) (i)** $E = \text{stress} / \text{strain}$ C1
 $E = 0.17 \times 10^{12}$ C1
stress = $0.17 \times 10^{12} \times 0.095 / 100$ C1
 $= 1.6(2) \times 10^8 \text{ Pa}$ A1 [4]
- (ii)** force = (stress $\times$ area) = $1.615 \times 10^8 \times 0.18 \times 10^{-6}$ C1
 $= 29(.1) \text{ N}$ A1 [2]

17 (a)	(Young modulus / E =) stress / strain	B1	[1]
(b) (i)	stress = F/A or = $F/(\pi d^2/4)$ or = $F/(\pi d^2)$ ratio = 4 (or 4:1)	M1 A1	[2]
(ii)	E is the same for both wires (as same material) [e.g. $E_p = E_o$] strain = stress / E ratio = 4 (or 4:1) [must be same as (i)]	M1 A1	[2]
18 (a)	<u>add small mass</u> to cause extension then remove mass to see if spring returns to original length repeat for larger masses and note maximum mass for which, when load is removed, the spring does return to original length	M1 A1	[2]
(b)	Hooke's law requires force proportional to extension graph shows a straight line, hence obeys Hooke's law	B1 M1	[2]
(c)	k = force / extension = $(0.42 \times 9.81) / [(30 - 21.2) \times 10^{-2}]$ = 47 (46.8) Nm^{-1}	C1 C1 A1	[3]
19 (a)	(i) solid: (molecules) vibrate no translational motion / fixed position, liquid: translational motion (ii) gas: molecules have random (and translational) motion	B1 B1 B1	[2] [1]
(b)	(i) ductile: straight line through origin then curving towards x-axis (ii) brittle: straight line through origin with no or negligible curved region	B1 B1	[1] [1]
(c)	similarity: obey Hooke's law / $F \propto x$ or have elastic regions difference: brittle no or (very) little plastic region ductile has (large(r)) plastic region	B1 B1	[2]
20 (a)	(i) two sets of co-ordinates taken to determine a constant value (F/x) F/x constant hence obeys Hooke's law or gradient calculated and one point on line used to show no intercept hence obeys Hooke's law (ii) gradient or one point on line used e.g. $4.5 / 1.8 \times 10^{-2}$ (k =) 250 Nm^{-1} (iii) work done or E_p = area under graph or $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$ = $0.5 \times 4.5 \times 1.8 \times 10^{-2}$ or $0.5 \times 250 \times (1.8 \times 10^{-2})^2$ = 0.041 (0.0405) J	M1 A1 (M1) (A1) C1 A1 C1 A1	[2] [2] [3]
(b)	$\text{KE} = \frac{1}{2}mv^2$ $\frac{1}{2}mv^2 = 0.0405$ or $\text{KE} = 0.0405$ (J) ($v = [2 \times 0.0405 / 1.7]^{1/2} =$) 0.22 (0.218) ms^{-1}	C1 A1	[2]

21 (a) extension is proportional to force (for small extensions)	B1	[1]
(b) (i) point beyond which (the spring) does not return to its original length when the load is removed	B1	[1]
(ii) gradient of graph = 80 N m^{-1}	A1	[1]
(iii) work done is area under graph / $\frac{1}{2} Fx$ / $\frac{1}{2} kx^2$ $= 0.5 \times 6.4 \times 0.08 = 0.256 \text{ J}$ (allow 0.26 J)	C1	A1 [2]
22 (a) force/load is proportional to extension/compression (provided proportionality limit is not exceeded)	B1	[1]
(b) (i) $k = F/x$ or $k = \text{gradient}$ $k = 600 \text{ N m}^{-1}$	C1	A1 [2]
(ii) $(W =) \frac{1}{2} kx^2$ or $(W =) \frac{1}{2} Fx$ or $(W =) \text{area under graph}$ $(W =) 0.5 \times 600 \times (0.040)^2 = 0.48 \text{ J}$ or $(W =) 0.5 \times 24 \times 0.040 = 0.48 \text{ J}$	C1	A1 [2]
(iii) 1. $(E_k =) \frac{1}{2} mv^2$ $= \frac{1}{2} \times 0.025 \times 6.0^2$ $= 0.45 \text{ J}$	C1	A1 [2]
2. (work done against resistive force $=$) $0.48 - 0.45 [= 0.03(0) \text{ J}]$ average resistive force $= 0.030 / 0.040$ $= 0.75 \text{ N}$	C1	A1 [3]
(iv) efficiency $= [\text{useful energy out} / \text{total energy in}] (\times 100)$ $= [0.45 / 0.48] (\times 100)$ $= 0.94$ or 94%	C1	A1 [2]
23 (a) $E = \text{stress} / \text{strain}$ or $(F/A) / (e/l)$	C1	
$= [\text{gradient} \times 3.5] / [\pi \times (0.19 \times 10^{-3})^2]$	C1	
e.g. $E = [(40 - 5) / (11.6 - 3.2) \times 10^{-3}] \times 3.5 / [\pi \times (0.19 \times 10^{-3})^2]$ or		
$[4170 \times 3.5] / [\pi \times (0.19 \times 10^{-3})^2]$		
$E (= 1.3 \times 10^{11}) = 0.13 \text{ TPa}$ (allow answers in range 0.120–0.136 TPa)	A1	
(b) a larger <u>range</u> of F required or <u>range</u> greater than 35 N	B1	

24 (a) (strain =) extension / original length B1

(b) (i) $E = \sigma / \epsilon$ C1

$$\begin{aligned}\text{maximum stress} &= 2.1 \times 10^{11} \times 4.0 \times 10^{-4} \\ &= 8.4 \times 10^7 \text{ Pa}\end{aligned}\quad \text{A1}$$

(ii) $\sigma = F/A$ C1

$$\begin{aligned}\text{minimum area} &= 8.0 \times 10^3 / 8.4 \times 10^7 \\ &= 9.5 \times 10^{-5} \text{ m}^2\end{aligned}\quad \text{A1}$$

25 (a) $p = 1000 \times 9.81 \times 7.0 \times 10^{-2}$ or $1000 \times 9.81 \times 1.9 \times 10^{-2}$ C1

$$\begin{aligned}\Delta p &= 1000 \times 9.81 \times (7.0 \times 10^{-2} - 1.9 \times 10^{-2}) \text{ or } 686 - 186 \\ &= 500 \text{ Pa}\end{aligned}\quad \text{A1}$$

(b) $F = pA$ or $(\Delta)F = \Delta p \times A$ C1

$$\begin{aligned}\text{upthrust} &= 500 \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N} \\ \text{or}\end{aligned}\quad \text{A1}$$

$$\begin{aligned}\text{upthrust} &= (686 - 186) \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N} \\ \text{or}\end{aligned}$$

$$\text{upthrust} = 1000 \times 9.81 \times 5.1 \times 10^{-2} \times (5.1 \times 10^{-2})^2 = 1.3 \text{ N}$$

(c) force = 4.0 – 1.3 A1
= 2.7 N

(d) extension/x/e = 2.7/30 C1

$$= 0.09 \text{ (m) or } 9 \text{ (cm)} \quad \text{C1}$$

$$\begin{aligned}\text{height above surface} &= 9 - 7 \\ &= 2 \text{ cm}\end{aligned}\quad \text{A1}$$

(e) (i) mass = 4.0/9.81 C1

$$\begin{aligned}\text{acceleration} &= 2.7 / (4.0/9.81) \\ &= 6.6 \text{ m s}^{-2}\end{aligned}\quad \text{A1}$$

(ii) viscous force increases (and then becomes constant) M1

(weight and upthrust constant so) acceleration decreases (to zero) A1

26 (a)	$\rho = m/V$	C1
	$= (560/9.81)/(1.2 \times 0.018)$	A1
	$= 2600 \text{ kg m}^{-3}$	
(b)	$(\Delta)p = 940 \times 9.81 \times 1.2$	C1
	(upthrust $\Rightarrow$) $940 \times 9.81 \times 1.2 \times 0.018 = 200 \text{ N}$	A1
(c) (i)	tension $= 560 - 200$	A1
	$= 360 \text{ N}$	
(ii)	$P = Fv$	C1
	$= 360 \times 0.020$	A1
	$= 7.2 \text{ W}$	
(d) (i)	upthrust decreases	B1
	tension (in wire) increases	M1
	power (output of motor) increases	A1
	(ii) there is work done (on the cylinder) by the upthrust	B1
	or GPE of oil decreases (as it fills the space left by cylinder and so total energy is conserved)	
27 (a)	(Young modulus $\Rightarrow$) stress / strain	B1
(b) (i)	$k = F/\Delta L$ or 1/gradient	C1
	$= 90 \times 10^3 / (2 \times 10^{-3})$ (or other point on line)	A1
	$= 4.5 \times 10^7 \text{ Nm}^{-1}$	
(ii)	$E = \frac{1}{2}F\Delta L$ or $E = \frac{1}{2}k(\Delta L)^2$	C1
	$= \frac{1}{2} \times 90 \times 10^3 \times 2 \times 10^{-3}$ or $\frac{1}{2} \times 4.5 \times 10^7 \times (2 \times 10^{-3})^2$	C1
	$= 90 \text{ J}$	A1
(c)	straight line starting from (0, 150) and below original line	M1
	line ends at (90, 147)	A1

28 (a)	current temperature (allow amount of substance, luminous intensity) any two correct answers, 1 mark each	B2
(b) (i)	$W = 2 \times (150 \times \sin 17^\circ)$ or $2 \times (150 \times \cos 73^\circ)$ $W = 88 \text{ N}$	C1 A1
(ii)	1. $\sigma = F/A$ $= 150 / (7.5 \times 10^{-6})$ $= 2.0 \times 10^7 \text{ Pa}$	C1 A1
	2. $\varepsilon = \sigma / E$ $= 2.0 \times 10^7 / (2.1 \times 10^{11})$ $= 9.5 \times 10^{-5}$	C1 A1
29 (a) (i)	$p = mv$ $= 0.2(00) \times 6.(00) \times \sin 60(.0)^\circ$ or $0.2(00) \times 6.(00) \times \cos 30(.0)^\circ$ $= 1.04 \text{ kg m s}^{-1}$	C1 A1
(ii)	$0.300 \times v_x \times \sin 60.0^\circ = 1.04$ $v_x = 4.00 \text{ m s}^{-1}$	A1
(iii)	$0.30 \times 4.0 \times \cos 60^\circ$ or $0.20 \times 6.0 \times \cos 60^\circ$ or $(0.30 + 0.20)v$ or $0.50v$ $0.30 \times 4.0 \times \cos 60^\circ + 0.20 \times 6.0 \times \cos 60^\circ = (0.30 + 0.20)v$ or $0.50v$ so $v = 2.4 \text{ m s}^{-1}$	C1 A1
(b) (i)	$E = \frac{1}{2}mv^2$ $\frac{1}{2} \times 0.50 \times 2.4^2 = \frac{1}{2} \times 72 \times x^2$ $x = 0.20 \text{ m}$	C1 C1 A1
(ii)	1. straight line from the origin sloping upwards 2. line drawn from a positive value of E_k at $x = 0$ to a positive value of x at $E_k = 0$ line has an increasing downwards slope	B1 M1 A1

30 (a)	Hooke's (law)	B1
(b)(i)	$\sigma = F/A$	C1
	$= 36 / (4.1 \times 10^{-7})$	A1
	$= 8.8 \times 10^7 \text{ Pa}$	
	(ii) Young modulus $= \sigma / \epsilon$ or $F/A\epsilon$	C1
	$\epsilon = 8.8 \times 10^7 / (1.7 \times 10^{11})$	A1
	$= 5.2 \times 10^{-4}$	
(c)	$R = \rho L / A$	C1
	$\Delta R = \rho \Delta x / A$	C1
	$= 3.7 \times 10^{-7} \times 0.12 \times 10^{-3} / (4.1 \times 10^{-7})$	
	$= 1.1 \times 10^{-4} \Omega$	A1
(d)	remove the force/ F and wire returns to original length	B1

31 (a)	for a body in (rotational) equilibrium	B1
	sum/total of clockwise moments about a point = sum/total of anticlockwise moments about the (same) point	B1
(b)(i)	$(W \times 0.45)$ or (19×1.3) or $(W \times 1.85)$ or (22×2.6)	C1
	$(W \times 0.45) + (19 \times 1.3) + (W \times 1.85) = (22 \times 2.6)$ so $W = 14 \text{ N}$	A1
(ii)	magnitude $= 19 + 14 + 14 - 22$	A1
	$= 25 \text{ N}$	
	direction: vertically upwards	A1
(c)(i)	the extension is zero when the force is zero	B1
	graph is a straight line and (so) Hooke's law obeyed	B1
(ii)	$k = F/x$ or $k = \text{gradient}$	C1
	e.g. $k = 60 / (1.00 - 0.25)$	A1
	$k = 80 \text{ N m}^{-1}$	
(iii)	area shaded below graph line between $L = 0.25 \text{ m}$ and $L = 0.75 \text{ m}$	B1

32.

(a)(i)	(stress =) force / cross-sectional area
(a)(ii)	(strain =) extension / original length
(b)(i)	$E = FL / Ax$ $= GL / A$
(b)(ii)	straight line from origin above the original line line ends at point (4 small squares, F_1).
(b)(iii)	1. shaded area below the graph line and between the two vertical dashed lines 2. remove the force/ F/F_2 and the wire goes back to original length/zero extension
(b)(iv)	values have a large range

33.

(a)(i)	$F = kx$ $F_1 = 800 \times 0.045$ $= 36 \text{ N}$
(a)(ii)	$(E =) \frac{1}{2}kx^2$ or $\frac{1}{2}Fx$ or area under graph $\frac{1}{2} \times 800 \times (0.045)^2$ or $\frac{1}{2} \times 36 \times 0.045 = 0.81 \text{ (J)}$
(b)(i)	efficiency = $(0.72 / 0.81) \times 100$ $= 89\%$
(b)(ii)	$E = \frac{1}{2}mv^2$ $p = mv$ $0.72 = \frac{1}{2} \times 0.020 \times v^2$ and $p = 0.020 \times v$ $p = 0.17 \text{ N s}$
(c)(i)	$(\Delta)E = mg(\Delta)h$ $h = 0.60 / (0.020 \times 9.81) = 3.1 \text{ m}$
(c)(ii)	$F = (0.72 - 0.60) / 3.1$ $= 0.039 \text{ N}$
(c)(iii)	resultant force on ball is less (than that with air resistance) so time (taken) is more (than T)

34.

(a)	compression/extension is proportional to force (provided limit of proportionality is not exceeded)
(b)	$(E) = \frac{1}{2}Fx$ or $\frac{1}{2}kx^2$ or area under graph $= \frac{1}{2} \times 8 \times 16 \times 10^{-2} = 0.64 \text{ (J)}$ or $= \frac{1}{2} \times 50 \times (16 \times 10^{-2})^2 = 0.64 \text{ (J)}$
(c)(i)	$(E) = \frac{1}{2}mv^2$ $0.64 = \frac{1}{2} \times 0.076 \times v^2$ $v = 4.1 \text{ m s}^{-1}$
(c)(ii)	$(\Delta)(E) = mg(\Delta)h$ $= 0.076 \times 9.81 \times 0.24$ $(= 0.18 \text{ (J)})$ kinetic energy = $0.64 - 0.18$ $= 0.46 \text{ J}$
c)(iii)	$v = 4.1 \text{ m s}^{-1}$
(d)	$W = Fs$ $d = 0.30 + (2\pi \times 0.12) + 0.25 (= 1.3 \text{ m})$ $F = 0.23 / 1.3$ $= 0.18 \text{ N}$

35.

(a)	Hooke's (law)
(b)(i)	$k = F / x$ or $k = \text{gradient}$ e.g. $k = 7.0 / 5.0 \times 10^{-2}$ $= 140 \text{ N m}^{-1}$
(b)(ii)	$E = \frac{1}{2} F x$ or $E = \frac{1}{2} k x^2$ or $E = \text{area under graph}$ $= \frac{1}{2} \times 5.6 \times 4.0 \times 10^{-2}$ or $\frac{1}{2} \times 140 \times (4.0 \times 10^{-2})^2$ $= 0.11 \text{ J}$
(c)(i)	(upthrust =) $6.20 - 5.60 = 0.60 \text{ (N)}$
(c)(ii)	$\Delta p = \Delta F / A$ $= 0.60 / 1.2 \times 10^{-3}$ $= 500 \text{ Pa}$

(c)(iii)	$(\Delta)p = \rho g(\Delta)h$ $\rho = 500 / (9.81 \times 5.8 \times 10^{-2})$ $= 880 \text{ kg m}^{-3}$
(d)(i)	(upthrust) increases
(d)(ii)	(extension) decreases

36.

(a)	force $\times$ distance
	perpendicular distance of (line of action of) force from the point
(b)(i)	distance moved by pointer = $123 - 86 (= 37 \text{ mm})$ (extension =) $37 \times (1.8 / 52.6) = 1.3 \text{ (mm)}$ or sin or tan $\theta = 37 / 526$ (so $\theta = 4.0^\circ$ so extension =) sin or tan $\theta \times 18 = 1.3 \text{ (mm)}$
(b)(ii)	moment = $0.472 \times 9.81 \times 6.2 \times 10^{-2}$ $= 0.29 \text{ N m}$
(b)(iii)	$(\Delta)F \times 1.8 \times 10^{-2} = 0.29$ $\Delta F = 16 \text{ N}$
(b)(iv)	$k = F / x$ $= 16 / (1.3 \times 10^{-3})$ $= 1.2 \times 10^4 \text{ N m}^{-1}$

37.

(a)(i)	$\sigma = F / xy$
(a)(ii)	$\varepsilon = (z - w) / w$
(a)(iii)	$E = \sigma / \varepsilon$ $= Fw / xy(z - w)$
(b)(i)	extension = 2.2 mm (allow $2.0\text{--}2.4 \text{ mm}$)
(b)(ii)	strain energy = area under graph/line or $\frac{1}{2}Fx$ or $\frac{1}{2}kx^2$ $= \frac{1}{2} \times 120 \times 1.4 \times 10^{-3}$ or $\frac{1}{2} \times 8.6 \times 10^4 \times (1.4 \times 10^{-3})^2$ $= 0.084 \text{ J}$
(b)(iii)	(some of the) deformation of the wire is plastic/permanent/not elastic or wire goes past the elastic limit/enters plastic region energy (that cannot be recovered) is dissipated as thermal energy/becomes internal energy

38.

(a)	$k = F / \Delta L$ or F / x or gradient
	= e.g. $30 / (0.60 - 0.20)$
	= 75 N m^{-1}
(b)	$E = \frac{1}{2} F \Delta L$ or $\frac{1}{2} F x$ or $\frac{1}{2} k (\Delta L)^2$ or $\frac{1}{2} k x^2$ or area under graph
	= $\frac{1}{2} \times 15 \times 0.20$ or $\frac{1}{2} \times 75 \times 0.20^2$
	= 1.5 J

39.

(a)(i)	$t = 1.8 / 4.9$
	= 0.37 s
(a)(ii)	$v = u + at$
	= 9.81×0.37
a)(iii)	= 3.6 m s^{-1}
	$v^2 = 3.6^2 + 4.9^2$
	$v = 6.1 \text{ m s}^{-1}$
	$\theta = \tan^{-1} (3.6 / 4.9)$
(b)(i)	= 36°
	$\rho = m / V$
	$V = \frac{4}{3} \pi r^3$
	$\rho = 0.017 / [\frac{4}{3} \pi \times (0.016 / 2)^3]$
(b)(ii)	= 7900 kg m^{-3}
	$(E =) \frac{1}{2} m v^2$
(b)(ii)	$(E =) \frac{1}{2} \times 0.017 \times 4.9^2 = 0.20 \text{ (J)}$

(b)(iii)	$k = F / x$ or $k = \text{gradient}$
	e.g. $k = 6.4 / 10 \times 10^{-2}$ $= 64 \text{ N m}^{-1}$ (allow 63–65 N m^{-1})
(b)(iv)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$
	$x_0 = [(2 \times 0.20) / 64]^{0.5}$ $= 0.079 \text{ m}$ or 0.080 m
(c)(i)	same elastic potential energy / same (initial) kinetic energy and (polystyrene ball has) smaller mass (so greater speed) or same (average) force and (polystyrene ball has) smaller mass, (so greater average acceleration so greater speed)
(c)(ii)	(for the polystyrene ball there is) less (average vertical) acceleration / smaller (average vertical component of) <u>resultant</u> force (so takes longer time to reach ground)

40.

(a)	$E = \frac{1}{2}mv^2$
	$= \frac{1}{2} \times 0.25 \times 2.3^2$ $= 0.66 \text{ J}$
(b)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ and $F = kx$
	$x = [(2 \times 0.66) / 420]^{0.5}$ $= 0.056 \text{ m}$
	or
	$E = \frac{1}{2}kx^2$
	$\frac{1}{2}mv^2 = \frac{1}{2}kx^2$
	$x = (0.25 \times 2.3^2 / 420)^{0.5}$ $= 0.056 \text{ m}$
(c)(i)	($p =$) 0.25×2.3 or 0.25×1.5
	change in momentum = $0.25 (2.3 + 1.5)$ $= 0.95 \text{ N s}$
(c)(ii)	resultant force = $0.95 / 0.086$ or $0.25 \times (2.3 + 1.5) / 0.086$ $= 11 \text{ N}$
(d)	curved line from the origin with an increasing gradient

41.

(a)	sum of CW moments = sum of ACW moments
	about the same point for (an object in rotational) equilibrium
(b)	moment = $0.3(0) \times 0.29 \cos 40^\circ$ or $0.3(0) \times 0.222$
	= 0.067 Nm
(c)(i)	$k = F/x$ or $k = \text{gradient}$
	e.g. $k = 21/10 \times 10^{-3}$ $k = 2100 \text{ N m}^{-1}$
c)(ii)	$V_{\text{(sphere)}} = \frac{4}{3} \times \pi \times (0.0480)^3$
	$F = \rho g V$ (upthrust =) $1000 \times 9.81 \times (\frac{4}{3} \times \pi \times (0.048)^3) \times 0.26(0) = 1.18 \text{ (N)}$
c)(iii)	1.18×0.29 or 0.30×0.29 or $F \times 0.017$
	$(1.18 \times 0.29) = (0.30 \times 0.29) + (F \times 0.017)$ $F = 15 \text{ N}$
(c)(iv)	$E_{(P)} = \frac{1}{2} k x^2$ or $E_{(P)} = \frac{1}{2} F x$
	$x = F/k = 15/2100$ or x determined from graph for $F = 15.0 \text{ N}$ $E_P = \frac{1}{2} \times 2100 \times (15/2100)^2$ or $E_P = \frac{1}{2} \times 15 \times (15/2100)$ $E_P = 0.054 \text{ J}$
(d)	the <u>sphere</u> has gained gravitational potential energy

42.

(a)	Hooke's (law)
(b)	$k = F/x$ or $k = \text{gradient}$
	= e.g. $12.0 / (0.240 - 0.08)$ $= 75 \text{ N m}^{-1}$
(c)	$E = \frac{1}{2} F x$ or $E = \frac{1}{2} k x^2$ or $E = \text{area under graph}$
	$E = \frac{1}{2} \times 6.0 \times 0.080$ or $\frac{1}{2} \times 75 \times 0.08^2$ $= 0.24 \text{ J}$

43.

(a)	$E_{(P)} = \frac{1}{2}kx^2$ or $E_{(P)} = \frac{1}{2}Fx$ and $F = kx$
	$0.048 = \frac{1}{2} \times k \times (2.1 \times 10^{-2})^2$
	$k = 220 \text{ N m}^{-1}$
(b)	$E_{(K)} = \frac{1}{2}mv^2$
	$0.048 = \frac{1}{2} \times 7.5 \times 10^{-3} \times v^2$
	$v = 3.6 \text{ m s}^{-1}$
(c)(i)	$(\Delta)E = mg(\Delta)h$
	$0.039 = 7.5 \times 10^{-3} \times 9.81 \times (\Delta)h$
	$\Delta h = 0.53 \text{ m}$
(c)(ii)	$F \times 0.53 = 0.048 - 0.039$
	$F = 0.02 \text{ N}$
(d)	sketch: curved line from the origin
	curved line has increasing gradient

44.

(a)(i)	spring constant
(a)(ii)	area represents the work done (to extend the wire)
	work done (to extend the wire) is equal to elastic potential energy
(b)(i)	$x_G = 0.39 \text{ mm}$ and $x_H = 0.29 \text{ mm}$
(b)(ii)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ and $F = kx$ or $E = \text{area under graph}$
	for G: $E = \frac{1}{2} \times 2.0 \times (0.39 \times 10^{-3})$ or $\frac{1}{2} \times 5.1 \times 10^3 \times (0.39 \times 10^{-3})^2$
	for H: $E = \frac{1}{2} \times 2.0 \times (0.29 \times 10^{-3})$ or $\frac{1}{2} \times 6.9 \times 10^3 \times (0.29 \times 10^{-3})^2$
	$E_P = 3.9 \times 10^{-4} + 2.9 \times 10^{-4}$
	$= 6.8 \times 10^{-4} \text{ J}$
(b)(iii)	$E = FL / Ax$ or stress / strain = FL / Ax
	$A_G / A_H = 1 \times (0.29 \times 10^{-3}) / [1.5 \times (0.39 \times 10^{-3})]$
	ratio = 0.50